



EVIDENCE-BASED EDITION · 2026

The Trader's Strategy Book

*100 Scalping & Intraday Strategies with Rules,
Statistics and Backtesting Blueprints*

100

FOREX · CRYPTO · INDICES · GOLD · FUTURES

How to Use This Book

This book is a working reference, not a reading book. Each of the 100 strategies occupies exactly one page and follows the same structure so you can compare strategies like-for-like and lift the rules straight into backtesting code.

The page anatomy. Every strategy page shows: (1) a header with market, timeframe, style and an evidence grade; (2) a quick-stats table with win rate, reward-to-risk, profit factor, maximum drawdown and expectancy; (3) the concept and the source of the edge; (4) exact setup, entry and exit rules; (5) risk management; (6) the published backtest evidence or typical practitioner ranges; (7) when the strategy works and when it fails; (8) backtesting notes so you can code it; (9) common variations.

Evidence grades. Each strategy carries one of three grades, and you should treat them very differently:

- **Grade A — Published backtest.** Specific statistics from a named public backtest, academic paper or book are quoted, with the source named on the page. You can verify the numbers yourself.
- **Grade B — Documented practitioner ranges.** The strategy is widely documented and traded, and the statistics shown are consensus ranges from multiple practitioner sources rather than one verified study.
- **Grade C — Framework strategy.** The logic is sound and widely used professionally, but reliable public statistics are scarce. Treat the numbers as hypotheses to validate, not results.

The brutal honesty clause. A backtest is a hypothesis, not a promise. Published results are frequently produced on idealized fills, without realistic spread, slippage, latency or fees — and retail scalping strategies are the most sensitive of all to these costs. Research on breakout systems has shown that 3 points of slippage can turn a winning DAX strategy into a loser, and that realistic fees can erase the edge of 5-minute crypto systems entirely. Before you trade any page in this book with real money: (1) re-run the backtest yourself on your broker's data with your real costs; (2) demand out-of-sample and walk-forward validation; (3) paper-trade it for at least 30 sessions. No page in this book guarantees profit. Roughly 70–90% of retail day traders lose money in regulatory studies — discipline and cost control, not any single strategy, are what separate the minority who survive.

Reading the statistics. *Win rate* alone is meaningless: a 40% win rate at 2:1 reward-to-risk (+0.35R expectancy per trade) beats a 70% win rate at 0.5:1 (+0.05R). *Profit factor* = gross wins ÷ gross losses; above 1.3 is viable, above 1.75 is strong. *Expectancy* = (win% × avg win) – (loss% × avg loss); anything above +0.2R after costs is a real edge. *Max drawdown* is the worst peak-to-trough equity decline; size positions so you can psychologically survive 1.5× the backtested drawdown.

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PART I

Index & Equity Mean-Reversion Strategies

Mean reversion is the single most robust documented edge in equity indices: since S&P 500 futures began trading in 1982, broad US indices have shown a persistent tendency to bounce after 1–3 day sell-offs. These 13 strategies harvest that tendency at different speeds.

S01	RSI(2) Mean Reversion — The Connors Classic
S02	RSI(2) with Trend Filter — Quality Over Quantity
S03	Cumulative RSI — Smoothed Panic Detection
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S12	Three Down Days Reversal
S13	Double 7s (Connors)

S01 RSI(2) Mean Reversion — The Connors Classic

Market: S&P 500 (SPY, ES), Nasdaq (QQQ) • **Timeframe:** Daily signal, intraday execution • **Style:** Mean reversion • **Evidence:** Grade A (Connors & Alvarez, *Short Term Trading Strategies That Work*, 2008; QuantifiedStrategies backtest, SPY 1993–2025)

Win rate	Avg gain/trade	Profit factor	Max DD	Time in market
70–85% (76% documented)	0.5–0.95%	~1.9	15–34%	18–28%

Concept & edge. A 2-period RSI is hyper-sensitive: one or two panic days slam it below 10. Because equity indices mean-revert, buying that panic inside a long-term uptrend has produced win rates above 70% for three decades. The edge is behavioral — you are buying forced liquidation from weak hands.

Setup. Daily chart: 200-day SMA, 5-day SMA, RSI(2).

Entry rules. (1) Close above the 200-day SMA (uptrend filter). (2) RSI(2) closes below 10 (aggressive: below 5). (3) Buy at the close, or at next day's open for realistic fills.

Exit rules. Exit when price closes above the 5-day SMA. No fixed stop — the original research found stops hurt this system; risk is controlled by the trend filter and position size.

Risk management. Risk 1–2% of equity per trade via position sizing. Optional disaster stop at $2 \times \text{ATR}(10)$. Skip signals when VIX > 30 or ahead of FOMC/CPI.

Backtest evidence. On SPY 1993–2025: buying RSI(2) < 10, selling above RSI(2) > 80 returned ~9%/yr with a 34% max DD (invested 28% of the time); the exit variant "close above yesterday's high" cut drawdown to 15% with a 76% win rate. Independent 34-year replications confirm the high win rate and much lower drawdown than buy-and-hold, though CAGR trails buy-and-hold because exposure is low.

Works / fails. Works on broad equity indices and large-cap ETFs in bull and sideways regimes. Fails badly in crash phases (2008, March 2020 win rates dropped under 60%) and on single volatile stocks or commodities.

Backtesting notes. Trivial to code: ~30 lines in Python. Model next-open fills, not close fills, to avoid look-ahead. Test thresholds 4–12 to confirm robustness (avoid single-point curve fit). Include 0.02% slippage.

Variations. RSI(2) < 5 with scale-in (buy 25% at <10, add at <5, <3, <1); intraday RSI(2) on 30–60 min bars with thresholds 10/90; dual-filter version requiring price above both 50- and 200-day SMA (boosts win rate 5–10%).

S02 RSI(2) with Trend Filter — Quality Over Quantity

Market: SPY, QQQ, large-cap ETFs • **Timeframe:** Daily • **Style:** Mean reversion • **Evidence:** Grade A (QuantifiedStrategies, 1993–2025)

Win rate	Avg gain/trade	Max DD	CAGR	Trades/yr
75–79%	~0.95%	~31% → 15% with exit tweak	6.8–9%	8–15

Concept & edge. The plain RSI(2) occasionally buys falling knives. Requiring price above the 200-day SMA sacrifices some trades but removes the worst regimes. This is the version most professionals quote: the filter converts a raw signal into a tradeable system.

Setup. 200-day SMA, 5-day SMA, RSI(2) on daily bars.

Entry rules. (1) Close > 200 SMA. (2) RSI(2) < 5. (3) Enter at close (backtest) or next open (live).

Exit rules. Close above 5-day SMA. Optional secondary exit: RSI(2) > 70.

Risk management. 1% risk sizing; max 2 concurrent positions if running on multiple ETFs; accept that some years produce only a handful of trades — that is the price of quality.

Backtest evidence. Adding the 200-SMA filter raised average gain per trade from 0.9% to 0.95% and cut max drawdown from 34% to 31% while cutting exposure from 28% to 18%. Published screeners show 75–79% win rates on S&P 500 stocks with the filter applied.

Works / fails. Works in persistent uptrends; underperforms in V-bottom recoveries because the filter keeps you out after crashes (price is below the 200 SMA precisely when rebounds are strongest). Fails on instruments without an upward drift.

Backtesting notes. Compare filter on/off over identical periods; measure exposure-adjusted return (CAGR ÷ time in market), not raw CAGR. Watch for the long flat stretches — 2015–2016 and 2022 produce few signals.

Variations. Require RSI(2) < 10 for first tranche and < 5 for second; use 100-day SMA for more signals; add "close below 5-day SMA for 2 consecutive days" as a pullback-quality filter.

S03 Cumulative RSI — Smoothed Panic Detection

Market: S&P 500 futures (ES), also NQ, YM • **Timeframe:** Daily • **Style:** Mean reversion • **Evidence:** Grade A (StatOasis/Ali Casey backtest, ES 2006–2025, published 2021 with out-of-sample tracking)

Win rate	Trades	Avg trade	Losing years (2006–2025)	OOS result
67%	542	\$367/contract	1	+\$23,997 (2021–2025)

Concept & edge. RSI(2) became crowded after Connors popularized it; edge decays when everyone watches the same number. Cumulative RSI sums the last 2–3 RSI(2) readings, demanding *persistent* oversold conditions instead of a one-bar extreme — fewer whipsaws, smoother equity.

Setup. Daily bars. Compute RSI(2); Cumulative RSI = sum of the last 2 RSI(2) values.

Entry rules. Buy when Cumulative RSI(2,2) < 40. The 3D-optimized sweet spot across 3,900 parameter combinations was RSI period 2, sum bars 2, buy level 40.

Exit rules. Sell when Cumulative RSI > 25 (quick exit), or on close above the 5-day SMA.

Risk management. 1–1.5% risk per trade; no stop in the published version (exit is signal-based), but a 3× ATR disaster stop is a sane addition.

Backtest evidence. 542 trades on ES, 2006–2025: 67% win rate, \$367 average trade, only one losing year. Out-of-sample since publication (2021–2025): +\$23,997 per contract net, with 2022 and 2025 the only negative years — exactly the durability profile you want.

Works / fails. Works on US index futures through volatile regimes (2022 bear included). Like all mean reversion, it suffers in straight-line crashes; the summation delays entry slightly versus raw RSI(2).

Backtesting notes. Optimize over a *grid* and demand a plateau of profitable parameters, not a peak. The published optimization shows a stable region around (2, 2, 40/25) — replicate that plateau before trusting it. Weighted variants (recent RSI counts double) are worth testing.

Variations. Asymmetric summation (3 bars for entry, 1 for exit); combine with a 200-SMA regime filter; apply to QQQ/SPY ETFs instead of futures.

S04 IBS — Internal Bar Strength

Market: SPY, QQQ (best on equity indices) • **Timeframe:** Daily close signal • **Style:** Mean reversion • **Evidence:** Grade A (QuantifiedStrategies, SPY/QQQ 1993–2024)

Win rate	Avg gain/trade	CAGR	Max DD	Profit factor
68–74%	0.41–0.56%	12.5–16.6%	22–26%	1.9–2.7

Concept & edge. $IBS = (Close - Low) \div (High - Low)$. When a day closes near its low ($IBS < 0.2$), short-term sellers are exhausted; indices bounce. It is the simplest complete edge ever published — one formula, no parameters to overfit, profitable for 30 years.

Setup. Daily bars. Compute IBS from the just-closed day.

Entry rules. Buy at the close when $IBS < 0.2$.

Exit rules. Sell at the close when $IBS > 0.8$ (typically 1–5 days later).

Risk management. Full-equity or fixed-fraction sizing; a 10% disaster stop is optional but historically unused. Avoid scaling up after losses.

Backtest evidence. SPY 1993–2024: 919 trades, 68% win rate, 0.41% average gain, 12.5% CAGR vs 9.9% buy-and-hold, max DD –26%, profit factor 1.9. QQQ: 742 trades, 0.56%/trade, 16.6% CAGR. Only three losing years (1994, 2002, 2018). A two-indicator combination lifted the win rate to 78% (SPY) and 75% (QQQ) with 0.8–1.33% per trade.

Works / fails. Works on equity indices, *especially* in volatile/bear periods (2020 was its best year ever). Fails on commodity-linked markets (Norwegian/Australian stocks, gold, metals) and in dead-low-volatility grinds (2017).

Backtesting notes. Signals fire at the close — model MOC (market-on-close) orders or next-open fills and compare. Watch the exposure (~25–36%); risk-adjusted returns beat buy-and-hold even when raw CAGR is similar.

Variations. 2-day average $IBS < 0.25$ entry; $IBS + RSI(3)$ combo; short side ($IBS > 0.9$) — historically weak long-only edge, short side barely profitable, use as a filter not a signal.

S05 IBS + RSI Confirmation Combo

Market: SPY, QQQ, XLP • **Timeframe:** Daily • **Style:** Mean reversion • **Evidence:** Grade A (QuantifiedStrategies strategy #4, inception–2024)

Win rate	Avg gain/trade	Days/trade	Max DD	Profit factor
78% (SPY), 75% (QQQ)	0.8% (SPY), 1.33% (QQQ)	~4.8	–19.5 to –23.8%	2.0–2.9

Concept & edge. Two independent oversold gauges agreeing is rarer and stronger than either alone. IBS measures *where* the day closed; RSI measures *how fast* it fell. Agreement filters out slow grinds that keep grinding.

Setup. IBS on daily bars plus RSI(3).

Entry rules. Buy at close when $IBS < 0.2$ **and** $RSI(3) < 20$ (both conditions same day).

Exit rules. Sell at close when $IBS > 0.8$ or $RSI(3) > 70$, whichever first.

Risk management. Fewer trades (~278 on SPY since 1993) means each matters more; risk 1–2% sizing, no averaging down beyond one add.

Backtest evidence. SPY: 278 trades, 78% win rate, 0.8%/trade, profit factor 2.0, max DD –23.75%. QQQ: 232 trades, 75% win rate, 1.33%/trade, profit factor 2.9, max DD –19.5%. Average winner 1.7–2.4% vs average loser 1.75–2.3% — the win rate, not the asymmetry, carries the edge.

Works / fails. Best on indices and defensive sectors (XLP works well). Trade overlap between SPY and QQQ is very high — pick one. Fails during prolonged distribution phases (2000–2002 style).

Backtesting notes. Test each component alone first, then the combo — the combo must beat *both* on expectancy per unit of exposure to justify its lower frequency. Beware confirmation bias: with two conditions it's easy to keep tweaking thresholds until the backtest looks pretty (curve fitting).

Variations. IBS 2-day average < 0.3 + $RSI(3) < 30$ (more trades); add $VIX < 25$ regime filter; XLP-specific version with 2-day IBS average.

S06 Fade the Small Gap (SPY)

Market: SPY / ES • **Timeframe:** Daily open, intraday hold • **Style:** Opening-gap mean reversion • **Evidence:** Grade A (QuantifiedStrategies, SPY 2005–2012; verified on 30-min IQFeed data)

Win rate	Avg gain/fill	Target	Trades/yr	Style risk
80–89% (small samples)	0.19–0.22% (long)	75% of gap	40–60	Occasional large loser

Concept & edge. Small overnight gaps (–0.15% to –0.6%) in SPY are usually noise, not news; the opening auction overreacts and the gap fills during the session. Big gaps (>1%) are information — they trend, so don't fade those.

Setup. Yesterday's OHLC + today's open. Compute gap % and yesterday's close position in range: $(C-L)/(H-L)$.

Entry rules. (1) Gap down between –0.15% and –0.6%. (2) Yesterday's close in the lower quartile of its range (< 0.25). (3) Buy the opening print. (Mirror for gap-ups, though longs work better.)

Exit rules. Target = 75% of the gap size. If not reached, exit at the close. No other stop in the original — a $1\times$ gap-size stop is the common live addition.

Risk management. The distribution is "many small wins, occasional big loser" — cap size so one bad day costs $< 1.5\%$. Best later in the week (Wednesday–Friday statistically strongest) and late in the month.

Backtest evidence. 2010–2012 EOD test: 110 fills, 98 winners, +0.19% average. Intraday verification: 89 fills, +0.22% long, +0.08% short. Gaps outside yesterday's range averaged 0.111%; gaps after high-range days 0.185%.

Works / fails. Works in normal-volatility regimes on index ETFs. Fails on news-heavy mornings (CPI, FOMC, NFP), in crash regimes, and the edge has narrowed as algos arb it — re-validate on recent data.

Backtesting notes. EOD data overstates results (bad high/low quotes); use intraday data. Model the opening auction fill realistically — you will not always get the print.

Variations. Trade only Wednesday–Friday; only gaps outside yesterday's range; combine with $RSI(2) < 10$ for higher average per fill.

S07 ES Micro-Gap Fill (0.05–0.2% Gaps)

Market: E-mini S&P (ES/MES) • **Timeframe:** Session open • **Style:** Gap-fill intraday • **Evidence:** Grade A (Edgeful 6-month statistics + TradeThatSwing 3-month live algo run, 2026)

Win rate	Gap-fill rate	Profit factor	Max DD	3-month return
61–71%	88% (0.05–0.2% gaps)	1.75	8.5–9%	+29–44% (one contract, \$10k)

Concept & edge. The smallest overnight gaps on ES are pure overnight inventory adjustment and statistically fill ~88% of the time — for both gap-ups and gap-downs. This is one of the few scalping edges documented with live (not just backtested) algo results in 2026.

Setup. Previous RTH close vs today's RTH open; gap size filter 0.05–0.2%.

Entry rules. At the open, enter in the direction of the fill (short a small gap-up, long a small gap-down). Skip gaps > 0.2% — fill rates collapse.

Exit rules. Target: full gap fill (yesterday's close). Stop: beyond the open by $\sim 1\times$ gap size or a fixed dollar cap (\$500/contract in the live run). Time stop: flat by lunch if unfilled.

Risk management. Largest documented loser was \$512 on a \$10k account (5%) — the publisher himself recommends \$20k per contract to halve that. One trade per day maximum.

Backtest evidence. Live algo, 3 months: 14 trades, 10 winners (71%), +44% on \$10k; subsequent quarter +29% with 61% win rate, PF 1.75, 9% max DD. The underlying 88% fill statistic comes from 6 months of Edgeful data.

Works / fails. Works in quiet, range-bound overnight environments. Fails when gaps are news-driven (the filter excludes most of these by size, but not all — skip CPI/FOMC mornings). Tiny edge per trade means costs and fill quality decide everything.

Backtesting notes. You need RTH open/close data (not 24h session data). Model the fill at open + 1 tick, not at the open print. Track fill-rate statistics monthly — when fill rates decay, stop trading it.

Variations. MES micro contracts for small accounts; partial target at 50% fill with breakeven stop; apply to NQ with a wider size band (NQ gaps are larger in %).

S08 Turnaround Tuesday

Market: SPY / S&P 500 • **Timeframe:** Daily (Mon → Tue) • **Style:** Calendar mean reversion • **Evidence:** Grade A (QuantifiedStrategies, SPY 1993–2024)

Win rate	Avg gain/trade	CAGR	Exposure	Trades
56–69% (variant dependent)	0.3–0.46%	1.8–7%	2.5–10%	~212

Concept & edge. After a down Monday, Tuesday rebounds with remarkable persistence — documented for decades. Suspected drivers: weekend-news overreaction, Monday retail panic, institutional re-positioning. Buying Monday *strength* fails — the edge is strictly in buying weakness.

Setup. Daily bars; Monday return; optional IBS filter.

Entry rules. (1) Monday closes down (baseline) — refined version: Monday down AND IBS < 0.3. (2) Buy Monday's close.

Exit rules. Baseline: sell Tuesday's close. Extended variant: hold up to 3–4 days or exit on IBS > 0.8 — this lifted CAGR to 6.5–7% with 60–69% win rates.

Risk management. Tiny edge per trade (~0.3%) — costs matter; use commission-free ETF trading or futures. 1% risk sizing.

Backtest evidence. Raw rule: 212 trades, 56% win rate, +0.3%/trade, 1.8% CAGR at 2.5% exposure. With IBS filter: 57% WR, 2.7% CAGR. Extended holding: 60–69% WR, 6.5–7% CAGR. Flipped (buy Monday strength): negative — confirming the anomaly.

Works / fails. Works on SPY and other equity indices; robust across decades. Fails in strong downtrends (Mondays that start crash weeks keep falling); the raw edge is modest and needs filters.

Backtesting notes. 25 lines of code. Test day-of-week interactions: is Tuesday special, or does any down day rebound? (The latter is mostly true — Tuesday is the cleanest calendar instance.) Watch the small sample: ~50 Mondays/year.

Variations. Buy Monday close, sell Wednesday close; only trade when Monday's drop > 0.5%; combine with RSI(2) < 10 for rare but high-quality signals.

S09 The Overnight Edge (Buy Close, Sell Open)

Market: IWM, QQQ, DAX, SPY • **Timeframe:** Close → next open • **Style:** Overnight premium • **Evidence:** Grade A (QuantifiedStrategies published edges, 2023–2024; academic "overnight returns" literature)

Win rate	Avg gain/trade	Profit factor	Max DD	Exposure
60–77%	0.15–0.31%	1.5–4.0	–2 to –10%	2.5–7.3%

Concept & edge. Academically, essentially *all* of the S&P 500's long-term return has accrued overnight; the intraday session nets to roughly zero over decades. Structural causes: closing-auction imbalances, overnight risk premium, retail market-on-open orders. You hold ~14 hours to capture the premium while avoiding intraday chop.

Setup. Any liquid index ETF/future. The published IWM variant adds one short-term condition (a down day / low IBS filter).

Entry rules. Buy at the close (MOC order). The filtered versions require a same-day weakness condition (e.g., close in lower half of the day's range).

Exit rules. Sell at next morning's open (MOO order). Never hold past the first 5 minutes.

Risk management. Overnight gap risk is the whole risk — size for a 2–3% adverse gap. Avoid holding overnight before CPI, FOMC, or earnings-heavy sessions.

Backtest evidence. Published IWM edge: 152 trades, 77% win rate, +0.31%/trade, profit factor 4, max DD –2% at 2.5% exposure. QQQ variants: 60–61% WR, PF 1.5–1.9. DAX: 202 trades, 66% WR, PF 2.4, max DD –4%.

Works / fails. Works on equity indices (US and European); far weaker or negative on forex and commodities. Fails in crash regimes when overnight gaps go both ways violently.

Backtesting notes. MOC/MOO fills are realistically achievable (auction participation), so this backtests honestly. Check the short side too — historically ~flat, so long-only is fine. Split results by day of week; Monday-overnight and Friday-overnight differ.

Variations. Overnight + RSI(2) < 10; "close below open today" trigger only; DAX version buying the 17:30 CET close, selling the 09:00 open.

S10 5-Day Moving Average Snapback

Market: S&P 500 • **Timeframe:** Daily • **Style:** Mean reversion • **Evidence:** Grade A (QuantifiedStrategies moving-average study, S&P 500)

Win rate	CAGR	Max DD	Profile	Best MA length
~65% (typical)	8.18% (5-day)	−27.9%	High-frequency MR	5 days

Concept & edge. A rigorous sweep of MA lengths 5–200 showed a clean pattern: *short* averages work as mean-reversion triggers (buy below, sell above), *long* averages work as trend filters. The 5-day is the sweet spot: near buy-and-hold returns with far less exposure.

Setup. 5-day SMA on daily bars.

Entry rules. Buy at close when price closes below the 5-day SMA.

Exit rules. Sell at close when price closes above the 5-day SMA.

Risk management. No stop in the study; add a 3× ATR disaster stop for live use. Whipsaw risk in sideways markets is the main cost.

Backtest evidence. S&P 500 long-term: 5-day version CAGR 8.18% vs 10-day 7.8%, 25-day 5.09%, 200-day 2.93% (mean-reversion direction). Max DD −27.9% for the 5-day vs −49 to −62% for the trend-direction variants. The shorter the MA, the better the mean-reversion performance — monotonic relationship.

Works / fails. Works on equity indices with upward drift. Fails in trending bear markets (you keep buying weakness) and on instruments without mean-reverting character (commodities, many forex pairs).

Backtesting notes. This is a teaching-system as much as a trading system: use it to learn how exposure-adjusted returns work. Code it in 20 lines, then test MA lengths 3–10 to confirm the plateau.

Variations. Buy only if close < 5-day SMA by > 0.5% (depth filter); exit on close > 5-day SMA for 2 consecutive days; combine with 200-SMA trend filter (buys only in uptrends).

S11 Bollinger %B Reversal

Market: SPY, QQQ, EUR/USD H1 • **Timeframe:** Daily / H1 • **Style:** Band mean reversion • **Evidence:** Grade B (documented across multiple practitioner backtests; mean-reversion meta-studies show 65–75% WR for band systems)

Win rate	Typical RR	Profit factor	Max DD	Best regime
60–75%	1:1 to 1:1.5	1.3–1.6	10–20%	Range-bound

Concept & edge. Price spends ~95% of its time inside 2-standard-deviation Bollinger Bands. Touches of the outer band in *ranging* markets are statistically likely to revert toward the mean (the 20 SMA). The edge is strongest when the bands are flat, not expanding.

Setup. Bollinger Bands (20, 2). Optional %B indicator (0 = lower band, 1 = upper band). ADX(14) < 20 as range filter.

Entry rules. Long: close below the lower band with %B < -0.05 AND ADX < 20; enter on the first close back inside the band. Short: mirror.

Exit rules. Target: the 20 SMA (band midline). Aggressive: opposite band. Stop: $1 \times \text{ATR}(14)$ beyond the entry bar's extreme, or a close beyond the band for 2 consecutive bars.

Risk management. Band rides are the killer — in trends price walks the band for days. The ADX filter is non-negotiable. Risk 0.5–1% per trade.

Backtest evidence. Documented band-reversion systems on stock indices show 60–75% win rates with modest per-trade gains and 10–20% drawdowns; 2025 out-of-sample tests of refined versions reported +30.4% with 72% WR and -10.2% max DD vs Nasdaq's +24.4%. Meta-analyses place mean-reversion systems at 65–75% win rate vs 10–40% for trend systems.

Works / fails. Works in low-ADX, range-bound conditions on indices and major forex pairs. Fails catastrophically in breakouts/trends without the regime filter — the single most common way beginners blow up with bands.

Backtesting notes. The regime filter (ADX or band width percentile) is part of the strategy, not an optional extra — test with and without. Use %B rather than raw price so the code is instrument-agnostic.

Variations. Double-band (1.5σ entry, 2.5σ stop); %B divergence entry; Keltner-Bollinger combination (trade reversion only when KC is inside BB).

S12 Three Down Days Reversal

Market: S&P 500, QQQ • **Timeframe:** Daily • **Style:** Streak mean reversion • **Evidence:** Grade B (documented in Connors research and multiple quant blogs)

Win rate	Avg gain/trade	Hold	Max DD	Frequency
60–68%	0.4–0.7%	1–3 days	15–25%	15–30 signals/yr

Concept & edge. Three consecutive lower closes in an uptrend is a statistically stretched state; the fourth day has historically been positive well above chance. Streaks trigger capitulation psychology — weak hands exit exactly when odds improve.

Setup. Daily bars; 200-SMA trend filter; count consecutive down closes.

Entry rules. (1) Price above 200 SMA. (2) Three consecutive down closes (stronger: each day closes in its lower half). (3) Buy day-3 close.

Exit rules. Sell on the first up close, or after a fixed 3 days. Exit all if price closes below the 200 SMA.

Risk management. 1% risk sizing; streaks of 5–6 down days happen in real sell-offs, so never add to a 4th/5th down day without the trend filter intact.

Backtest evidence. Connors-group research on 3-day streaks in the S&P 500 documented win rates in the low-to-mid 60s with ~0.5% average gains per trade over multi-decade samples, degrading below the 200 SMA. Consistent with the broader 2–3 day panic literature (RSI-2, IBS).

Works / fails. Works on equity indices in uptrends. Fails in bear markets (streaks extend) and on low-liquidity instruments where 3 down days is normal noise.

Backtesting notes. Count the conditional probabilities yourself: P(up day | 3 down days, above 200 SMA) vs unconditional P(up day). The difference is the edge; demand it exceed 8–10 points. Compare 2, 3, 4-day streaks.

Variations. Require cumulative 3-day loss > 2%; scale-in across days 3–5; pair with IBS < 0.2 on day 3.

S13 Double 7s (Connors)

Market: SPY / S&P 500 • **Timeframe:** Daily • **Style:** Pattern mean reversion • **Evidence:** Grade A (Connors & Alvarez, *Short Term Trading Strategies That Work*, 2008)

Win rate	Avg gain/trade	Profit factor	Trades	Hold
~80%+ (published)	~0.6–0.8%	~2	~100+ over decades	3–7 days

Concept & edge. A simple two-condition pattern: price above the 200-day SMA, and a close at a 7-day low. Buying 7-day lows in an uptrend and exiting at 7-day highs captured the index's bid with a very high win rate in the original published tests.

Setup. 200-day SMA; 7-day highest high / lowest low.

Entry rules. (1) Close above 200 SMA. (2) Today closes at the lowest close of the last 7 days. (3) Buy at close.

Exit rules. Sell at close when price closes at a 7-day high.

Risk management. No stop in the original (signal exit only); position-size for a 2–3% adverse move. Adding a stop historically reduced performance — document this trade-off in your own test.

Backtest evidence. Published in 2008 on S&P 500 data: win rates around 80% with average gains near 0.7% per trade over hundreds of historical occurrences, materially better than random-entry baselines. The pattern is a close cousin of RSI(2)/IBS and shares their regime dependence.

Works / fails. Works on equity indices in structural uptrends; the "7" generalizes (5–10 day lookbacks all work, showing robustness). Fails in bear regimes — the 200-SMA filter is load-bearing.

Backtesting notes. Test lookbacks 5–10 to confirm the plateau. As with all on-close systems, compare close fills vs next-open fills. Expect long quiet stretches with zero signals.

Variations. Scale-in: buy $\frac{1}{3}$ at 7-day low, $\frac{1}{3}$ at 10-day low, $\frac{1}{3}$ at 14-day low; exit at 5-day high for faster turnover; ETF-rotation version across SPY/QQQ/DIA/IWM.

PART II

Opening Range & Session Breakout Strategies

The open is the highest-volume, highest-information window of any session. Toby Crabel's *Day Trading with Short-Term Price Patterns and Opening Range Breakout* (1990) first documented the edge statistically. Modern mega-backtests (1.1M+ trades) give us unusually solid numbers for this family.

S14	ORB — 5-Minute Opening Range Breakout
S15	ORB — 15-Minute Opening Range Breakout
S16	ORB — 30-Minute Opening Range Breakout
S17	ORB with Daily Trend Filter (NQ Variant)
S18	DAX Opening Range Breakout (14-Year Study)
S19	London Breakout (Asian Range Break)
S20	Asian Session Range Fade
S21	New York Open Breakout (Equities)
S22	London–NY Overlap Momentum
S23	Initial Balance Breakout (Futures)
S24	The 10 O'Clock Reversal (NYSE)
S25	Opening Drive Continuation (Gap-and-Go)

S14 ORB — 5-Minute Opening Range Breakout

Market: US stocks & index futures (ES, NQ) • **Timeframe:** 1-min/5-min • **Style:** Momentum breakout • **Evidence:** Grade A (ORB Setups mega-backtest: 1,178,668 trades, 600+ symbols, 2025–26)

Win rate	Expectancy	Best RR target	Trades/day	Edge source
52.2–53.8%	+0.028R/trade	0.5–1R	1–3	Trend-day capture

Concept & edge. The first 5 minutes define a battle zone. A decisive break of that zone on volume signals that the opening price discovery resolved in one direction — and the first directional move predicts the session's trend ~58% of the time on the S&P 500 (Journal of Technical Analysis).

Setup. Mark the high/low of 09:30–09:35 ET. Volume average for confirmation; VWAP for bias.

Entry rules. Enter on a candle *close* beyond the 5-min range (not a wick), with breakout-bar volume above the 20-bar average. Optional: only in the direction of the overnight gap.

Exit rules. Target 1 = 50% of range height (highest documented win rate on ES: 72% with optimized settings). Target 2 = 100–200% of range. Stop at the opposite range edge; trail after 1R.

Risk management. Max 2 attempts per day. Skip if range > 0.55% of price (documented filter) — wide ranges have broken risk/reward. Risk 0.5–1% per attempt.

Backtest evidence. Across 591 symbols: 5-min ORB average win rate 52.2–53.8% (highest of all ORB timeframes) with +0.028R expectancy. An optimized ES variant (50% range target, 0.55% max range) returned 108% in 6 months with 72.17% WR and PF 1.62. NQ longs-only variant: 114 trades, 74.56% WR, PF 2.51, ~12% DD.

Works / fails. Works on high-relative-volume names and index futures on catalyst days. Fails on choppy, narrow-range days and low-float manipulation. Slippage at the open is the #1 killer — see S18's slippage table.

Backtesting notes. You need 1-minute data with realistic volume. Model fills at range edge + 1–2 ticks slippage. Test target/stop combinations systematically: target choice swings results by 30 percentage points.

Variations. Retest entry (wait for pullback to range edge); half-size first entry, add on 15-min confirmation; longs-only in bull regimes.

S15 ORB — 15-Minute Opening Range Breakout

Market: Stocks, ES/NQ futures, forex session opens • **Timeframe:** 5-min execution • **Style:** Momentum breakout • **Evidence:** Grade A (ORB Setups, 605 symbols; TradeAlgo 10-yr S&P study)

Win rate	Expectancy	Avg RR	Signal quality	Best use
46–56%	+0.004R to +0.35R	1.8:1	Balanced	All-round default

Concept & edge. The 15-minute range is the professional default: long enough to filter opening noise, early enough to catch the morning leg. The mega-backtest shows it has the lowest raw win rate of the three classic ORBs — but the *highest expectancy per trade* when paired with full-range targets, because breakouts after 15 minutes of balance tend to travel further.

Setup. High/low of 09:30–09:45 ET; VWAP; relative volume vs the same time yesterday.

Entry rules. Close beyond the range on $\geq 1.5\times$ average volume; bias long if price holds above VWAP, short below. Require the opening range to be a sensible fraction of 14-day ATR.

Exit rules. Stop at opposite range edge (or mid-range for tighter risk). Targets at 1R and 2R; let the final third trail for trend days — a few trend days per month pay for all the small losses.

Risk management. The 15-min ORB is a reward-to-risk strategy, not an accuracy strategy: expect to lose ~half your trades. Daily loss limit of 2 failed breakouts.

Backtest evidence. 6,748 symbol-records: average WR 52.5%, expectancy +0.004R raw — but the full-target configuration earned the highest per-trade EV in the 1.17M-trade study despite only 46% WR. Ten-year S&P data shows 56% WR at 1.8:1 average RR with proper filters.

Works / fails. Works on stocks with catalysts (earnings, upgrades) and trending index days. Fails in low-volatility balance and on Fed-day mornings before the announcement.

Backtesting notes. Segment results by relative volume quintile and gap size — the edge concentrates in the top quintiles. Compare stop-at-edge vs stop-at-midpoint: midpoint stops lift WR but cut RR.

Variations. Enter on the first pullback after breakout; "ORB + relative strength vs SPY" filter (trade only the day's strongest/weakest names); apply to London open on GBP pairs.

S16 ORB — 30-Minute Opening Range Breakout

Market: Index futures, large-cap stocks • **Timeframe:** 5-min/15-min execution • **Style:** Momentum breakout • **Evidence:** Grade A (ORB Setups, 608 symbols; OptionAlpha 0DTE backtests)

Win rate	Expectancy	Signal count	Character	Best use
49.4–52.9%	+0.019R	Fewest/day	Highest conviction	Trend days

Concept & edge. Thirty minutes of balance compresses more energy; breaks of the 30-min range are fewer but cleaner, and they shine on genuine trend days. This is the patient trader's ORB: one decision by 10:00 ET, then manage.

Setup. High/low of 09:30–10:00 ET; VWAP; daily ATR context.

Entry rules. Close beyond the 30-min range with volume expansion. Strong filter: trade only in the direction of the higher-timeframe trend (above/below 20-day SMA).

Exit rules. Stop at the opposite edge. Scale out at 1R and 2R; trail the remainder below 15-min swing lows. Flat by 15:45 ET.

Risk management. The wider range means wider stops — size down accordingly so risk stays at 0.5–1%. If the 30-min range is > 40% of daily ATR, the move is likely exhausted: skip.

Backtest evidence. 6,982 symbol-records: average WR 52.9% (highest of the three ORBs), +0.019R expectancy. Related 60-min ORB variant on SPX 0DTE spreads: 89.4% WR, PF 1.44 (options structure, high-WR/low-RR profile).

Works / fails. Works on macro-driven trend days and heavy-news mornings. Fails in summer chop and quad-witching weirdness; produces nothing on quiet days (that's a feature).

Backtesting notes. Test exit-time sensitivity — the edge should be robust to exiting at 12:00 vs 15:30. If it isn't, you've curve-fit. Compare longs vs shorts separately; short ORBs are historically weaker on indices.

Variations. 60-min range for maximum conviction; two-stage entry (half at break, half at retest); pair with 0DTE iron structures for defined-risk versions.

S17 ORB with Daily Trend Filter (NQ Variant)

Market: Nasdaq futures (NQ/MNQ) • **Timeframe:** 1-min signal + daily filter • **Style:** Filtered breakout • **Evidence:** Grade A
(QuantifiedStrategies live-traded strategy, NQ)

Win rate	Avg gain/trade	Profit factor	Trades	Structure
65%	0.27%	2.0	198	1-min ORB + 1 daily filter

Concept & edge. Raw intraday ORB on index futures is barely profitable (best case ~0.04%/trade in unfiltered tests — not tradeable after costs). Adding a single daily-bar filter (trend/mean-reversion context) transformed the same breakout into a 65%-win-rate, PF-2.0 system. The lesson generalizes: intraday signals need daily context.

Setup. 1-minute bars for the opening range; one daily filter (e.g., only long after an oversold daily reading, or only in the direction of the 20-day trend).

Entry rules. Define the opening range over the first n minutes; buy the breakout above it only when the daily filter agrees. One trade per day.

Exit rules. Exit at the daily session close (15:00 Chicago) — time exit, no intraday stop in the published version. Live traders add a stop at the range low.

Risk management. One trade/day keeps commission drag minimal. Risk max 1% per day; consider the micro (MNQ) for accounts under \$25k.

Backtest evidence. NQ, 198 trades: 65% win rate, 0.27% average gain per trade, profit factor 2.0 — traded live by the publisher. Unfiltered baseline: ~0.04%/trade, sub-50% win rate (not viable).

Works / fails. Works on volatile index futures (NQ > ES) in trending years. Fails in mean-reverting chop regimes and whenever the daily filter and intraday action conflict persistently.

Backtesting notes. The filter is proprietary in the source; test your own daily filters systematically (RSI(2) < 10, above/below 20-SMA, yesterday's IBS). Measure how each filter changes frequency and expectancy — you want a filter that cuts trades by half and doubles expectancy.

Variations. Short side with its own daily filter (weaker historically); apply the same structure to RTY or YM; exit half at 1R, half at close.

S18 DAX Opening Range Breakout (14-Year Study)

Market: DAX-40 index (FDAX/CFD) • **Timeframe:** H1 range definition • **Style:** Session breakout • **Evidence:** Grade A (FXVPS 14-year study, 2,199–2,899 trades, 2011–2025)

Win rate	Profit factor	Total	Max DD	Sharpe
49.4% (H1)	1.25	+206R	−21.4R	1.46

Concept & edge. The DAX cash open (09:00 CET) concentrates European institutional flow. A break of the pre-open range, targeted at 2R with the stop one range-width away, compounded steadily for 14 years — positive in 9 of 14 years, worst year under −10R. This is one of the most honestly reported ORB studies available.

Setup. Overnight/pre-open range; H1 (60-min) opening range performed best; realistic spread included.

Entry rules. Resting stop orders at the range edges; first break triggers entry. Stop at the opposite edge; target at 2 range-widths.

Exit rules. Time-exit flexibility is documented: flattening anywhere from 11:00 to the 17:00 cash close keeps PF in a tight 1.20–1.27 band — holding to the close captures more (+243R total). Every weekday profitable; Tuesday/Thursday strongest (PF 1.35/1.39).

Risk management. *Execution is the strategy.* Slippage sensitivity is published: +1 pt slippage → PF 1.20; +3 pts → M15 version goes negative. Use a low-latency VPS near the exchange, and stop orders (not market orders after the fact).

Backtest evidence. H1: 2,199 trades, 49.4% WR, PF 1.25, +206R, Sharpe 1.46. M15: 2,899 trades, PF 1.11 — viable but fragile to slippage. Expectancy ~0.1R/trade: thin but extremely consistent.

Works / fails. Works because it's robust across exit times and weekdays (anti-curve-fit shape). Fails with retail-grade latency; the edge is ~0.1R and costs can eat all of it.

Backtesting notes. Model spread + slippage as the *primary* variable. Run the slippage stress table yourself. If your backtest can't model open-auction fills, assume 2 pts.

Variations. Trade only Tue/Thu (PF 1.35–1.39); half target at 1R + runner; apply the identical structure to the EuroStoxx 50 open.

S19 London Breakout (Asian Range Break)

Market: GBP/USD, GBP/JPY, EUR/USD • **Timeframe:** M15–H1 • **Style:** Session breakout • **Evidence:** Grade B (multiple independent 300+ trade backtests; typical WR 48–60%, PF 1.3–1.74)

Win rate	Profit factor	Typical RR	Max DD	Monthly target
48–60%	1.3–1.74	1.5:1	15–20%	2–5% (realistic)

Concept & edge. The Asian session builds a tight range; London's liquidity arrival (08:00 London time) breaks it with genuine directional intent. You're trading the day's first institutional repricing.

Setup. Mark the Asian range (roughly 00:00–07:00 London / 02:00–07:00 UTC depending on convention; one academic EUR/USD study found the 05:00–07:00 GMT window optimal).

Entry rules. Enter on a close beyond the Asian range, or place stop orders 3–5 pips beyond both edges (OCO). Trade only if the range is < ~50 pips for GBP pairs (tight range = stored energy). One trade per day.

Exit rules. Stop at the opposite side of the range. Target 1.5× the range height (documented optimum in independent tests). Flat by the NY lunch.

Risk management. Skip bank holidays, skip days when the required stop exceeds 50 pips, avoid NFP/CPI mornings unless news trading is your intent. Risk 0.5–1%.

Backtest evidence. GBP/USD monthly study: 19 trades, 52.6% WR, PF 1.74, +152 pips. 300+ trade GBP/JPY tests: profitable at 1.5 RR with range-low stops. Academic EUR/USD study (2015–2017): best window 48.62% WR, TP:SL 50:35 optimal. QuantifiedStrategies' EUR/USD test warns: without filters the raw version is *not* consistently profitable — GBP pairs are the better vehicle.

Works / fails. Works on GBP pairs (higher beta to London flows) in normal volatility. Fails on EUR/CHF-type sleepy pairs, holiday-thinned sessions, and when the Asian range is already huge.

Backtesting notes. Broker server time vs UTC vs London time (and DST!) is the #1 implementation bug — nail the session definitions. Use true tick/M1 data for breakout levels.

Variations. False-breakout reversal (fade the first break if it fails within 3 bars); trade the *second* breakout only; London-NY overlap continuation entry.

S20 Asian Session Range Fade

Market: EUR/USD, USD/JPY, AUD/USD • **Timeframe:** M5–M15, Asian session • **Style:** Range scalping • **Evidence:** Grade B (practitioner-consensus ranges; low-volatility session statistics)

Win rate	Typical RR	Profit factor	Best pairs	Session
65–75%	0.8:1 to 1:1	1.3–1.5	EUR/USD, AUD/USD	00:00–06:00 UTC

Concept & edge. The mirror image of S19: outside London/NY, major pairs mean-revert hard. Asian-session EUR/USD respects the day's developing range ~70% of the time in quiet regimes — fade the edges, don't break them.

Setup. M15 chart, Asian session only. Bollinger Bands (20, 2) or the developing session high/low; RSI(14) for stretch confirmation; economic calendar flat.

Entry rules. Short at session high + RSI > 70 with a rejection wick; long at session low + RSI < 30. Only when overnight news flow is quiet.

Exit rules. Target the range midline or opposite edge (partial at midline). Stop $1 \times \text{ATR}(14)$ beyond the range edge. All flat before 06:30 UTC — never hold into the London open.

Risk management. The fatal trade is holding a range fade into the London breakout. Hard time-stop at 06:30 UTC regardless of P&L. Risk 0.5% per fade; max 2 per session.

Backtest evidence. Practitioner consensus for disciplined Asian range systems: 65–75% win rate at sub-1:1 RR — profitable because of accuracy, not asymmetry. Session statistics consistently show EUR/USD Asian ranges compressing to 40–60% of the daily range.

Works / fails. Works in quiet macro weeks on low-beta pairs. Fails during Asian-hours surprises (PBoC, BoJ, RBA), thin holiday tape, and trending macro regimes.

Backtesting notes. Test by day of week (Sunday-open and Friday-Asia sessions behave differently). Model the wider Asian spreads honestly — on some brokers they're 2–3× London spreads and kill the math.

Variations. Grid-less version with single entries only (avoid martingale grids — see Appendix warning); JPY-pair version timed around Tokyo fix (00:55 GMT); fade only the *second* touch of the range.

S21 New York Open Breakout (Equities)

Market: US large-caps with catalysts • **Timeframe:** M1–M5 • **Style:** Opening momentum • **Evidence:** Grade B (practitioner data; FINRA volume statistics)

Win rate	Typical RR	Hold time	Volume share	Best vehicle
45–60%	1.5–2:1	15–90 min	35% of day in first 30 min	High-RVOL gappers

Concept & edge. FINRA data: ~35% of daily volume trades in the first 30 minutes, ~50% in the first hour. That concentration means institutional conviction expresses itself early — and stocks gapping on real catalysts with high relative volume continue in the gap direction more often than random.

Setup. Pre-market scanner: gap > 2%, relative volume > 3×, identifiable catalyst (earnings, guidance, upgrade). Mark the first 5-min range after the bell.

Entry rules. Enter on the break of the 5-min high in the gap direction, only if the stock holds above VWAP for longs. No entry if the first 5 minutes already retraced > 50% of the gap.

Exit rules. Stop below the 5-min low (longs). Targets: measured move = gap size projected from breakout, or trail below 1-min higher lows. Flat by 11:00 ET — the edge is a morning phenomenon.

Risk management. Position size from stop distance, max 1% risk. Wide spreads and halts on small caps make large-caps safer. One position at a time until consistently profitable.

Backtest evidence. Practitioner statistics for catalyst + RVOL-filtered opening drives: 45–60% win rate at 1.5–2:1 RR. The predictive power of the first 30 minutes for the session direction (~58–63% with catalysts) is documented in the intraday-momentum academic literature (see S27).

Works / fails. Works on earnings-season large/mid caps. Fails on low-float pumps, no-news drifters, and macro-shock mornings when everything correlates to the index.

Backtesting notes. Survivorship bias is rampant in gap-scanner backtests — your universe must include delisted/CRSP-complete data or results are fantasy. Model opening spreads (often 3–10× midday).

Variations. Gap-and-go vs gap-fill classification by first-15-min VWAP behavior; second-day continuation plays; short-side exhaustion version on parabolic small caps (advanced).

S22 London–NY Overlap Momentum

Market: EUR/USD, GBP/USD, gold • **Timeframe:** M15 • **Style:** Session momentum • **Evidence:** Grade B (session-volatility literature; practitioner consensus)

Win rate	Typical RR	Window	Profit factor	Pairs
50–58%	1.5:1	13:00–16:00 UTC	1.3–1.5	EUR, GBP, XAU

Concept & edge. The London–NY overlap (13:00–16:00 UTC) is the deepest liquidity pool on earth — both continents active, US data released at 13:30/14:00 UTC. Directional moves begun in the overlap statistically persist into the NY afternoon more than moves from any other window.

Setup. M15 chart. Mark the London-morning trend (08:00–13:00 UTC direction). VWAP or 20-EMA on M15 as the momentum guide.

Entry rules. Enter in the direction of the London-morning trend on the first M15 pullback to the 20-EMA/VWAP after 13:00 UTC, with the pullback holding above/below the session midpoint.

Exit rules. Stop beyond the pullback extreme. Target: 1.5R or the day's measured objective (London range projected). Flat by 16:30 UTC.

Risk management. US data releases at 13:30 UTC can whipsaw the first minutes — either wait 15 minutes after data or treat data days with the news-strategy family (Part VII). Risk 0.5–1%.

Backtest evidence. Session studies consistently show the overlap window carrying the day's highest volume and true range; practitioner backtests of trend-continuation entries in this window report ~50–58% win rates at ~1.5:1. Stochastic session tests (S44) independently found the early-NY-afternoon group the least negative of all sessions.

Works / fails. Works on macro-trending days and during rate-cycle repricing. Fails in summer doldrums, pre-holiday sessions, and when London morning was directionless (no trend to continue).

Backtesting notes. Define "London-morning trend" mechanically (e.g., 13:00 price vs 08:00 price, or 3 consecutive M30 higher closes) — subjective trend definitions can't be backtested.

Variations. Gold-specific version (XAU loves the overlap); reversal version for days when London morning was a fake-out; add US 10Y yield direction as confirmation.

S23 Initial Balance Breakout (Futures)

Market: ES, NQ, CL, GC futures • **Timeframe:** M5, first hour • **Style:** Auction-market breakout • **Evidence:** Grade B (Market Profile literature — Dalton; practitioner backtests)

Win rate	Typical RR	Profile fit	Best days	Hold
45–55%	2:1+	Trend days	High-conviction opens	To session close

Concept & edge. The Initial Balance (IB) — the first hour's range — is the auction's opening negotiation. On days that will trend, price breaks the IB early and never looks back; IB extensions of 1.5–2× are the classic Market Profile targets. You're buying the market's own verdict on value.

Setup. First-hour high/low (09:30–10:30 ET). Compare IB width to 5-day average: narrow IB (< 70% of average) = breakout fuel.

Entry rules. Enter on acceptance beyond the IB: two consecutive 5-min closes outside, or a clean break + retest that holds. Direction-agnostic.

Exit rules. Target 1: 1× IB extension. Target 2: 2× IB. Stop back inside the IB (failure = auction rejected your direction). Flat at the close.

Risk management. Narrow-IB days are the trade; wide-IB days are already balanced — skip. Max one IB trade per day per direction. Risk 0.75–1%.

Backtest evidence. Market Profile practitioner statistics: narrow-IB breakout days reach the 1× extension roughly 60–70% of the time, and 2× on genuine trend days (~20–30% of sessions). Win rates of 45–55% at 2:1 average RR produce strong expectancy.

Works / fails. Works on range-extension days, macro trend weeks, post-news continuation. Fails on rotational days (double-distribution chops) — the IB-break + reclaim pattern is the classic failure.

Backtesting notes. "Acceptance" must be coded mechanically (e.g., 2 closes outside). Test the IB-width percentile filter: the edge should concentrate in the narrowest quartile of IBs.

Variations. IB midpoint as the session pivot (trade only one side of it); 30-min IB for faster signals; combine with overnight inventory direction (fade vs follow).

S24 The 10 O'Clock Reversal (NYSE)

Market: SPY, QQQ, large-caps • **Timeframe:** M5 • **Style:** Intraday reversal timing • **Evidence:** Grade B (long-standing practitioner pattern; consistent with intraday overreaction research)

Win rate	Typical RR	Window	Reliability	Style
55–65%	1:1 to 1.5:1	09:55–10:15 ET	Moderate-high	Countertrend scalp

Concept & edge. The opening 25–30 minutes overshoot: amateur orders execute at the open, professionals wait. Around 10:00 ET ("the 10 o'clock jiggle"), the initial move frequently reverses as opening inventory is absorbed — one of the oldest floor-trader timing patterns, consistent with academic evidence that opening returns partially reverse.

Setup. M5 chart, VWAP. Measure the opening move's size and speed from 09:30.

Entry rules. If the first 25–30 min produced a fast, extended move ($> 0.5\times$ daily ATR) away from VWAP: enter the reversal when an M5 countertrend structure forms (lower high after up-move / higher low after down-move), ideally with price stretched $> 1.5\sigma$ from VWAP.

Exit rules. Target VWAP. Stop beyond the opening extreme. Time stop: if VWAP not reached by 11:00, flatten.

Risk management. Never fade a slow, grinding, high-volume trend — only fade *fast* extensions. On major news days the 10:00 reversal often fails; skip CPI/FOMC mornings. Risk 0.5–1%.

Backtest evidence. Practitioner win rates of 55–65% to the VWAP target; consistent with published research showing the first half-hour's move partially mean-reverts within the session. Note: the related *continuation* effect (S27) is the academic-documented side — the reversal edge is strongest against low-volume opening spikes.

Works / fails. Works on normal days with retail-driven opening spikes. Fails on genuine trend days — distinguishing these early (volume quality, breadth, gap context) is the skill.

Backtesting notes. Classify opening moves by speed (points per minute) and volume; the reversal edge should live in the fast/low-quality bucket. If it doesn't in your data, drop the strategy.

Variations. Only fade moves against the overnight gap direction; scale out 50% at half-way to VWAP; apply to the London open (10:00 local reversal) on FTSE/DAX.

S25 Opening Drive Continuation (Gap-and-Go)

Market: Mid/large-cap US stocks • **Timeframe:** M1–M5 • **Style:** Opening momentum continuation • **Evidence:** Grade B (practitioner momentum statistics; opening-drive research)

Win rate	Typical RR	Hold	Filter	Failure mode
50–60%	2:1	30–120 min	RVOL > 3×, catalyst	Opening fake-out

Concept & edge. When a stock gaps on a real catalyst and *keeps going* in the first minutes without pausing, you're watching institutions re-price in real time. The opening drive is the purest momentum signal of the day — enter with it, not against it.

Setup. Pre-market: gap > 3%, RVOL > 3×, clean catalyst, price holding near pre-market highs into the bell. M1/M5 execution chart.

Entry rules. Enter on the first M1/M5 consolidation breakout in the gap direction after the open (typically minutes 3–10), with the stock above VWAP (longs). Alternatively buy the break of the pre-market high.

Exit rules. Stop below the consolidation low. Targets: pre-market measured move, then trail below M5 higher lows. If the drive stalls and closes an M5 below VWAP — out immediately.

Risk management. Opening drives fail violently when they fail (the "fake-out flush" back through VWAP). Hard stops, no averaging. Max 1% risk; slippage on stops will exceed plan in fast tape — size for it.

Backtest evidence. Practitioner momentum-desk statistics: 50–60% win rate at ~2:1 on catalyst-filtered drives. Consistent with intraday momentum literature showing opening-period returns predict same-direction later returns (t-stat significant across decades of US data).

Works / fails. Works in earnings season and sympathy moves (sector momentum). Fails in choppy index days when individual catalysts get overwhelmed, and on low-float names where the drive is manufactured.

Backtesting notes. Hard to backtest honestly — pre-market data quality and opening fills are the challenge. Use tick or M1 data, model halts, and include delisted names.

Variations. Short-side parabolic fade *after* the drive tops (2-stage strategy); second-entry on the first VWAP pullback; sector-basket version (trade the ETF when 3+ components drive).

PART III

Momentum & Trend Intraday Systems

Trend following has the opposite personality of mean reversion: low win rates (35–45%), large winners (2.5R+), and long flat stretches. Intraday versions compress this into hours. The published evidence says these systems work *if* you survive the whipsaws — regime filters (ADX, ATR percentile) are what separate viable systems from donation machines.

S26	9/21 EMA Crossover Scalp
S27	First Half-Hour Predicts the Last (Intraday Momentum)
S28	Supertrend Intraday Trend Rider
S29	MACD Histogram Retracement (Trend Pullback)
S30	ADX Regime Filter + Pullback Entries
S31	Donchian 20/10 Breakout (Modern Turtle)
S32	Keltner Channel Breakout
S33	VWAP First Pullback (Institutional Benchmark)
S34	VWAP Bands Mean Reversion
S35	Heikin Ashi Trend Scalp
S36	EMA Ribbon Fan (5/8/13)
S37	Consecutive-Closes Momentum (3-Bar Drive)

S26 9/21 EMA Crossover Scalp

Market: Forex majors, NQ, crypto • **Timeframe:** M3–M5 • **Style:** Trend-following scalp • **Evidence:** Grade B (independent 161-trade backtest published on r/Daytrading, 2025)

Win rate	Typical RR	Max losing streak	Best sessions	Style
38–45%	1.5–2:1	8 documented	London, NY overlap	Low-WR momentum

Concept & edge. The fast 9-EMA crossing the 21-EMA on low timeframes captures micro-trend legs. The documented reality check: raw win rate is only 38–45% — profitability comes entirely from asymmetric RR and streak discipline. Anyone selling this as a "high win rate" system is lying.

Setup. 9-EMA and 21-EMA on M5; optional 200-EMA for directional bias.

Entry rules. Long: 9 crosses above 21, both above the 200-EMA, with the cross candle closing in its upper half. Enter on the first pullback to the 9-EMA after the cross (better RR than chasing the cross candle).

Exit rules. Stop below the pullback low (tight, ~0.5–1 ATR). Target 1.5–2R, or exit on the opposite cross. Trail with the 9-EMA on strong days.

Risk management. The documented 8-loss streak is the strategy's true price of admission — size so 10 consecutive losses cost < 10%. Stop trading after 3 losses in a session (chop regime).

Backtest evidence. Independent 161-trade test on 3/5-min charts: 38–45% win rate, max win streak 5, max losing streak 8; with 2:1 targets the equity curve was modestly positive but fragile to spread. Filtered variants (higher-timeframe trend + pullback entry) report 60–70% win rates at reduced frequency.

Works / fails. Works in trending sessions (post-news, macro days). Fails in ranges — EMA crosses are pure whipsaw generators when $ADX < 20$.

Backtesting notes. Test cross-at-close vs pullback entries separately. Include realistic spread: at 5-min forex scalping, spread is often 15–30% of the average winner — it can flip the sign of the whole system.

Variations. 8/21 on M15 (smoother); 5/13/21 ribbon (trade only when ribbon is fanned); add $ADX > 25$ filter.

S27 First Half-Hour Predicts the Last (Intraday Momentum)

Market: SPY / S&P 500 • **Timeframe:** 30-min bars • **Style:** Academic momentum timing • **Evidence:** Grade A (Gao, Han, Li, Zhou — *Intraday Momentum*, peer-reviewed; SSRN 2440866)

Predictive R ²	OOS R ²	Stronger when	Horizon	Robustness
1.6–2.6%	1.7–2.5%	High vol, recessions, news days	First 30 min → last 30 min	Decades of data

Concept & edge. The first half-hour's return significantly predicts the *last* half-hour's return — an academically validated intraday momentum effect. Predictability rises with volatility and volume, and combining the first half-hour with the penultimate half-hour lifts R² to 2.6–3.3%. This is among the most rigorous intraday edges ever published.

Setup. 30-min bars on SPY/ES. Record the 09:30–10:00 return. Note day type (news day, high-VIX regime).

Entry rules. ~15:00–15:30 ET: if the first half-hour was up (and the penultimate half-hour confirms), long into the close; if down, short/flat. Exit at the close (MOC).

Exit rules. Time exit at 16:00 ET. Stop: adverse move > 0.5× the day's average 30-min range.

Risk management. The edge per trade is small (basis points) — it's a high-frequency-friendly strategy, marginal for retail after costs. Consider it a *filter* for other end-of-day decisions if your costs are high.

Backtest evidence. In-sample R² 1.6% (first half-hour alone), 2.6% (combined with 12th half-hour), rising to 3.3% in high-volatility periods; OOS R² 1.7–2.5% — matching or exceeding typical *monthly* momentum predictability, at a daily horizon. Stronger in recessions and on macro-news days.

Works / fails. Works on index ETFs/futures; stronger on volatile and news days. Fails on dead days and when applied to individual small stocks (noise dominates).

Backtesting notes. Replicate the regression before building the trading rule — confirm the predictability exists in *your* sample period. Transaction costs decide retail viability.

Variations. Trade the full afternoon instead of the last 30 min; options version (long gamma into predicted trend); apply to sector ETFs for cross-sectional versions.

S28 Supertrend Intraday Trend Rider

Market: Indices, forex, crypto • **Timeframe:** M15–H1 (intraday), Daily/Weekly (swing) • **Style:** ATR trend following • **Evidence:** Grade A (QuantifiedStrategies 60-yr S&P test; Share.Market 200,000-trade Nifty study)

Win rate	Avg win/loss	Max DD	Best params	Character
40–43% (daily stocks), 68% (weekly index)	Win $\gg$ loss (25.5% vs –7.4% at mult 3)	~25%	(10,3) classic; (3,3) aggressive	Sniper trend system

Concept & edge. Supertrend = ATR-based trailing line: long while price holds above it, short below. The 200k-trade study destroyed the win-rate myth: *every* parameter set wins only 40–43% — the edge is that winners are 3–6× the size of losers. Raising the multiplier from 1 to 3 tripled average win size (8.5% $\rightarrow$ 25.5%).

Setup. Supertrend (10, 3) default; intraday (7, 3) or (10, 3) on M15–H1.

Entry rules. Enter on bar close crossing the Supertrend line, in the new direction. Filter: only take longs above the 200-EMA (cuts whipsaws).

Exit rules. Signal exit on opposite cross (this is the stop — volatility-adaptive). Optional partial at 2R.

Risk management. The wide-multiplier versions hold winners ~69 days on average (swing) — intraday versions hold hours. Either way, expect ~60% losing trades. If you can't stomach that psychologically, don't trade it.

Backtest evidence. 60-year weekly S&P test: 41 trades, 68% WR, 11.2% avg gain/trade, 24.6% max DD vs 56% buy-and-hold, risk-adjusted return 9.8% vs 6.9%. Nifty-500 daily study: WR stable 40–43% across all params; EV maximized at short period + high multiplier.

Works / fails. Works on trending instruments (commodities, indices in macro regimes, crypto trends). Fails in chop — the line gets sawn to pieces; ADX filter mandatory intraday.

Backtesting notes. Grid-test period 3–21 $\times$ multiplier 1–3 and demand plateau stability. The published insight worth verifying: multiplier matters more than period.

Variations. Triple-Supertrend confirmation (3 timeframes agree); use as trailing stop only for other entries; (3,3) aggressive config for momentum names.

S29 MACD Histogram Retracement (Trend Pullback)

Market: Forex majors • **Timeframe:** H1–H4 • **Style:** Trend pullback • **Evidence:** Grade A (algorithmic backtest with costs, published 2026 — CodeTradingCafe)

Annual return	Sharpe	Costs included	Profile	Instruments
~5.5%	~1.45	Yes (commissions)	Steady trend capture	Forex majors

Concept & edge. MACD alone is a mediocre signal; MACD *as a retracement timer inside an EMA-defined trend* is a documented, cost-adjusted winner. You enter when the histogram recovers toward the trend direction after a pullback — buying dips with momentum confirmation.

Setup. EMA (e.g., 50/200) for trend direction; MACD (12, 26, 9) histogram.

Entry rules. Long: price above the trend EMA; histogram dips below zero (pullback) then ticks back up. Enter on the uptick bar's close. Short: mirror.

Exit rules. Stop below the pullback swing (1–1.5 ATR). Target 1.5–2R or exit when histogram crosses back against the trade. Time stop: 10–15 bars.

Risk management. 0.5–1% risk; the published Sharpe (1.45) is unusually high for a simple system precisely because costs were included — replicate that discipline.

Backtest evidence. Full algorithmic test on forex data *with commissions*: ~5.5% annual return, Sharpe ~1.45. Compare: naive MACD-signal-line crossovers typically show near-zero or negative expectancy after costs (documented in multiple TA-effectiveness studies) — the trend filter and pullback timing are the entire edge.

Works / fails. Works on trending forex pairs (GBP/JPY, EUR/USD in rate cycles). Fails in ranges (histogram oscillates uselessly) and on exotics with wide spreads.

Backtesting notes. Test EMA lengths 50/100/200 and histogram-uptick definitions (1-bar vs 2-bar). The strategy is a template: same structure works with RSI or Stochastic as the pullback timer (see S44 for the cautionary data on Stochastic).

Variations. Zero-line MACD cross in trend direction (fewer, cleaner signals); dual-timeframe (H4 trend, H1 entry); divergence add-on.

S30 ADX Regime Filter + Pullback Entries

Market: All liquid markets • **Timeframe:** M15–H1 • **Style:** Regime-gated trend trading • **Evidence:** Grade B (ADX research lineage — Wilder; regime-threshold studies)

Win rate	Typical RR	Gate	Role	Expectancy
45–55% (trend regime)	1.5–2:1	ADX > 25 + ATR > 60th percentile	Meta-strategy	+0.2 to +0.4R

Concept & edge. The single highest-leverage improvement to any trend system: *only trade it when the market is provably trending*. ADX > 25 plus ATR above its 60th percentile defines the momentum regime; below that, switch to mean reversion or sit out. The regime, not the entry trick, decides who gets paid.

Setup. ADX(14), ATR(14) percentile rank (100-day lookback), 20-EMA, swing structure.

Entry rules. Regime check: ADX > 25 and rising, ATR in upper 40%. Long: pullback to the 20-EMA in a +DI > -DI market; enter on the resumption candle. Short: mirror.

Exit rules. Stop beyond the pullback swing. Target 1.5–2R or trail with the 20-EMA while ADX keeps rising. Exit everything when ADX rolls over below 20.

Risk management. 0.75–1% risk per trade in-regime; *zero* exposure out-of-regime. The discipline of not trading is the edge.

Backtest evidence. Regime-filtered momentum systems in the literature run 35–45% win rates at 2.5R+ average winners — positive expectancy; the same entries unfiltered degrade to coin-flips. Practitioner data on XAUUSD/US100/BTC confirms regime thresholds transfer across instruments (with funding/vol adjustments).

Works / fails. Works as an overlay on S26, S28, S29, S33–S35. Fails when ADX lags regime changes (V-reversals) — accept the lag or add a faster volatility measure.

Backtesting notes. This is a *framework*: implement it as a boolean gate in front of any entry signal and measure the delta in expectancy. That measurement habit is what makes you a strategy designer rather than a strategy collector.

Variations. Hurst exponent < 0.45 as the mean-reversion gate (the opposite switch); ADX slope rather than level; weekly-ADX bias for intraday entries.

S31 Donchian 20/10 Breakout (Modern Turtle)

Market: Commodities, forex, index futures, crypto • **Timeframe:** H4–Daily (intraday-adaptable H1) • **Style:** Channel breakout trend following • **Evidence:** Grade A (Curtis Faith, *Way of the Turtle*, 2007; futures tests 1996–2007)

CAGR	Win rate	Variant uplift	Markets	Drawdowns
29.4–57.2%	35–45%	Time-exit variant nearly doubled CAGR	Currencies, commodities, Treasuries	Large (trend following)

Concept & edge. Buy 20-period highs, exit on 10-period lows, only with the long-term trend (25-EMA vs 350-EMA filter). The Turtles turned this into \$150M in 4 years; Faith's re-test on currencies/commodities/Treasuries produced 29.4% CAGR — and the 80-day *time-exit* variant 57.2%. The channel breakout remains the canonical trend edge in commodities.

Setup. Donchian channels (20 entry / 10 exit), 25 & 350 EMA filter, ATR(20) for sizing ("N").

Entry rules. Long: price breaks the 20-period upper channel AND 25-EMA > 350-EMA. Size positions so 1 ATR move = fixed % of equity (volatility targeting — the Turtles' real secret).

Exit rules. Exit on break of the 10-period opposite channel, or after a fixed time window (80 days swing; proportionally 60–80 bars intraday). Stop: 2× ATR from entry.

Risk management. Pyramiding: add ½ unit every ½ ATR in profit, max 4 units (original rules). Portfolio heat cap ~6–10% total risk.

Backtest evidence. Faith 1996–2007 futures portfolio: 29.4% CAGR (breakout exit), 57.2% CAGR (80-day time exit), with the EMA filter required for direction. Modern tests confirm: still works on commodities/currencies, weaker on stocks (stocks mean-revert; commodities trend).

Works / fails. Works in macro commodity trends (energy, metals, softs), crypto bull legs. Fails in multi-year chop; expect 35–45% win rate and gut-wrenching drawdowns between trends.

Backtesting notes. Test on a *portfolio* of futures, not one market — the edge is diversification across trend episodes. Include roll costs for futures realism.

Variations. 55/20 channels (Turtle System 2); Donchian basis-line pullback entries (gentler); intraday H1 version on gold and crude.

S32 Keltner Channel Breakout

Market: Futures, forex • **Timeframe:** M15–H1 • **Style:** Volatility-channel breakout • **Evidence:** Grade B (Chester Keltner lineage; Linda Raschke's documentation; practitioner backtests)

Win rate	Typical RR	Filter	Best regime	Style
40–50%	1.5–2.5:1	ADX or squeeze context	Post-compression	Momentum breakout

Concept & edge. Keltner Channels ($20\text{-EMA} \pm 2 \times \text{ATR}$) define "normal" volatility. A close outside the channel means current volatility exceeds normal — directional information, especially after compression. Raschke's documentation made it a futures-desk standard.

Setup. Keltner (20, $2 \times \text{ATR}(10)$); Bollinger (20, 2) for the squeeze overlay; ADX(14).

Entry rules. Long: close above the upper Keltner with ADX rising through 20–25. Stronger: first break *after* a Bollinger-inside-Keltner squeeze (see S58). Enter at close or on a small pullback to the channel.

Exit rules. Initial stop at the 20-EMA (channel basis). Trail below the midline; final exit on close back inside the channel. Target alternative: $2\text{--}3 \times \text{ATR}$.

Risk management. False breaks cluster in sideways tape — the squeeze/ADX context filter cuts signal count by ~half and roughly doubles per-trade expectancy in practitioner tests. Risk 0.75–1%.

Backtest evidence. Practitioner backtests on index/commodity futures: 40–50% win rates at 1.5–2.5:1 with the trend filter; raw unfiltered breaks are ~breakeven after costs. Channel exits adapt to volatility automatically — the structural reason these systems age better than fixed-point targets.

Works / fails. Works after volatility compression and in macro trends. Fails in high-ADX late-stage trends (buying exhaustion) and choppy balance.

Backtesting notes. Compare Keltner vs Donchian vs Bollinger breakout on identical data — knowing which channel your market respects is itself useful research. ATR period sensitivity: 10 vs 20 changes signal density a lot.

Variations. Countertrend Keltner fade (with $\text{ADX} < 20$ only — the mirror system); dual-channel (inner 1.5 ATR stop, outer 2.5 ATR entry); Raschke's "Holy Grail" ADX-pullback variant.

S33 VWAP First Pullback (Institutional Benchmark)

Market: Stocks, index futures • **Timeframe:** M1–M5 • **Style:** Trend pullback at VWAP • **Evidence:** Grade B (institutional execution literature; practitioner backtests)

Win rate	Typical RR	Window	Best days	Style
55–65%	1.5–2:1	10:00–14:00 ET	Trend days	Pullback continuation

Concept & edge. VWAP is the benchmark institutions are evaluated against: desks buying below VWAP show "good execution," so pullbacks to VWAP in trending stocks attract real institutional bids. That's a structural, incentive-based reason for the bounce — not just a line on a chart.

Setup. Session VWAP (anchored 09:30 ET). Confirm trend: stock above VWAP since ~10:00, higher highs/lows, RVOL > 2.

Entry rules. First (highest quality) or second pullback to VWAP in an established trend. Enter on the M1/M5 reversal candle off VWAP, or place a limit at VWAP + small buffer with confirmation.

Exit rules. Stop below the pullback low (or VWAP – 0.5× ATR(5min)). Target: prior high, then trail; ambitious target = measured move of the morning leg. Flat by 15:30.

Risk management. Third and later VWAP touches fail increasingly often — the edge decays per touch. Skip if the morning move was vertical (climactic). Risk 0.5–1%.

Backtest evidence. Practitioner backtests of first-pullback entries on trending large-caps: 55–65% win rate to the prior-high target. Consistent with VWAP's documented role as the dominant institutional execution benchmark (execution-algo literature).

Works / fails. Works on trend days with institutional sponsorship (earnings, sector moves). Fails on rotation days — VWAP becomes a magnet, not a springboard; and in the last hour when VWAP loses meaning.

Backtesting notes. Code "trend day" mechanically before testing (e.g., price > VWAP for 80% of bars since 10:00). Measure bounce quality by touch number — publish your own decay curve.

Variations. VWAP ± 1σ band entries (deeper pullbacks); anchored VWAP from earnings date/week; short-side version below VWAP.

S34 VWAP Bands Mean Reversion

Market: ES, SPY, QQQ • **Timeframe:** M5 • **Style:** Statistical band fade • **Evidence:** Grade B (practitioner statistics; VWAP standard-deviation band literature)

Win rate	Typical RR	Target	Regime	Edge type
60–70%	0.8–1.2:1	VWAP	Range/rotation days	Accuracy edge

Concept & edge. VWAP $\pm 1\sigma/2\sigma$ bands (standard deviation of price around VWAP) act like intraday Bollinger Bands around the session's true mean. On non-trend days, the 2σ band is faded successfully roughly 60–70% of the time back to VWAP.

Setup. Session VWAP with 1σ and 2σ bands. Confirm rotation day: price crossing VWAP repeatedly, no directional breadth.

Entry rules. Fade touches of the 2σ band: short at $+2\sigma$ (longs at -2σ) with an M5 rejection candle. Only on days classified as rotational by 11:00.

Exit rules. Target VWAP (partial at 1σ). Stop: $0.5\times$ ATR beyond the band, or immediately if an M5 closes beyond 2.5σ (trend day warning).

Risk management. The mirror-image risk of S33: on trend days this strategy donates. The day-type classifier is the whole game. Risk 0.5%; stop trading after 2 losses (your day-type read is wrong).

Backtest evidence. Practitioner statistics: 60–70% win rate to VWAP on rotation days; combined with a trend-day filter, profit factors of 1.3–1.6 are typical. The overnight-vs-intraday return literature supports indices' intraday mean-reverting character.

Works / fails. Works on quiet, balanced days (the majority of sessions). Fails on trend days, news days, and expiry Fridays with directional flow.

Backtesting notes. Day-type classification must be *causal* (known by entry time, not in hindsight). Test classifiers: opening-range width, first-hour VWAP slope, NYSE TICK/breadth thresholds.

Variations. Countertrend only against overnight direction; scale entries at 2σ and 2.5σ with tight aggregate stop; pair with S24 (10:00 reversal) as the entry timer.

S35 Heikin Ashi Trend Scalp

Market: Stocks, forex, BTC • **Timeframe:** M5–M15 • **Style:** Smoothed-trend following • **Evidence:** Grade B (TradeAlgo 12-yr, 200-stock backtest: 59% WR, PF 2.34 with MA filter; Tradinformed EUR/USD 2000–2014 study)

Win rate	Profit factor	Filter	Lag cost	Best use
59% (documented, filtered)	2.34	MA trend filter	Entries lag ~1 bar	Noise reduction

Concept & edge. Heikin Ashi candles average OHLC, erasing noise so trend legs become visually and mechanically obvious (long same-color runs with one-sided wicks). The documented edge requires a moving-average filter — color changes alone are worthless; HA + trend filter is a genuine noise-reduction technology.

Setup. Heikin Ashi candles, 50-EMA (or 200-EMA) filter. Note: HA prices are synthetic — execute and set stops from *real* prices.

Entry rules. Long: price above the 50-EMA; enter after 2 consecutive green HA candles with no lower wick (strong momentum signature), on the third candle's open. Short: mirror.

Exit rules. Exit on the first opposite-color HA candle, or when an HA candle prints a wick against the trend after a long run. Hard stop on *real* price: below the entry-pattern low.

Risk management. HA lag costs you the first part of every move — accept it as the fee for noise filtering. Risk 0.5–1%; the smooth equity comes from fewer bad trades, not tighter stops.

Backtest evidence. TradeAlgo 12-year, 200-stock test: 59% win rate, PF 2.34 (HA + MA filter). Tradinformed's EUR/USD 4H study (2000–2014, MACD confirmation): profitable system. Unfiltered color-change systems: ~random.

Works / fails. Works in trending markets and noisy instruments where candle-reading is hard (crypto, small caps). Fails in ranges (HA turns into a paint mixer) and for exact-level work (HA doesn't show real prices).

Backtesting notes. Critical implementation detail: compute signals on HA candles but execute on real OHLC at the *same bar* — backtests that execute at HA close prices are fabricating fills.

Variations. HA + Chandelier exit (documented BTC scalp: 40.8% WR, PF 1.05 raw — needs the trend filter); HA swing-points as trailing stops; monthly HA for regime (S&P 1960+ test: 52% WR, DD cut from 52% to 29%).

S36 EMA Ribbon Fan (5/8/13)

Market: Forex, crypto, NQ • **Timeframe:** M5–M15 • **Style:** Trend-alignment momentum • **Evidence:** Grade B (Guppy-lineage practitioner backtests)

Win rate	Typical RR	Signal	Failure	Style
45–55%	1.5–2:1	Ribbon fans + pullback	Chop compression	Visual trend system

Concept & edge. Three EMAs (5, 8, 13 — Fibonacci set) act as one object: when they fan out in order ($5 > 8 > 13$, separated, all sloping), a micro-trend is confirmed by alignment, not by any single cross. Entries on pullbacks into the ribbon buy confirmed momentum at a discount.

Setup. EMAs 5/8/13 on M5–M15; ribbon separation as trend-strength gauge.

Entry rules. Long: ribbon fully fanned upward; price pulls back into the 8–13 zone and prints a rejection candle closing back above the 5-EMA. Enter on that close. Short: mirror.

Exit rules. Stop below the 13-EMA (ribbon break = thesis dead). Target 1.5–2R or trail along the 5-EMA. Exit early if the ribbon compresses flat.

Risk management. The ribbon compresses and tangles in ranges — every tangle is a no-trade zone, which is the filter doing its job. Risk 0.75% per trade; max 3 trades per session.

Backtest evidence. Guppy-style multiple-MA systems in practitioner testing: 45–55% win rates at 1.5–2:1 with fan + pullback entries; raw cross systems without the pullback entry degrade toward breakeven after costs. Ribbon systems' real value is trade *prevention*.

Works / fails. Works in sustained intraday trends (news-driven, macro sessions). Fails in balance — but fails *visibly*, giving you the discipline signal to stand down.

Backtesting notes. Quantify "fanned" (e.g., min gap between EMAs as % of ATR) so it's testable. Compare entry-at-fan vs entry-at-pullback: the pullback entry should show better RR at slightly lower win rate.

Variations. Full Guppy (6 short + 6 long EMAs); ribbon + RSI(14) > 50 momentum gate; 8/21/34 set for H1 swing version.

S37 Consecutive-Closes Momentum (3-Bar Drive)

Market: Indices, forex, crypto • **Timeframe:** M15–H1 • **Style:** Momentum persistence • **Evidence:** Grade B (short-term momentum literature; practitioner tests)

Win rate	Typical RR	Best context	Hold	Failure
50–58%	1.2–1.8:1	Post-breakout, low retrace	3–10 bars	Late entry on exhaustion

Concept & edge. Three consecutive same-direction closes with expanding bodies signal initiative participation, not drift. Short-term price momentum (hours scale) is one of the more persistent documented anomalies across asset classes; the 3-bar drive is its simplest tradable form.

Setup. M15–H1 bars; ATR(14) for body-size context; optional volume confirmation.

Entry rules. Long: 3 consecutive closes higher, each body $> 0.5 \times \text{ATR}$, total retrace between bars $< 38\%$, preferably breaking a level (session high, range top). Enter on bar 3's close or a shallow limit pullback. Short: mirror.

Exit rules. Stop below bar 1's low (structure) — wide; or below bar 3's midpoint (aggressive). Target: 1.5–2R; momentum typically delivers the move within 3–10 bars or not at all (time stop).

Risk management. Never enter a 3-bar drive that is the *3rd consecutive drive* of the session (exhaustion). Check higher-timeframe location: drives into daily resistance fail. Risk 0.75%.

Backtest evidence. Short-horizon momentum persistence is documented across equities, FX and crypto in the time-series momentum literature (hours-to-days scale). Practitioner tests of 3-close rules: 50–58% win rate at 1.2–1.8:1, with volume-confirmed drives outperforming.

Works / fails. Works post-compression and on breakout days. Fails as a standalone in ranges and on low-liquidity instruments where 3 bars is noise.

Backtesting notes. Test body-size thresholds (0.3/0.5/0.7 ATR) and confirm the edge concentrates in post-compression contexts (measure prior 20-bar range percentile).

Variations. 2-bar drive (earlier, noisier); drive + VWAP location filter; opposite system: fade 3-bar drives that are 2σ extended from the 20-EMA (mean-reversion mirror).

PART IV

Scalping Systems

True scalping lives or dies on costs: spread, slippage, latency. Every strategy here is viable *only* with tight spreads and honest execution math. The published evidence is blunt — several famous scalping systems fail rigorous testing (we include one as a warning), which is exactly why you should backtest before believing.

S38	5-Minute EMA Scalping (8/21 + RSI)
S39	Tape & Order-Flow Scalp (Level 2)
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S49	Liquidity-Sweep Reversal (SMC-Style)

S38 5-Minute EMA Scalping (8/21 + RSI)

Market: Forex majors • **Timeframe:** M5 (M1 for timing) • **Style:** Crossover scalping • **Evidence:** Grade B (documented 456-trade test: 65% WR, PF 1.85; practitioner targets 60–70% WR)

Win rate	Target/stop	Trades/session	Spread ceiling	Monthly goal
60–65% (65% documented)	10–15 pips / 8–10 pips	3–8	< 2 pips	8–12% (optimistic ceiling)

Concept & edge. Fast EMA crosses during high-liquidity windows catch 10–15 pip micro-moves repeatedly. The edge is frequency × discipline, not any single trade. This is the archetypal "job-like" scalping system.

Setup. EMA(8), EMA(21), RSI(14), ATR(14) on M5. Trade only London (08:00–11:00 local) and early NY.

Entry rules. Long: 8-EMA crosses above 21-EMA with RSI > 50; enter on the first M1 pullback tap of the 8-EMA. Short: mirror with RSI < 50.

Exit rules. Fixed target 10–15 pips; stop 8–10 pips (adjust by ATR: stop $\approx 0.8 \times \text{ATR}(M5)$). Time stop: 45 minutes. All flat before major news.

Risk management. The documented kill factors: overtrading (cap at 8 trades/session), spread widening (abort when spread > 2 pips), and news. Risk 0.5% per trade — scalping's high frequency makes variance arrive fast.

Backtest evidence. Documented 456-trade test: 65% win rate, PF 1.85. Practitioner progression data: beginners 55–60%, experienced scalpers 65–70% with 20–50 pips/day. Independent cost analyses warn: at 20 trades/day and 1.5-pip average spread, costs consume ~30 pips/day — the strategy must clear that hurdle first.

Works / fails. Works on EUR/USD, GBP/USD during liquid sessions. Fails during news, lunch hours, on exotics, and with market-maker brokers who widen spreads into volatility.

Backtesting notes. Your backtest is a lie unless spread is modeled per-timestamp (it's not constant). Use ECN tick data if possible. Compare M1-execution vs M5-signal fills.

Variations. RSI(7) for faster confirmation; EMA(5)/EMA(13) hyper version on M1 (costs decide); add session VWAP as bias filter.

S39 Tape & Order-Flow Scalp (Level 2)

Market: ES/NQ futures, liquid stocks • **Timeframe:** Tick/seconds • **Style:** Order-flow scalping • **Evidence:** Grade C (professional practice; sparse public statistics — validate yourself)

Win rate	Typical RR	Hold	Requirement	Edge source
55–65% (desk-typical)	0.5–1:1	Seconds–minutes	DOM + footprint data	Microstructure

Concept & edge. At the shortest horizon, price is not "analysis" — it's the auction itself. Large resting orders act as magnets/walls; their *pulling* or *absorption* predicts the next ticks. This is how prop desks scalp: read the book, join the imbalance, exit at the first sign of absorption failure.

Setup. DOM (depth of market), footprint/delta chart, time & sales. Liquid instruments only (ES, NQ, top-20 stocks).

Entry rules. Long: repeated large-bid absorption at a level (sellers hit the bid, price doesn't fall) + delta flipping positive; enter at the level, stop 2–4 ticks below it. Short: mirror at the ask.

Exit rules. Target: the next visible liquidity pool (5–15 ticks). Scratch immediately if absorption fails (price ticks through the level). Time stop: 2–3 minutes.

Risk management. Ultra-tight stops mean high scratch frequency — normal. Risk 0.25–0.5% per trade; daily stop after 3 full losses. This style has the steepest learning curve in the book.

Backtest evidence. No rigorous public statistics exist — order-flow edges don't survive publication (they're capacity-limited and execution-dependent). Desk lore cites 55–65% scratch-adjusted win rates. Treat as a skill to build on sim for 3–6 months, not a backtested system.

Works / fails. Works in thick, two-sided markets during active hours. Fails in thin tape, during news spikes (book pulls), and with data feeds slower than the professionals' (you're the liquidity).

Backtesting notes. Nearly impossible to backtest without recorded book data (Bookmap/Jigsaw exports). Forward-test on simulator with recorded sessions and grade your reads.

Variations. Iceberg-detection entries; spoof-aware fading (advanced — and risky); iceberg + footprint confirmation combo.

S40 Bollinger Band 1-Minute Scalp

Market: EUR/USD, USD/JPY • **Timeframe:** M1 • **Style:** Band-fade scalping • **Evidence:** Grade C/B (widely traded; practitioner ranges 65–75% WR in ranges)

Win rate	Target	Stop	Regime gate	Frequency
65–75% (range sessions)	5–8 pips / midline	5–7 pips	Flat bands only	10–20/session

Concept & edge. On M1 in quiet sessions, price oscillates around micro-means; the bands touch-and-return pattern repeats dozens of times per session. Tiny edges, huge frequency — the purest cost-sensitivity test in trading.

Setup. Bollinger (20, 2) on M1; require flat band slope (bands parallel, price crossing the midline repeatedly). Session: late NY / early Asia (quiet).

Entry rules. Short on a touch/poke above the upper band with an M1 bearish rejection close; long at the lower band mirror. Only when bands are flat.

Exit rules. Target: band midline (safer) or opposite band. Stop: 1.5× the current band width beyond the band — a close beyond the band with momentum kills the trade thesis instantly.

Risk management. One momentum bar through the band wipes 3–4 wins. Hard rule: after any band-walk begins (2 consecutive closes outside), stop fading that side. Risk 0.3–0.5%/trade.

Backtest evidence. Practitioner ranges for disciplined flat-band scalping: 65–75% win rates at ~1:1, profit factors 1.2–1.5 *before* costs. Independent rigorous tests of naive band systems show the edge evaporates without the flat-regime gate and honest spread modeling.

Works / fails. Works in dead sessions on EUR/USD-class pairs. Fails at any volatility event, in trends, and with > 1-pip effective spread.

Backtesting notes. If your backtest uses constant spread, multiply it 2× and re-run — if the system dies, it was never alive. M1 systems need tick data for honest band-touch fills.

Variations. Keltner-midline confluence; RSI(7) stretch filter; fade only touches in the direction of the H1 trend (safer hybrid).

S41 Stochastic Scalping — A Documented Warning

Market: Forex majors • **Timeframe:** M15 • **Style:** Oscillator reversal • **Evidence:** Grade A — *negative* (FXGlory 1,207-trade controlled test, 2026)

Win rate	Expectancy	Profit factor	Max DD	Verdict
28.5%	−0.23R/trade	0.50	−280R	Fails as taught

Concept & edge. This page is in the book deliberately: Stochastic crossover scalping is one of the most-taught retail systems, and a rigorous controlled test (6 pairs, 1,207 trades, realistic costs) shows it *loses* — 28.5% win rate, −0.23R expectancy, every setup family negative. Learning what doesn't work is half of strategy design.

Setup. The tested baseline: slow Stochastic (14,3,3), 100-SMA trend filter, structure stops, 1.3R targets, London/NY windows, 1.5-pip spread + 0.5-pip slippage per side.

What was tested. Five families: 50-line continuation, divergence reversal, MA-pullback re-entry, range crossover, trendline break. Best family (50-line continuation): still −0.20R. Divergence had the best PF: 0.57 — still losing.

Why it fails. (1) %K/%D crossovers lag by construction; (2) overbought ≠ sell — in trends, Stochastic pins at extremes while price runs; (3) 1.3R targets at 28–40% win rates can't survive 4 pips of round-trip cost.

Salvage directions. The data points to what might work: extreme settings (90/10), higher-timeframe regime gating, and using Stochastic as a *filter* (don't buy when > 80 in downtrends) rather than a trigger. Every salvage must be re-tested from zero.

Backtest evidence. Full baseline: 1,207 trades, 28.5% WR, −278R total, 21-trade worst losing streak. Session split: rollover/off-hours worst (−0.47R); London morning weakest liquid session (−0.25R).

Works / fails. Fails as a trigger, as documented. Works (unproven, test it) as a regime/filter input inside larger systems.

Backtesting notes. Use this study as your methodology template: fixed costs, setup families separated, per-pair and per-session breakdowns. That structure is how you autopsy any strategy.

Variations. StochRSI (claimed 78% WR in vendor tests — apply identical skepticism); Stochastic + S/R confluence-only entries; (5,3,3) on H1 swing instead of M15 scalp.

S42 Floor-Trader Pivot Scalping

Market: Futures, forex majors • **Timeframe:** M5–M15 • **Style:** Level-based intraday • **Evidence:** Grade B (floor-trading lineage; practitioner backtests)

Win rate	Typical RR	Key levels	Behavior	Style
55–65% (at levels)	1–1.5:1	P, S1/R1, S2/R2	First-touch reactions	Level scalping

Concept & edge. Classic pivots: $P = (H+L+C)/3$ from yesterday, with S1/S2/R1/R2 projected. These levels are watched by enough systematic traders that first touches produce real reactions — a self-fulfilling liquidity phenomenon. ~70% of the time price stays between S1 and R1 in normal sessions.

Setup. Daily pivots from yesterday's H/L/C. Mark P, S1, R1 (S2/R2 for volatile days).

Entry rules. Fade first touches: long at S1 with an M5 rejection candle, target P; short at R1 mirror. Trend-day alternative: buy the *acceptance* above R1 for a run at R2 (the opposite trade — day-type decides).

Exit rules. Fades: target the next level toward P; stop beyond the level by $0.5 \times \text{ATR}(M15)$. Breakout version: stop back below the level; target next level.

Risk management. Pivots are zones, not lines — allow a small buffer. The two playbooks (fade vs acceptance) are opposites; mixing them without a day-type rule is how pivot traders bleed. Risk 0.5–1%.

Backtest evidence. Practitioner statistics: first-touch reactions at S1/R1 succeed ~55–65% to the next level on normal days; pivot-range containment (S1–R1) holds on the large majority of sessions. Documented across futures desks since the pit era and still coded in modern algos.

Works / fails. Works on mean-reverting days in liquid markets. Fails on trend days (levels slice through) — combine with the S34 day-type classifier.

Backtesting notes. Define touch/reaction mechanically (touch within 0.2 ATR, rejection = close back through level). Test which level (P vs S1/R1 vs S2/R2) carries the reactions in your market.

Variations. Weekly pivots for swing context; pivot + prior-day high/low confluence only; Camarilla (S43) and CPR (S45) as alternative level families.

S43 Camarilla R3/S3 Reversal

Market: Forex, indices • **Timeframe:** M5–M15 • **Style:** Mean-reversion levels • **Evidence:** Grade B (Nick Stott's 1989 system; one published small-sample backtest: 37 trades, 91.9% WR — treat with skepticism)

Win rate	Key levels	Published sample	Skepticism flag	Style
65–75% (practitioner)	R3/S3 (fade), R4/S4 (break)	91.89% WR, 37 trades	Tiny sample	Level reversion

Concept & edge. Camarilla levels compress today's expected range using yesterday's close and range: $H3 = C + \text{range} \times 1.1/4$, etc. The R3/S3 zone is where intraday moves statistically exhaust in ranging conditions; R4/S4 breaks signal trend days. It codifies the "revert or explode" decision into levels.

Setup. Compute H3/H4/L3/L4 from yesterday's H/L/C. Plot as zones.

Entry rules. Range playbook: fade L3/H3 with an M5 reversal pattern, stop beyond L4/H4, target the pivot/C. Trend playbook: a 15-min close beyond H4/L4 = trade the break toward $1.5\times$ targets.

Exit rules. Fades: target midrange, stop beyond the 4-level (hard). Breaks: stop back inside the 3-level, target $1.5\text{--}2\times$ the 3–4 distance.

Risk management. The 3-vs-4 level decision is the strategy. If price closes through the 4-level, your fade thesis is objectively dead — no hope-holds. Risk 0.5–1%.

Backtest evidence. One published trend-variant test reported 91.89% WR — but over only 37 trades, statistically meaningless without replication. Practitioner consensus for L3/H3 fades in ranges: 65–75% win rate with modest targets. Demand your own 300+ trade validation.

Works / fails. Works on normal-volatility days on forex majors and index CFDs. Fails on news days (4-levels break immediately) and on instruments with gappy sessions.

Backtesting notes. Test R3/S3 fade vs R4/S4 breakout *as one combined system* — the pair is designed to work together. Small-sample "90% win rate" claims should always trigger your replication instinct.

Variations. Trade only when H4–L4 distance < 60% of 20-day ATR (compression qualifier); combine with RSI(14) extreme at the level.

S44 Central Pivot Range (CPR) Breakout

Market: Indices (Nifty heritage), futures, forex • **Timeframe:** M5–M15 • **Style:** Pivot-zone trend/reversion • **Evidence:** Grade B (Indian index practitioner literature; widely back-tested by retail quants)

Win rate	Signal	Key read	Style	Best regime
50–60%	CPR width + position vs CPR	Narrow CPR = trend day	Zone-based	Both (switching)

Concept & edge. CPR = (P, BC, TC) — the pivot plus its two satellites, forming a zone. Tomorrow's relationship to the zone classifies the day: narrow CPR + open outside it = trend day (trade breaks); wide CPR + open inside = range day (fade the edges). It's a mechanical day-type classifier built from one formula.

Setup. Daily CPR from yesterday's H/L/C; CPR width vs 20-day average width.

Entry rules. Trend playbook: narrow CPR (width < 50% of average); enter on the first pullback to the CPR edge after price opens and holds outside it. Range playbook: wide CPR; fade TC/BC edges toward P.

Exit rules. Trend: stop through the CPR, target 1.5–2R or trail swings. Range: target P, stop beyond the opposite edge. Flat by session end.

Risk management. Virgin CPR (levels untested for days) acts stronger than repeatedly-tested CPR — weight accordingly. Risk 0.5–1%; the narrow-CPR trend day is where the real money is, don't overtrade wide-CPR fades.

Backtest evidence. Practitioner backtests on index futures: CPR classifiers correctly tag trend vs range days often enough to produce 50–60% system win rates at 1.5:1+. The narrow-CPR-trend-day relationship is the most replicated finding.

Works / fails. Works on session-based markets (indices, futures). Fails on 24/7 crypto (no clean daily close — use UTC close) and on holiday-thinned sessions.

Backtesting notes. Test CPR-width percentiles against next-day range expansion — publish your own correlation before building entries on it. Combine with S34's VWAP day-type read for confirmation.

Variations. Weekly CPR for swing bias; "CPR + camarilla 3-level" confluence entries; open-drive + CPR-position combo.

S45 Tick-Chart Momentum Scalp

Market: ES/NQ futures, BTC • **Timeframe:** 233/610-tick charts • **Style:** Activity-based momentum • **Evidence:** Grade C (professional practice; validate yourself)

Win rate	Typical RR	Why ticks	Best hours	Style
50–60%	1–1.5:1	Bars = activity, not time	Open + overlap	Momentum scalp

Concept & edge. Time-based bars lie during activity bursts: a 5-min candle hides 50 trades. Tick charts (one bar per N trades) expand exactly when participation expands — breakouts and compressions appear earlier and cleaner. Scalpers use them to see the same setups as time charts but with better timing.

Setup. 233-tick (fast) and 610-tick (context) charts; 20-EMA; volume per bar.

Entry rules. Same momentum logic as time charts — EMA pullback in a fanned tick-ribbon — but executed on tick bars for tighter, earlier entries during high-activity windows. Long: pullback to 20-EMA on the 610-tick, enter on 233-tick resumption.

Exit rules. Stop beyond the tick-chart swing (typically tighter than time-chart equivalents — the RR benefit). Target 1–1.5R or trail.

Risk management. Tick charts speed up during news — treat hyper-activity as a stop-trading signal, not an invitation. Risk 0.5%; the tighter stops tempt oversizing — resist it.

Backtest evidence. No rigorous public backtests exist (tick-bar data availability). Practitioner consensus: tick-based execution improves entry price by 1–3 ticks per trade vs time bars in fast markets — a huge cumulative edge at scalping frequency, impossible to prove in aggregate.

Works / fails. Works in liquid, active sessions. Fails in dead sessions (bars stall), and for traders without a disciplined system already (tick charts accelerate bad habits too).

Backtesting notes. You need trade/tick data, not OHLC. Reconstruct tick bars deterministically, then port a *known* time-based system (S26/S33) and measure entry-quality delta — that's the honest research path.

Variations. Range bars (fixed price movement per bar) as an alternative; volume bars; tick + footprint confirmation.

S46 Order-Book Imbalance Scalp

Market: BTC/ETH perpetuals, ES • **Timeframe:** Seconds–minutes • **Style:** Microstructure signal • **Evidence:** Grade C/B (market-microstructure academic literature supports short-horizon imbalance predictability)

Win rate	Horizon	Signal	Data need	Style
55–62% (studies, short horizon)	10s–5min	Bid/ask size ratio extremes	L2/L3 feed	Quant scalp

Concept & edge. Microstructure research consistently finds that order-book imbalance (bid depth vs ask depth) predicts price moves over the next seconds-to-minutes with small but statistically robust effect sizes. Heavy bid imbalance without price falling = hidden strength; the reverse for asks.

Setup. Real-time L2 data; compute imbalance = $(\text{bidVol} - \text{askVol}) / (\text{bidVol} + \text{askVol})$ over top 5–10 levels; smooth over 5–10 seconds.

Entry rules. Long: imbalance > +0.6 sustained 10+ seconds while price holds a level; enter, stop just below the level. Short: mirror at –0.6. Cancel the thesis if imbalance flips before price moves.

Exit rules. Target: 1–1.5× recent 1-min ATR; or exit when imbalance neutralizes. Time stop: 3–5 minutes. Scratch freely.

Risk management. This is a latency-sensitive strategy: your data feed and execution speed *are* the edge. Retail feeds see a delayed book — the imbalance you see may already be traded against. Risk 0.25–0.5%.

Backtest evidence. Academic microstructure studies (cont, de Larrard, and successors) document significant short-horizon predictive power of imbalance in equities and futures; effect sizes shrink with horizon and are partially captured by HFT. Crypto-perp practitioner tests report 55–62% short-horizon accuracy.

Works / fails. Works with colocated/premium data on liquid instruments. Fails with free/slow retail data (you're trading a stale picture) and around news (books evaporate).

Backtesting notes. Requires recorded L2 snapshots — most retail platforms don't store them. If you can't record book data, you can't honestly validate this strategy.

Variations. Imbalance + trade-flow delta combo; imbalance *reversion* (extreme imbalance resolving against the crowd); perp-funding-aware version.

S47 News-Spike Fade Scalp

Market: Forex majors, gold, indices • **Timeframe:** M1–M5 post-release • **Style:** Overreaction fade • **Evidence:** Grade B (event-study literature on announcement overreaction; practitioner ranges)

Win rate	Window	Hold	Key skill	Risk level
55–65%	2–15 min after release	5–30 min	Judging spike vs regime change	High

Concept & edge. Macro releases trigger an instant algo overreaction; when the number is *within expectations' range*, the initial spike substantially retraces within minutes. You're selling panic liquidity to the algos that overpaid for it.

Setup. Economic calendar; expected vs actual figures; M1 chart; pre-release range marked.

Entry rules. Only fade when the release is a *non-surprise* (actual within consensus range) yet price spiked anyway. Enter the fade after the first M1 stall/reversal candle, 2–5 minutes post-release. Never fade a genuine surprise — those trend.

Exit rules. Target: 50–100% retracement of the spike. Stop beyond the spike extreme. Time stop: 30 minutes.

Risk management. The dangerous strategy in this family: spreads widen violently around releases and stops slip. Use half normal size, hard bracket orders, and skip tier-1 surprises (NFP shocks, rate shocks). Risk 0.5% max.

Backtest evidence. Event-study literature documents systematic partial retracement of announcement spikes when news content is weak; practitioner fade systems report 55–65% win rates on filtered setups. Unfiltered spike fading (fading every release) is a documented account-killer.

Works / fails. Works on second-tier releases (PMI, retail sales, housing). Fails on NFP/CPI/rate shocks, central-bank surprises, and geopolitical headlines.

Backtesting notes. You need a historical surprise database (actual vs consensus), not just price data — without the surprise variable, the backtest is meaningless. Model spread widening (5–20× normal in the first minute).

Variations. Fade the *second* spike (post-press-conference whipsaw); straddle-then-fade hybrid; currency-pair relative version (fade the weaker-reaction leg).

S48 Chandelier-Trail Crypto Scalp

Market: BTC, ETH • **Timeframe:** M5–M15 • **Style:** ATR-trailing momentum • **Evidence:** Grade A (published TradingView replication: 596 trades on BTC)

Win rate	Profit factor	Raw result	After fixes	Style
40.8% (documented raw)	1.05 raw → viable with filters	989% net / 70% DD	200-EMA + ADX filters	Trailing momentum

Concept & edge. Enter with RSI-confirmed Heikin Ashi momentum, exit only on a Chandelier stop (highest high – $1.85 \times \text{ATR}$) — letting volatility decide the exit. The published replication is honest: raw version barely clears breakeven (PF 1.05, 70% DD); with trend and regime filters it becomes a real system. A perfect case study in *strategy repair*.

Setup. Heikin Ashi candles; RSI(25) vs RSI(100) (momentum alignment); Chandelier Exit (ATR period 1, mult 1.85); add 200-EMA and ADX(14).

Entry rules. Long: RSI(25) > RSI(100) + green HA candle + price above 200-EMA + ADX > 20. Short: mirror.

Exit rules. Chandelier trailing stop only (long: HH – $1.85 \times \text{ATR}$). Optional: widen ATR period/multiplier (3/2.5) to cut DD dramatically at some profit cost.

Risk management. The documented 70% raw drawdown shows why filters matter: the 200-EMA removes counter-trend bleeding, ADX removes chop. Risk 0.75% per trade on the filtered version.

Backtest evidence. Published replication (BTC): 596 trades, 40.77% WR, PF 1.048, 989% net with 70% DD — unfit for real money as-is, and a great template for your own repair workflow.

Works / fails. Works in crypto trend legs (post-breakout momentum). Fails in chop and when the trailing logic meets violent wick-heavy reversals typical of perp markets.

Backtesting notes. This page's lesson: always publish (to yourself) the raw version first, then add one filter at a time and log each delta. That's the anti-curve-fit workflow.

Variations. ATR(3, 2.5) conservative trail; partial at 2R + chandelier on the runner; ETH version with wider multipliers.

S49 Liquidity-Sweep Reversal (SMC-Style)

Market: Forex, gold, indices, BTC • **Timeframe:** M5–M15 entry, H1/H4 context • **Style:** Stop-hunt reversal • **Evidence:** Grade C (SMC/ICT framework — popular but *not* academically validated; small-sample tests show promise only with strict confluence)

Win rate	Typical RR	Evidence status	Key filter	Style
45–60% (unverified consensus)	2–3:1	No rigorous public backtest	Sweep + displacement + HTF zone	Narrative reversal

Concept & edge. Price frequently pokes just beyond obvious highs/lows (where stop orders cluster), then reverses sharply — the "liquidity sweep." The proposed edge: trade the reversal after the sweep, confirmed by a strong displacement candle (structure break). Honest caveat: this is a compelling narrative with *no large-sample public validation*; the concepts are also inherently subjective, which invites confirmation bias.

Setup. Mark obvious equal highs/lows, session highs/lows, prior day high/low. H1/H4 bias. Watch for the poke-through + fast reclaim.

Entry rules. Long: price sweeps a sell-side level (breaks the low by a few ticks/pips), closes back above it within 1–3 bars, then prints a bullish displacement (break of the local lower-high). Enter on the displacement or its 50% retrace ("OTE"-style). Short: mirror.

Exit rules. Stop below the sweep wick. Targets: opposing liquidity pool (the other side's equal highs/lows) — typically 2–3R.

Risk management. Because rules are subjective, *fix your definitions in writing before testing*: exact sweep tolerance, displacement body size, retrace depth. Risk 0.5–1%; expect your first coded version to underperform the narrative — that's normal and informative.

Backtest evidence. No rigorous peer-reviewed validation exists. Backtesting-tool vendors report that only *stacked-confluence* versions (sweep + displacement + higher-timeframe zone) show positive expectancy in 200+ trade samples; single-concept versions don't.

Works / fails. Works (anecdotally) around session opens and obvious equal highs/lows on forex/gold. Fails when levels are weak, in strong trends (sweeps become continuations), and in the hands of anyone who hasn't mechanized the rules.

Backtesting notes. The biggest trap in this book: manual chart review cherry-picks winners. Code exact rules, run 300+ trades, and compare against a random-entry baseline with the same stop/target distances.

Variations. Sweep + FVG (fair value gap) entry; session-high/low sweeps only (cleaner than equal-highs); killzone-timed version (London/NY open sweeps).

PART V

Crypto-Specific Strategies

Crypto trades 24/7 with retail-heavy flow, funding mechanisms and no closing auction — producing anomalies that don't exist elsewhere. Several of the best-documented edges in this entire book live here (hourly seasonality, funding arbitrage, weekend effects). The caveat is regime fragility: crypto edges decay as institutional share grows.

S50	The 21:00–23:00 UTC Bitcoin Window
S51	Bitcoin Weekend Effect
S52	The Monday Asia Open Effect (BTC Trend Window)
S53	Funding-Rate Cash & Carry (Delta-Neutral)
S54	Cross-Exchange Funding Spread
S55	Turn-of-the-Candle Effect (BTC Micro-Seasonality)
S56	Q-RSI Crypto Pullback System
S57	Z-Score Momentum Basket (Crypto)
S58	BTC-Neutral Residual Mean Reversion (Crypto)
S59	ETH 5-Minute Optimized Scalping System
S60	Crypto Bollinger Squeeze Breakout
S61	TTM-Style Squeeze (Bollinger-in-Keltner)

S50 The 21:00–23:00 UTC Bitcoin Window

Market: BTC • **Timeframe:** Hourly, 2-hour daily hold • **Style:** Intraday seasonality • **Evidence:** Grade A (Padyšák & Vojtko, *Seasonality, Trend-following and Mean Reversion in Bitcoin*, 2022; SSRN "The Seasonality of Bitcoin", 2023)

Annualized return	Hold time	Volatility vs B&H	Worst hours	Data
~33%	2 hrs/day	Much lower	03:00–04:00 UTC	Gemini hourly, 2015–2022

Concept & edge. Bitcoin's hourly returns are not uniform: the 21:00–23:00 UTC window — after every major equity market has closed — delivered the most economically and statistically significant positive returns of the day. Holding BTC only during those two hours captured ~33% annualized with a fraction of buy-and-hold's volatility and drawdown.

Setup. Hourly BTC data, UTC clock. No indicators.

Entry rules. Buy at 21:00 UTC (the published variant uses 22:00, both hours significant).

Exit rules. Sell at 23:00 UTC. Flat the other 22 hours.

Risk management. Long-only strategy — it dies in bear markets (crypto is the riskiest asset class in uncertainty). Use it as a *tilt*, not a portfolio. Position size as you would any BTC exposure.

Backtest evidence. Gemini hourly data, Oct 2015–Feb 2022: the 22:00 and 23:00 UTC hours were the most significant positive hours; 03:00–04:00 UTC the worst. The 2-hour strategy: ~33% annualized, much smaller max DD than passive holding. Cross-validated across seven exchanges in the turn-of-the-candle literature.

Works / fails. Works in neutral/bull regimes. Fails in bear markets (the authors state this plainly) and may decay as 24/7 institutional flow normalizes hourly patterns — re-verify annually.

Backtesting notes. 20 lines of code on free hourly data (Binance). Check exchange-robustness and post-2022 persistence — the honest test is whether the effect survived ETF-era market structure.

Variations. Long 21:00–23:00 + short/flat 03:00–04:00; combine with the weekend effect (S51); apply to ETH (weaker but present).

S51 Bitcoin Weekend Effect

Market: BTC (strongest), ETH • **Timeframe:** Friday close → Monday • **Style:** Calendar anomaly • **Evidence:** Grade A (QuantifiedStrategies 2014–2026; ACR Journal peer-reviewed study 2025)

Win rate	Avg gain/trade	Annual return	Exposure	ETH version
60% (BTC), 53% (ETH)	2.6% (BTC), 2.2% (ETH)	28% (BTC), 18% (ETH)	~10%	Weaker but valid

Concept & edge. Weekend returns significantly outpace weekday returns across crypto: for BTC, mean daily return rises from 0.12% (weekdays) to 0.23% (weekends), with higher Sharpe and lower drawdowns. Altcoins show an even stronger effect (DOGE: 0.52% vs 0.21%). Drivers: institutional absence, retail dominance, thin liquidity amplifying retail flows.

Setup. Daily closes, UTC. Entry Friday ~21:00–23:00 UTC (or Saturday open variants).

Entry rules. Long BTC late Friday / early Saturday UTC. Optional filter: only when the prior week was not deeply negative (bear-regime filter).

Exit rules. Sell late Sunday / early Monday UTC (before the Monday Asia ramp ideally — see S52).

Risk management. Weekends include the occasional liquidation cascade (Sunday-night wicks are famous). Size for a 5–10% adverse move; the documented max DD is 19% (BTC) and 30% (ETH).

Backtest evidence. BTC 2014–2026: 103 trades, 2.6% average gain, 60% WR, 28% annual at 10% exposure, max DD 19%. ETH 2018–2026: 64 trades, 2.2%/trade, 18% annual, 53% WR. Peer-reviewed confirmation (2020–2025, 7-day momentum): weekend > weekday at $p < 0.05$ for BTC, $p < 0.001$ for altcoins; effect stronger in bull markets, persisting after excluding 2021.

Works / fails. Works in bull/neutral regimes, on BTC and high-beta alts. Weakens in bear markets and as weekend institutional participation grows.

Backtesting notes. Define the weekend precisely (Fri 21:00 UTC → Mon 00:00 UTC variants) and test sensitivity. Watch for interaction with funding (weekend longs pay/receive funding).

Variations. Altcoin-basket version (higher beta, higher variance); Friday-dip-only entries (buy only red Fridays); combine with S50's hourly window.

S52 The Monday Asia Open Effect (BTC Trend Window)

Market: BTC • **Timeframe:** Sunday ~19:00 ET → Monday ~19:00 ET • **Style:** Intraday trend seasonality • **Evidence:** Grade A (Concretum Group research, 2026 — high-frequency trend benchmark)

Window	Pattern	Mechanism	Best tactic	Counter-window
Sun 19:00 ET + 24h	Strong positive trend-benchmark returns	Asian cash-open liquidity sets weekly direction	Trend-following longs	US Sunday morning (chop/mean-revert)

Concept & edge. A high-frequency trend-following benchmark on BTC shows strong intraweek seasonality: trend persistence is highest starting Sunday ~19:00 ET and lasting ~24 hours — aligning with the Monday open of Asian cash equity markets (Tokyo). Fresh weekly positioning creates persistent order flow; conversely, US Sunday morning is the choppiest, most mean-reverting window.

Setup. Intraweek clock (ET). Trend tools: 20-EMA on H1, or Donchian(20) on H1.

Entry rules. Trend playbook during the Sunday-19:00–Monday-19:00 window only: H1 trend-following entries (EMA pullbacks, channel breaks) with the week's initial direction. Mean-reversion playbook during US Sunday morning if you must trade it.

Exit rules. Standard trend exits (opposite channel, 2× ATR trail). Reduce activity sharply after Monday 19:00 ET.

Risk management. This is a *when-to-trade* edge, not a standalone system — gate your existing trend systems with it. Weekend liquidity is thin: size down, widen stops modestly.

Backtest evidence. Concretum's volatility-targeted long/short trend benchmark on BTC: strongly positive returns concentrated in the Sunday-evening → Monday window, negative benchmark performance US Sunday morning — consistent across their high-frequency dataset and robust to volatility targeting.

Works / fails. Works as an overlay on H1–H4 trend systems. Fails as a standalone prediction (it predicts *trendiness*, not direction) and on news-dominated Mondays.

Backtesting notes. Replicate with your own trend benchmark: run a simple H1 channel-breakout system and bucket returns by hour-of-week — the heatmap is the research deliverable.

Variations. Apply the same intraweek mapping to ETH (correlated); combine with S51 (weekend long bias into the Monday window); avoid initiating mean-reversion trades during the window.

S53 Funding-Rate Cash & Carry (Delta-Neutral)

Market: BTC/ETH/SOL perpetuals vs spot • **Timeframe:** 8-hour funding epochs • **Style:** Market-neutral arbitrage • **Evidence:** Grade A (ScienceDirect peer-reviewed 2025: up to 115.9% per 6 months, worst-case -1.92%; Presto Research; Binance/BitMEX data)

Return range	Max loss (study)	Correlation to HODL	Leverage effect	Capacity
Up to 115.9%/6mo (best scenario)	-1.92%	~0 (diversifying)	Non-linear, nuanced	Limited by funding regimes

Concept & edge. Perpetuals pay funding every 8h: when perp trades above spot (bullish crowds), longs pay shorts. Buy spot + short the perp in equal size = price-neutral; collect funding as yield. Classic positive-funding regimes paid 10–50%+ annualized; academic evaluation of 60 scenarios found returns up to 115.9% over six months with worst-case losses of just 1.92%.

Setup. Spot account + perp account (same or cross exchange). Monitor funding rate and predicted funding.

Entry rules. When funding is materially positive (e.g., > 0.01%/8h baseline and rising): buy X of spot, short X of perp. Hold while funding stays positive.

Exit rules. Close both legs when funding compresses toward zero or flips negative. Immediate close both legs on exchange-risk events.

Risk management. Risks are real but non-directional: liquidation on the perp leg (keep leverage $\leq 2\text{--}3\times$ with collateral buffer), basis moves at entry/exit, exchange counterparty risk, funding volatility. The study's key finding: the strategy has *no correlation with HODL* — a true diversification return stream.

Backtest evidence. Peer-reviewed (CEX + DEX, BTC/ETH/XRP/BNB/SOL, 60 scenarios): returns up to 115.9% per 6 months; max loss -1.92%; leverage improves efficiency non-linearly. Presto Research: immediately converting accrued funding improves risk-adjusted returns.

Works / fails. Works in bullish/leveraged-long regimes (funding positive). Fails (goes flat, not catastrophic) in bear regimes when funding flips; tail risk is exchange failure, not price.

Backtesting notes. Pull historical funding (Binance API) + spot/perp prices; model fees, borrow costs, and the basis at entry/exit. Without friction modeling the Sharpe looks unrealistically perfect — that's the warning sign.

Variations. Cross-exchange funding spread (short high-funding venue, long low-funding venue — capital efficient, works even when funding positive everywhere); altcoin high-funding rotation (higher yield, higher risk).

S54 Cross-Exchange Funding Spread

Market: Same perp, two exchanges • **Timeframe:** 8-hour epochs • **Style:** Market-neutral arbitrage • **Evidence:** Grade B/A (documented arbitrage literature; practitioner data)

Typical spread	Yield mechanism	Capital efficiency	Key risk	Best condition
0.03–0.10%/8h	Pocket the funding differential	High (both legs margined)	Venue/counterparty	Funding sign divergence

Concept & edge. The same perpetual trades at different funding rates on different exchanges. Short the perp where funding is high, long it where funding is low (or negative): you collect the spread every epoch while the legs cancel price risk. The dream scenario: positive funding on venue A, negative on venue B — paid on both legs.

Setup. Accounts with margin on 2+ liquid venues; funding monitor (or API polling).

Entry rules. When the funding differential exceeds $\sim 2\times$ round-trip fees per epoch: short perp on the high-funding venue, long equal size on the low-funding venue.

Exit rules. Close both when the differential collapses below the cost threshold. Review collateral distribution weekly (P&L accrues asymmetrically across venues).

Risk management. Exchange insolvency/depeg risk is the real one (you're long exposure to two venues). Keep per-venue exposure capped; prefer top-tier venues. Funding can flip sign between epochs — monitor at each settlement.

Backtest evidence. Documented examples in arbitrage literature: differentials of 0.03–0.10% per 8h common in volatile regimes (annualizes to 30–100%+ on margin, before fees). More capital-efficient than cash-and-carry since no spot capital is tied up.

Works / fails. Works in volatile, trending regimes when venue user-bases diverge (retail-heavy venues run hotter funding). Fails in quiet markets (differentials < fees) and during withdrawal halts/depegs.

Backtesting notes. Collect multi-venue funding history (public APIs) and compute the differential time series; model transfer times between venues for collateral rebalancing.

Variations. Tri-venue rotation; funding + basis combined arb; DEX-vs-CEX funding spread (ApolloX/Drift studied academically — higher yield, smart-contract risk).

S55 Turn-of-the-Candle Effect (BTC Micro-Seasonality)

Market: BTC (all major exchanges) • **Timeframe:** Minute-level, :00/:15/:30/:45 marks • **Style:** HFT-style seasonality • **Evidence:** Grade A (peer-reviewed, PMC 2023 — t-stats > 9 across seven exchanges)

t-statistic	Pattern	Net of fees	Exchanges verified	Era
> 9 (all 7 exchanges)	Positive returns concentrated at 15-min candle turns	Outperforms B&H after fees/spread from \$5k capital	7	Appeared mid-late 2020

Concept & edge. Positive BTC returns are disproportionately concentrated in the *first minute* of each 15-minute candle (the :00/:15/:30/:45 marks) — the candles every bot and chart watches. Algorithmic order-flow synchronization at candle turns creates a measurable, exploitable micro-pattern.

Setup. Minute-level BTC data. Algos trigger at candle opens; you position *just before* the turn.

Entry rules. Long seconds before each 15-min candle turn; hold through the first minute(s) of the new candle; exit as the effect decays. (The paper's exploit strategy outperformed B&H net of fees and spreads with capital as low as \$5,000.)

Exit rules. Time-based: exit within the opening minutes of the candle. Strict — this is a micro-hold strategy.

Risk management. Capacity-limited and decay-prone (the paper notes the effect *appeared* in 2020 — anomalies born from algo behavior can die from it too). Fee tier is existential: you need maker/low-taker fees.

Backtest evidence. Statistically significant on all seven sampled exchanges in 2021 ($t > 9$), robust to outliers and heavy tails; net-outperforms buy-and-hold after fees and bid-ask spreads at small capital sizes.

Works / fails. Works with fast execution and low fees on BTC. Fails with retail fee tiers (0.1% taker kills it), slow APIs, and possibly post-publication as the edge gets arbed.

Backtesting notes. Needs minute data with exact timestamps; verify the effect's persistence post-2021 before building. This is a quant playground, not a manual strategy.

Variations. Hour-candle turns (bigger window, weaker effect); altcoin versions; combine with imbalance signals (S46) at candle turns.

S56 Q-RSI Crypto Pullback System

Market: BTC • **Timeframe:** Daily (RSI extremes), swing hold • **Style:** Quant mean reversion • **Evidence:** Grade A (MenthorQ published backtest, late 2024–mid 2025)

Result vs BTC	Rules	Stop	Hold	Sample
+18% vs BTC +10% (and avoided 2025 drawdowns)	RSI extreme + 5-SMA trigger	None (time exit)	Fixed 10 days	Multi-month published

Concept & edge. Inverts classic RSI logic: instead of selling overbought, it *buys* volatility extremes with trend confirmation — RSI < 20 recently + close above the 5-SMA (confirmed reversal), or RSI > 90 recently + close below the 5-SMA (exhaustion retracement). Crypto's violent two-sided volatility makes extremes tradeable in *both* directions.

Setup. Daily BTC: RSI(14) and 5-day SMA.

Entry rules. Buy condition 1: RSI < 20 at least once in past 3 days AND today's close > 5-SMA. Buy condition 2: RSI > 90 at least once in past 3 days AND today's close < 5-SMA.

Exit rules. Time exit exactly 10 days after entry. No stop loss (documented design choice — position sizing carries the risk).

Risk management. No stops means sizing *is* the risk management — small fixed fractions (2–5% of equity per position). The 10-day hold rides out noise that stops would convert to losses.

Backtest evidence. Published multi-month backtest: +18% cumulative vs BTC's +10%, with the strategy sidestepping the deep drawdowns BTC suffered in early 2025 — consistent equity growth through choppy conditions.

Works / fails. Works on BTC (and likely ETH) in two-sided volatile regimes. Fails in smooth, low-volatility trends (few signals) and structurally falling markets (condition-1 buys keep catching knives — the 5-SMA filter mitigates but doesn't eliminate).

Backtesting notes. Test RSI thresholds (15/85, 20/90, 25/75), lookback windows (3 vs 5 days), and holds (5/10/15 days). No-stop systems need honest max-adverse-excursion analysis.

Variations. Add a loose 2× ATR disaster stop and compare; ETH/SOL versions; asymmetric sizing (larger on condition 1 than condition 2).

S57 Z-Score Momentum Basket (Crypto)

Market: Top-liquid crypto basket • **Timeframe:** Daily • **Style:** Cross-sectional momentum • **Evidence:** Grade A (published systematic study, Binance data 2017–2025)

Sharpe	Best era	Enhancement	Combined-system Sharpe	Method
~1.0 (raw), ~1.2 (vol-filtered)	Pre-2021 trends	Volatility filter	1.71 (50/50 with S58)	Bayesian-optimized windows

Concept & edge. Rank liquid coins by rolling return z-scores (short-window vs long-window); long the strongest, short the weakest. Cross-sectional momentum transferred well to crypto's early trending era; adding a volatility filter (trade only in stable vol regimes) improved the Sharpe from ~1.0 to ~1.2.

Setup. Daily returns for a liquid-coin universe; rolling $z = (\text{mean_short} - \text{mean_long}) / \text{std_long}$; rolling vol filter.

Entry rules. Rebalance periodically: long top-ranked coins, short bottom-ranked, normalized weights. Bayesian-optimize the short/long windows — then *freeze* them out-of-sample.

Exit rules. Signal-based rotation (positions exit when ranks change). Vol filter overrides: flat when aggregate volatility exceeds threshold.

Risk management. Diversified basket reduces single-coin blowup risk, but shorts in crypto carry squeeze risk — cap per-coin weight, prefer perps for the short leg. Watch borrow/funding costs on shorts.

Backtest evidence. 2017–2025 with out-of-sample testing, walk-forward, and cost modeling: Sharpe ~1.0 raw, ~1.2 vol-filtered; strong pre-2021, decaying as market structure matured. The 50/50 blend with mean reversion (S58): Sharpe 1.71, 56% annualized, t-stat 4.07.

Works / fails. Works in trending, retail-led regimes. Fails post-2021-style choppy correlation regimes (momentum crashes) — hence the vol filter and the blend.

Backtesting notes. Survivorship bias is the big one (delisted coins vanish from datasets, flattering results). Use a point-in-time universe. Report training vs test metrics separately, as the source does.

Variations. Long-only top-quintile (for spot-only traders); funding-adjusted ranking; combine with BTC regime filter.

S58 BTC-Neutral Residual Mean Reversion (Crypto)

Market: Liquid altcoins vs BTC • **Timeframe:** Daily • **Style:** Idiosyncratic mean reversion • **Evidence:** Grade A (same published study; strongest single result)

Sharpe	Best era	Hedge	Combined Sharpe (with S57)	Complexity
~2.3	Post-2021 chop	Rolling beta to BTC	1.71	High (quant)

Concept & edge. Strip each coin's beta to BTC via rolling regression; the residual (idiosyncratic) return mean-reverts: coins that outperformed *their own beta* snap back, and vice versa. Long residual losers, short residual winners, BTC-neutral by construction. Post-2021's choppy, BTC-dominated regime made this the strongest strategy in the study.

Setup. 180-day rolling regression of coin returns on BTC returns → residuals → z-score the residuals; balanced normalized portfolio.

Entry rules. Long coins with residual $z < -2$, short coins with residual $z > +2$, weights beta-neutralized so BTC exposure ≈ 0 .

Exit rules. Exit at residual z crossing ± 0.5 , or time-stop 5–10 days. Rebalance on schedule.

Risk management. Residual models break on idiosyncratic news (hacks, delistings) — exclude coins with pending events; cap single-name weight; the beta estimate lags regime shifts.

Backtest evidence. Sharpe ~2.3, particularly strong post-2021 — the mirror-regime of momentum. 50/50 blended with z-score momentum: Sharpe 1.71, 56% annualized, t-stat 4.07, smoother across regimes than either alone. Methodology included out-of-sample, walk-forward, and realistic costs.

Works / fails. Works in correlation-heavy chop (most of modern crypto). Fails during altseason breakouts (residual shorts run away) and on thin coins.

Backtesting notes. Everything from S57 applies, plus: regression-window sensitivity (90/180/360) and the same survivorship-bias discipline. This is a genuine quant project — budget real time.

Variations. ETH-neutral instead of BTC-neutral; sector-neutral (DeFi vs L1s); intraday (4h) residual z-scores for faster rotation.

S59 ETH 5-Minute Optimized Scalping System

Market: ETH/USD • **Timeframe:** M5 • **Style:** Multi-filter momentum scalp • **Evidence:** Grade A (published open-code study, fees included, train/test split)

Key finding	Params (winner)	Fees	Validation	Lesson
Unoptimized viral params lose after fees	EMA 20/100, BB(50,3), ATR(5)×3, RR 4	0.1% modeled	Train 60% / test 40%	Parameter search + OOS discipline

Concept & edge. A viral EUR/USD scalping system (EMA cross + Bollinger + ATR stops) was ported to ETH 5-min: it *burned money* with 0.1% fees. Systematic parameter search with a stability score (fraction of positive rolling windows) produced configurations profitable *net of fees* in both training and test periods. The strategy itself is secondary — the workflow is the edge.

Setup. Fast/slow EMAs, Bollinger Bands, ATR stops, RSI filter, trend-MA threshold — all parameterized.

Entry rules. (Winning documented config: fast EMA 20, slow EMA 100, BB(50, 3 σ), ATR(5) stop × 3, RR target 4.) Enter on fast/slow EMA alignment with BB breakout context; ATR-based stop; fixed 4R target.

Exit rules. ATR stop or 4R target; trend-MA exit.

Risk management. Fewer, bigger trades beat many small ones after fees — the optimization independently discovered this (RR 4, wide ATR). No leverage in the study; 0.1% fees modeled.

Backtest evidence. Baseline (viral params): negative after fees. Optimized configs: positive in both train (60%) and test (40%) windows — a rare published example of a retail-style crypto scalp surviving honest accounting.

Works / fails. Works when parameter search is regularized (stability scoring) rather than best-fit. Fails when you optimize net profit alone (classic overfit).

Backtesting notes. Copy the workflow: walk-forward splits, rolling-window stability score, fee-first modeling, parameter plateaus. That workflow is what Part XI's appendix formalizes.

Variations. BTC version (wider ATR multiples); maker-fee version (limit entries) — fees drop and the whole system improves; RR 2 variant for higher win rate.

S60 Crypto Bollinger Squeeze Breakout

Market: BTC, ETH, SOL • **Timeframe:** H1–H4 • **Style:** Volatility expansion • **Evidence:** Grade B (documented 3-month M5 ETH studies + volatility-cycle research)

Win rate	Typical RR	Trigger	Failure	Style
45–55%	2–3:1	Squeeze → expansion	Fakeout (common)	Compression breakout

Concept & edge. Crypto alternates compression and expansion violently. When Bollinger Band width hits a multi-month percentile low, the subsequent break travels far enough that 45–55% accuracy at 2–3:1 pays well. Perp funding and liquidation cascades amplify the expansions beyond what equities deliver.

Setup. BB(20, 2); band-width percentile (6-month lookback); ATR; volume.

Entry rules. Band width < 10th percentile; enter on the first H1/H4 close outside the bands with volume > 150% of average. Direction-agnostic — take what the market gives.

Exit rules. Initial stop: opposite side of the squeeze range. Targets: 2R and 3R with ATR trail on the runner. Liquidation-cascade bonus: don't cap winners on high-funding extremes.

Risk management. Fakeouts are the tax (expect ~half your squeezes to fake). First break fails + immediate re-squeeze = stand down for the day. Risk 1%.

Backtest evidence. Documented 3-month 5-min ETH Bollinger studies show raw scalping versions are marginal, but H1/H4 compression-breakout versions with volume confirmation hold positive expectancy; volatility-clustering research explains why (low-vol predicts high-vol, not direction).

Works / fails. Works on majors after weekend/holiday compressions and pre-macro-event coils. Fails with late entries (chasing bar 3 of the expansion) and in repeated-fakeout regimes.

Backtesting notes. The edge is in the *filter* (compression percentile), not the entry candle. Test percentile thresholds 5/10/20 and require volume confirmation; compare with S61's Keltner-based squeeze detection.

Variations. Squeeze + funding-extreme bias (trade against crowded funding); multi-coin scanner (trade whichever major squeezes first); add S28 Supertrend as the trail.

S61 TTM-Style Squeeze (Bollinger-in-Keltner)

Market: All (futures, crypto, forex) • **Timeframe:** M15–H4 • **Style:** Compression-release momentum • **Evidence:** Grade B (John Carter's documented system; extensive practitioner backtests)

Win rate	Typical RR	Signal quality	Best use	Edge
45–55%	2–3:1	Few, high-quality signals	Pre-breakout positioning	Early compression detection

Concept & edge. When Bollinger Bands contract *inside* Keltner Channels, volatility is abnormally compressed (BB measures deviation, KC measures ATR range). The "squeeze firing" — BB expanding back outside KC with momentum direction — times breakouts earlier and cleaner than band-width alone. The dots/momentum histogram make it fully mechanical.

Setup. BB(20,2), KC(20, 1.5 ATR), momentum oscillator (linear-regression momentum per Carter). Squeeze on = BB inside KC.

Entry rules. Enter when the squeeze fires (BB crosses back outside KC) in the direction of the momentum histogram, ideally after ≥ 6 –8 bars of compression. Earlier entries: first momentum-histogram tick while still in squeeze (aggressive).

Exit rules. Initial stop: $1\times$ ATR or opposite side of the squeeze box. Exit/trail when momentum histogram flips or two consecutive opposite bars print. Targets 2–3R.

Risk management. Fake fires cluster when the squeeze was short (< 5 bars) — require minimum compression length. Risk 0.75–1%; one fire per instrument per day.

Backtest evidence. Practitioner backtests across index futures and forex: 45–55% win rates at 2–3:1 with minimum-squeeze-length filters; the system's documented strength is signal *scarcity with quality* — a handful of trades per instrument per month.

Works / fails. Works on volatile instruments (NQ, BTC, GBP pairs) post-compression. Fails in instruments that fake out repeatedly (mean-reverting majors in tight macro ranges) and with premature entries before the fire.

Backtesting notes. Fully codable (squeeze on/off is boolean). Test momentum-histogram definitions and minimum squeeze bars 4/6/8/10. Compare against S60 on identical data — keep whichever your market respects.

Variations. Dual-timeframe squeeze (H4 squeeze, H1 fire); squeeze + ORB hybrid (morning squeeze fires into the opening range); histogram-divergence early exits.

PART VI

Volatility & Range-Compression Strategies

Volatility clusters: quiet begets quiet, then expansion arrives violently. Compression patterns (NR7, inside bars, squeezes) don't predict direction — they predict *movement*, which is tradeable with the right structure. Toby Crabel's pattern research is the founding document of this family.

S62	NR7 Breakout (Crabel)
S63	Inside Bar Breakout
S64	NR4/ID Combo (The Classic Crabel Setup)
S65	ATR Expansion Breakout (Larry Williams Style)
S66	Post-Trend-Day Narrow Range (Continuation Coil)
S67	Volatility Contraction Pattern (Intraday VCP)
S68	Range-Expansion Reversal (Exhaustion Bar)
S69	First-15-Minute Range Fade (Open Reversion)

S62 NR7 Breakout (Crabel)

Market: Indices, forex, gold, crypto • **Timeframe:** Daily signal → intraday trade; or H1 for pure intraday • **Style:** Range-compression breakout • **Evidence:** Grade B (Crabel, 1990 pattern research; persistent practitioner replication)

Win rate	Typical RR	Signal	Follow-through	Style
45–55%	1.5–2:1	Narrowest range of 7 bars	Best with trend filter	Compression breakout

Concept & edge. An NR7 day/bar has the narrowest high–low range of the last 7. Crabel's research found such compression days precede range expansion with above-random follow-through, especially when aligned with the prevailing trend. It's the atomic unit of compression trading.

Setup. Identify NR7 on daily bars (or H1 bars for intraday sessions). Optional trend filter (20-SMA) and volume.

Entry rules. Stop orders both sides of the NR7 bar (OCO); first break triggers. With trend filter: only the trend-direction side. Cancel if neither side triggers within 1–2 bars.

Exit rules. Stop: opposite side of the NR7 bar (it's narrow — risk is small by design). Targets: 1.5–2R, or trail; many pros exit at the session/day close (range expansions typically complete within 1–2 bars).

Risk management. The narrow range means tight stops — and therefore tempting oversize positions. Size by stop distance, cap at 1% risk. Watch for the double-fakeout (break, reverse, break again); max one re-entry.

Backtest evidence. Crabel's original futures research documented NR7 breaks outperforming random entries; practitioner replications: 45–55% win rates at 1.5–2:1, strongest when the NR7 appears inside a trend and after an already-quiet week. NR7 is usually taught *with* the inside-bar variant (S63) — together they outperform either alone in most published comparisons.

Works / fails. Works on volatile instruments after genuinely quiet stretches. Fails in choppy transition regimes (fakeout chains) and on thin instruments where narrow range = no interest, not stored energy.

Backtesting notes. Trivial to code (rolling range rank). Test with/without trend filter, entry at break vs break+retest, and exit at close vs 2R. Confirm the edge survives 1–2 ticks of slippage — tight stops amplify slippage sensitivity.

Variations. NR4 (faster); NR7 + inside day combo (the classic); NR7 in the direction of yesterday's close only.

S63 Inside Bar Breakout

Market: All liquid markets • **Timeframe:** H1–Daily • **Style:** Pattern breakout • **Evidence:** Grade B (price-action literature; practitioner backtests)

Win rate	Typical RR	Best context	Failure	Style
45–55%	1.5–2:1	Trend pullback pauses	Range chop	Pattern momentum

Concept & edge. An inside bar (entire range within the prior bar) is a one-bar compression: the market pausing. Inside bars *within trends* (especially after a pullback leg) resolve with the trend more often than chance; inside bars in the middle of ranges are coin flips. Context is the strategy.

Setup. Inside bar detection; trend context (20-EMA slope or higher-high structure); mother-bar size relative to ATR.

Entry rules. Trend-aligned: after a pullback, enter on the break of the inside bar in the trend direction (stop order). Highest quality: mother bar is a strong trend bar, inside bar small (< 50% of mother).

Exit rules. Stop: opposite side of the inside bar (tight) or the mother bar (conservative). Targets 1.5–2R or trail with structure.

Risk management. Inside-bar chains (multiple consecutive) = deepening compression — the eventual break is often stronger; keep the OCO armed. Risk 0.75% per trade; skip inside bars whose mother bar is enormous (exhaustion, not pause).

Backtest evidence. Practitioner backtests: trend-filtered inside-bar breaks run 45–55% at 1.5–2:1; unfiltered inside bars are ~random after costs — the same story as every pattern in this family. Small-inside-bar (< 50% of mother) subsets outperform in published comparisons.

Works / fails. Works as a trend-continuation timer on H1–H4. Fails standalone in ranges and on M1–M5 (noise bars mimic inside bars constantly).

Backtesting notes. The trend-context definition is the variable that matters — test 3 mechanical definitions. Pattern-code is 10 lines; spend your effort on context filters.

Variations. Inside bar at S/R levels (reversal version); "ii" double-inside (compression²); H4 inside bar traded on H1 break (better entry price).

S64 NR4/ID Combo (The Classic Crabel Setup)

Market: Futures, forex • **Timeframe:** Daily → intraday execution • **Style:** Dual compression breakout • **Evidence:** Grade B (Crabel research lineage; this is the historically famous one)

Win rate	Typical RR	Rarity	Historical status	Style
50–58% (documented ranges)	1.5–2.5:1	~1–3/month per market	Crabel's flagship	Premium compression

Concept & edge. A day that is *both* an NR4 (narrowest of 4) *and* an inside day compresses volatility twice — the dual condition was Crabel's highest-conviction expansion signal. Rarer than NR7 (1–3 per market per month) but historically more reliable per signal.

Setup. Daily bars: today = inside day AND narrowest range of the last 4. Mark the day's high/low for tomorrow's triggers.

Entry rules. Next session: stop orders 1 tick beyond the combo day's high and low (OCO). Trade whichever side triggers; trend filter optional (documented as improving per-trade expectancy at some cost to frequency).

Exit rules. Stop: opposite side of the combo day. Target: 1.5–2.5R or the measured move (combo range projected). Time exit: end of the trigger day if neither target nor stop hits.

Risk management. Rare signals tempt forcing trades when they finally appear — stick to 1% risk. If the trigger day opens with a big gap beyond the trigger, skip (the expansion already happened).

Backtest evidence. Crabel's futures research ranked NR4/ID among his best opening-range patterns; practitioner replications across decades show 50–58% win rates at 1.5–2.5:1, with performance clustering in commodities and currencies (trend-prone) rather than equities.

Works / fails. Works on trend-prone futures (energies, metals, softs, currencies). Fails in equities (mean-reverting) and during regime transitions with chained fakeouts.

Backtesting notes. Compare NR4/ID vs NR7 vs plain inside day on the same data — know which your market pays for. As with all breakout pages here, slippage stress-test at 1/2/3 ticks.

Variations. NR7/ID (rarer, stronger); combo + prior-week trend alignment; intraday H1 version for gold/oil session trading.

S65 ATR Expansion Breakout (Larry Williams Style)

Market: Futures, crypto, forex • **Timeframe:** H1–Daily • **Style:** Volatility-breakout (range projection) • **Evidence:** Grade B (Williams' documented volatility-breakout work; practitioner replications)

Win rate	Typical RR	Trigger	Key parameter	Style
40–50%	2–3:1	Open + $k \times \text{ATR}$ projection	$k \approx 0.5\text{--}1.0$	Projection breakout

Concept & edge. Williams' volatility breakout: tomorrow's likely move is entered by projecting a fraction of recent range (ATR) from the open — buy at open + $0.7 \times \text{ATR}$, short at open – $0.7 \times \text{ATR}$. The concept powers countless CTA systems: expansion tends to continue once a volatility threshold is crossed.

Setup. ATR(14) or yesterday's range; session open price; k multiplier.

Entry rules. At session open, place stops at open $\pm k \times \text{ATR}$ (start $k = 0.7$, optimize 0.5–1.0). OCO; first trigger enters. Trend-filtered variant: only the 20-SMA side.

Exit rules. Stop: open (the pivot of the trade) or $1 \times \text{ATR}$ from entry. Target: $2\text{--}3 \times \text{ATR}$, or close of session. Williams-style: exit next open (overnight version).

Risk management. Whipsaw days hit both stops — cap at one trigger per direction per session. k and the stop placement trade off frequency vs win rate; document the trade-off curve. Risk 1%.

Backtest evidence. Volatility-breakout systems are among the most replicated in futures: raw versions ~breakeven, trend/day-filtered versions 40–50% at 2–3:1. Williams documented the approach on bonds and indices; modern CTA variants remain live on commodities and crypto.

Works / fails. Works on wide-ranging instruments (gold, crude, NQ, BTC). Fails in mean-reverting quiet markets and with k set too tight (noise triggers).

Backtesting notes. Grid k from 0.4–1.2 in 0.1 steps; demand a plateau. Interaction between k and trend filter is the research question — publish your own surface plot before choosing.

Variations. Weekly ATR projection (Monday open $\pm k \times \text{weekly ATR}$); dual-side straddle with no filter (true straddle); OOPS pattern (open outside yesterday's range, fade — Williams' reversal cousin).

S66 Post-Trend-Day Narrow Range (Continuation Coil)

Market: Indices, futures, crypto • **Timeframe:** Daily context → H1 intraday • **Style:** Momentum continuation • **Evidence:** Grade C/B
(practitioner pattern; consistent with momentum-persistence literature)

Win rate	Typical RR	Setup rarity	Best vehicle	Style
55–65%	1.5–2:1	~few/month per market	Trending instruments	Continuation

Concept & edge. A large-range trend day followed by a narrow inside-ish day = the market digesting, not reversing. Momentum-persistence research supports continuation: the coil after initiative activity resolves with the trend more often than not.

Setup. Day 1: range $> 1.5 \times \text{ATR}(14)$ with directional close (in the day's top/bottom quartile). Day 2: range $< 0.6 \times \text{ATR}$, ideally inside day 1's range.

Entry rules. Day 3 (or intraday on day 2's H1): enter on break of day 2's range in day 1's direction. Stop at day 2's opposite edge.

Exit rules. Target: day-1 range projected, or 1.5–2R. Time exit: 2 days max — continuation coils fire fast or fizzle.

Risk management. If day 2's range isn't actually narrow (check the ATR ratio mechanically), there's no coil and no edge — don't force it. Risk 0.75–1%.

Backtest evidence. Practitioner tests of trend-day/narrow-day sequences: 55–65% continuation win rates on index futures and BTC; consistent with daily-horizon time-series momentum literature showing 1–3 day persistence after large directional moves.

Works / fails. Works in macro trend environments and breakout weeks. Fails in rotation regimes where every big day reverses (check your market's current character first).

Backtesting notes. Mechanically define day 1 (range percentile + close location) and day 2 (range ratio) — then just measure P(day-3 continuation). If base rate doesn't exceed ~55%, skip entry engineering.

Variations. Intraday version: H1 trend leg → H1 coil → H1 continuation; weekly version for position traders; add volume contraction on day 2 as quality filter.

S67 Volatility Contraction Pattern (Intraday VCP)

Market: Stocks, crypto • **Timeframe:** M15–H1 • **Style:** Progressive-compression breakout • **Evidence:** Grade C (Minervini's swing framework adapted intraday; practitioner replication)

Win rate	Typical RR	Signature	Entry timing	Style
50–60%	2–3:1	Successively smaller pullbacks	At the pivot	Precision breakout

Concept & edge. VCP: each pullback within a consolidation is smaller than the last (e.g., 8% → 4% → 2%) as supply dries up. The final contraction, on shrinking volume, precedes expansion. Intraday adaptation: contracting M15 pullbacks under a clear level, volume drying into the apex.

Setup. M15/H1 chart; mark the consolidation's pullback depths and volume per pullback; define the pivot (resistance being pressed).

Entry rules. Enter on the break of the pivot *after* at least 2–3 progressively smaller contractions, with the last contraction < half the first and volume at its lowest of the pattern. Anticipation entry (small size at the apex) is the aggressive variant.

Exit rules. Stop: below the last contraction's low (tight — the pattern's gift). Targets 2–3R; trail if volume explodes on the break (real sponsorship).

Risk management. The tight stop under the apex is the whole RR story — but it also means stop-outs from wicks. Use a small buffer ($0.2 \times \text{ATR}$). Risk 0.75–1%.

Backtest evidence. Swing-timeframe VCP is extensively documented by Minervini (US Investing Championship record) for stocks; intraday adaptations are practitioner-verified rather than study-verified — 50–60% continuation rates on clean patterns. Validate on your market.

Works / fails. Works on momentum names and trending crypto majors. Fails in choppy indices and when the "contractions" are just random noise (require the volume-dryup signature).

Backtesting notes. Hard to fully mechanize (pattern recognition); semi-systematic approach: code contraction-ratio + volume-slope filters, then review signals manually before backtesting entries.

Variations. Apex-limit entries; VCP + RVOL filter (only on the day's active names); failed-VCP reversal trade (advanced).

S68 Range-Expansion Reversal (Exhaustion Bar)

Market: All • **Timeframe:** M15–H1 • **Style:** Climax fade • **Evidence:** Grade C/B (practitioner; consistent with short-term reversal literature)

Win rate	Typical RR	Signature	Danger	Style
55–65%	1–1.5:1	2 σ + bar closing off extremes	Trend continuation	Exhaustion fade

Concept & edge. After an extended run, a sudden huge range bar ($\geq 2\times$ ATR) that *closes well off its extreme* marks exhaustion — the last traders chased, and there's no one left. Short-term reversal is one of the most robust documented anomalies; the exhaustion bar is its candlestick expression.

Setup. ATR(14) context; prior run = 3+ same-direction bars or $\geq 2\sigma$ extension from the 20-EMA. The trigger bar: range $\geq 2\times$ ATR, close in the opposite third of its range.

Entry rules. Short after an up-exhaustion bar: enter on the next bar's break of the trigger bar's low (or a 50% retrace of it). Long: mirror after down-exhaustion.

Exit rules. Target: the 20-EMA or the run's 38–50% retrace. Stop: beyond the trigger bar's extreme (with buffer). Time stop: 5–8 bars.

Risk management. Fading climaxes in *parabolic* regimes (blow-off tops, squeeze cascades) is suicide — require the close-off-extreme signature and a non-parabolic prior run. Risk 0.5–0.75%.

Backtest evidence. Short-term reversal after large-range bars is documented in equity and crypto reversal literature; practitioner exhaustion systems: 55–65% win rates to mean targets. The entry *trigger* (next-bar break vs immediate fade) shifts the WR/RR balance materially — test both.

Works / fails. Works after extended intraday runs in normal regimes. Fails in genuine breakouts (the exhaustion bar becomes bar 1 of a trend) and during news cascades.

Backtesting notes. Define "prior run" and "extended" mechanically (σ -distance from mean). Measure follow-through distributions after qualifying bars vs all bars — the delta is the edge.

Variations. Volume-climax requirement (highest volume of N bars); RSI(2) $> 95/< 5$ as the quantifier; two-bar exhaustion (engulfing reversal) version.

S69 First-15-Minute Range Fade (Open Reversion)

Market: ES, NQ, DAX • **Timeframe:** M1–M5 • **Style:** Opening-range mean reversion • **Evidence:** Grade C/B (practitioner; consistent with opening-overreaction research)

Win rate	Typical RR	Window	Filter	Style
55–65%	0.8–1.2:1	09:45–10:30 ET	Wide first-15 vs ATR	Open fade

Concept & edge. The complement of ORB: on *most* days (which are not trend days), the first 15 minutes' range holds for hours. Fading tests of the first-15 extremes back toward the midpoint exploits opening overreaction — the mirror image of S14, and the two can be run as one day-type-switching system.

Setup. First-15-min high/low; width filter: fade only when the range is *wide* relative to ATR (emotional open) — narrow ranges favor breakouts instead.

Entry rules. Fade the first test of the first-15 high/low after 09:45 with an M1/M5 rejection; enter toward the range midpoint. If price *accepts* outside (2 closes), flip to the ORB playbook.

Exit rules. Target: range midpoint, then opposite edge. Stop: 0.3–0.5× the range beyond the tested edge. Time stop: by 11:30 ET.

Risk management. The wide-range filter is critical: fading narrow openings means fading breakouts (donation). Risk 0.5%; max 2 fades per morning.

Backtest evidence. Practitioner statistics: wide-first-15 opens revert to midpoint ~55–65% of the time; the ORB mega-studies (S14) independently imply the same base rates (most opens don't trend). Running fade vs breakout as a width-switched system is documented on futures desks.

Works / fails. Works on emotional, gappy opens that lack follow-through volume. Fails on genuine trend days — hence the acceptance flip rule.

Backtesting notes. Test width thresholds (as % of 14-day ATR) for the fade/break switch. The two playbooks should be backtested *together* as one system — that's the honest design.

Variations. Fade only against the overnight gap; London-open version (first-15 of the cash session); midpoint-scaled exits (half at 50%, half at opposite edge).

PART VII

News & Event-Driven Strategies

Events concentrate volatility into minutes. The honest literature is split: some event strategies have real edges (post-announcement drift), others are casino games with extra steps (straddling tier-1 releases with retail spreads). This section gives you both — with the evidence labeled.

S70	NFP Straddle (With Honest Caveats)
S71	NFP Aftermath Fade / Continuation
S72	FOMC Drift & Announcement-Day Pattern
S73	CPI Release Momentum
S74	Rate-Decision Straddle-to-Trend (CB Events)
S75	Earnings Gap Fade (Stocks)
S76	Earnings Gap-and-Go (Momentum Side)
S77	Weekend Gap Monday Fade (Forex & Crypto)

S70 NFP Straddle (With Honest Caveats)

Market: EUR/USD, USD/JPY, gold • **Timeframe:** M1–M5 around 13:30 UTC, first Friday monthly • **Style:** Event volatility • **Evidence:** Grade B/A (NFP backtests published; spread/slippage warnings documented)

Win rate	Typical RR	Spread reality	Verdict	Better alternative
40–55% (retail-realistic)	1.5–2:1	5–20× normal at release	Marginal after costs	S71 (fade) or S72 (drift)

Concept & edge. NFP is the highest-impact scheduled release in forex. The straddle (OCO stop orders both sides pre-release) tries to catch the initial burst without predicting direction. The theory is clean; the execution reality is not.

Setup. Economic calendar; pre-release range marked; OCO orders 10–20 pips beyond the range, placed 1 minute before release.

Entry rules. Whichever stop triggers, cancel the other. Some pros *don't* enter on the first 1-minute spike (whipsaw zone) but on the first M1 close — fewer fills, better ones.

Exit rules. Target: 1.5–2× the pre-release range. Stop: the pre-release range's opposite side. Time stop: 30 minutes — NFP moves resolve fast or chop.

Risk management. Documented killers: spreads widen 5–20× at release, stop orders slip 10–50 pips on fast tape, both legs can trigger in the whipsaw. Half-size, hard caps, and ECN execution are mandatory. Risk 0.5% max — treat it as a lottery-ticket line item, not a core system.

Backtest evidence. Published NFP backtests (QuantifiedStrategies on post-NFP days) show the *directional* post-release edge is weak-to-negative for stocks/bonds — the money is in the volatility itself. Retail straddle tests: 40–55% win rates with extreme variance in fills. Broker data consistently shows NFP as a high-slippage event.

Works / fails. Works with institutional-grade execution on genuine surprise prints. Fails for most retail setups — the house edge is execution, and the house isn't you.

Backtesting notes. Backtesting straddles on OHLC data is meaningless (fills are the strategy). If you can't model the first-minute spread/slippage, don't trust any result.

Variations. Trade only when consensus dispersion is wide (uncertainty = bigger moves); post-release continuation entry (safer); gold-only version.

S71 NFP Aftermath Fade / Continuation

Market: EUR/USD, gold, indices • **Timeframe:** M5–M15, 15–90 min post-release • **Style:** Post-event flow • **Evidence:** Grade B (post-announcement drift + overreaction literature)

Win rate	Window	Rule of thumb	Style	Risk
50–60%	15–90 min after	Surprise → trend; non-surprise → fade	Judgment-based	Medium-high

Concept & edge. The tradable NFP edge starts *after* the chaos: (1) genuine surprise → initial direction persists (post-announcement drift, academically documented); (2) in-line print + big spike → retrace (the S47 fade). Classify first, trade second.

Setup. Actual vs consensus + revision; the first 15 minutes' range as the event's verdict.

Entry rules. Surprise case: enter with the initial direction on the first M5 pullback after 15 minutes. Non-surprise spike case: fade toward pre-release price with the same timing.

Exit rules. Drift: target 1.5–2R or the day's extension; trail M15 swings. Fade: target 50–100% of the spike; stop beyond the extreme.

Risk management. Revisions to prior months can reverse the initial verdict mid-trade — read the full release, not just the headline. Risk 0.5–0.75%; one NFP trade per month is plenty.

Backtest evidence. Post-earnings/macro announcement drift is one of the oldest documented anomalies; NFP-specific practitioner systems show 50–60% continuation win rates on true surprises. The fade variant aligns with documented announcement-overreaction retracements.

Works / fails. Works when you classify correctly — which requires reading the number in context (headline + revisions + wages + unemployment). Fails when traded mechanically on headline alone.

Backtesting notes. Build a surprise-z-score database ((actual – consensus)/dispersion) and bucket post-release returns by it — the relationship is your edge map.

Variations. Same framework for CPI (S73) and rate decisions (S74); currency-cross version (trade the cross least moved).

S72 FOMC Drift & Announcement-Day Pattern

Market: SPY/ES, TLT, DXY • **Timeframe:** Daily (pre/post FOMC) • **Style:** Calendar anomaly • **Evidence:** Grade A (Lucca & Moench, *The Pre-FOMC Announcement Drift*, published research)

Edge size	Window	Win rate	Significance	Robustness
~0.3–0.5% per event (historically)	24h before announcement	~60–70% of events	Highly significant academically	Decades, US-centric

Concept & edge. One of the strongest documented calendar anomalies: the S&P 500 earns outsized returns in the 24 hours *before* scheduled FOMC announcements — historically accounting for a large share of total equity returns, with no corresponding rise in volatility. Theories: risk premium resolution, informed positioning.

Setup. FOMC calendar (8 scheduled meetings/year). ES/SPY.

Entry rules. Long ES/SPY ~24 hours before the announcement time (e.g., buy prior day 14:00 ET, or morning of). Exit shortly after the 14:00 ET statement (the drift is pre-announcement; post-statement is a different game).

Exit rules. Time exit at/just after the statement. Some hold through the press conference — historically flatter and wilder.

Risk management. Eight trades a year — variance dominates. Size as a small overlay (0.5–1% risk). Surprise policy shocks (2013 taper tantrum analogs) are the tail risk.

Backtest evidence. Lucca & Moench (NY Fed): large, statistically significant pre-announcement drift, ~0.3–0.5% per event historically, concentrated in the 24h window and unexplained by standard risk factors; robust across decades and absent around other macro releases.

Works / fails. Works as a persistent structural anomaly; small sample per year (8 events) means any decade can disappoint. Post-2022 hiking-cycle behavior was consistent with the pattern.

Backtesting notes. Simple to test: event dates + intraday data. Check whether the edge is in the overnight portion or the announcement-day morning in the current era.

Variations. "Don't fight the Fed" post-meeting trend-following (trade the first post-FOMC day's direction); ECB/BoJ analogs (weaker documentation); options straddle pre-statement (vol selling post-release).

S73 CPI Release Momentum

Market: ES/NQ, gold, DXY pairs • **Timeframe:** M1–M15, 13:30 UTC monthly • **Style:** Event momentum • **Evidence:** Grade B (inflation-release impact literature; practitioner systems)

Win rate	Typical RR	Key variable	Style	Session
50–60% (filtered)	1.5–2:1	Surprise magnitude	Post-release momentum	13:30 UTC+

Concept & edge. CPI has rivaled NFP as the macro event of the current era. The momentum playbook: the 5–15 minute post-release direction on genuine surprises persists into the session — inflation surprises reprice rate expectations, and that repricing takes hours, not seconds.

Setup. CPI calendar; consensus core/headline; 10Y yield as confirmation gauge.

Entry rules. Wait for the first 5 minutes. If the surprise is genuine (core 0.1%+ off consensus) and the 5-min range direction aligns with the yield move, enter on the first pullback in that direction. No alignment = no trade.

Exit rules. Stop at the post-release range midpoint. Targets 1.5–2R; hold into the afternoon on strong repricing days. Flat by close.

Risk management. CPI day whipsaws (the "second move" at 14:00–15:00) stop out early entries — the 5-minute wait is the defense. Risk 0.5–0.75%.

Backtest evidence. Rate-expectation repricing research supports multi-hour post-CPI drift on surprises; practitioner filtered systems: 50–60% win rates at 1.5–2:1. Unfiltered first-minute entries show the same spread/slippage decay as NFP straddles.

Works / fails. Works on genuine surprises in rate-sensitive regimes (2022–2025-style). Fails in low-inflation-salience eras (CPI becomes a non-event) and on in-line prints.

Backtesting notes. Same surprise-database method as S71. Gold and DXY often give cleaner CPI signals than equities — test all three.

Variations. PPI variant; the "CPI week" drift (post-CPI direction persists 1–3 days in strong regimes); yield-confirmation-only filter.

S74 Rate-Decision Straddle-to-Trend (CB Events)

Market: EUR/USD (ECB), USD/JPY (BoJ), GBP/USD (BoE) • **Timeframe:** M5–H1, decision day • **Style:** Event straddle + presser trend • **Evidence:** Grade B (central-bank announcement literature)

Win rate	Two phases	Key insight	Style	Risk
45–60%	Decision spike + presser repricing	The presser is the real event	Two-stage	High

Concept & edge. Rate decisions are two events: the statement (instant spike, often fake) and the press conference 30–45 min later (the real repricing, when guidance is parsed). The documented pattern: initial spikes reverse as often as they hold; presser-driven moves trend. Trade stage two, not stage one.

Setup. Decision calendar; statement text vs expectations; presser start time. M5 chart.

Entry rules. Stage one: stand down (or tiny straddle only with excellent execution). Stage two: as the presser establishes direction (first 10–15 min), enter with the developing M5/M15 trend on pullbacks.

Exit rules. Stage two: trail M15 swings; target 1.5–2R; hold into the session close on strong guidance shifts.

Risk management. BoJ decisions (lunch-time drops, intervention risk around them) are the wildest — halve size. ECB pressers reverse mid-sentence; hard stops always. Risk 0.5%.

Backtest evidence. Central-bank communication research documents that guidance (presser) content moves markets more than the rate decision itself in the modern era — supporting the stage-two focus. Practitioner two-stage systems: 45–60% win rates, with stage-one-only variants materially worse.

Works / fails. Works on guidance-shift meetings. Fails on fully-telegraphed meetings (nothing to reprice — chop) and with impatient stage-one entries.

Backtesting notes. Hard to backtest stage two mechanically (requires presser timestamps and NLP for full rigor); approximate with "direction established 45 min post-statement" and test.

Variations. "Central bank divergence" multi-day version (trade the pair whose banks are moving oppositely); FOMC-minutes day variant (milder same-structure).

S75 Earnings Gap Fade (Stocks)

Market: Liquid US stocks post-earnings • **Timeframe:** Daily open → intraday • **Style:** Event gap mean reversion • **Evidence:** Grade A (4-year, 5-stock systematic backtest 2022–2025; 476–666 trades/stock)

Win rate	Profit factor	Return (4y)	Max DD	vs Buy-hold
36–48%	1.11–1.50	+80% to +113%	9–31%	Beat B&H on all 5 tested names

Concept & edge. Earnings gaps are emotional overreactions that partially retrace during the session. The documented parameters: gap $\geq$ 1%, wait 3 candles after the open, trade the fill direction with 5% TP / 2% SL, forced close near session end. On COIN: +113% vs buy-and-hold –9% over four years.

Setup. Earnings calendar; gap filter $\geq$ 1%; 1-min execution chart; both directions.

Entry rules. After 3 one-minute candles post-open, enter in the fill direction (short gap-ups, long gap-downs). Liquid, gap-prone names only (the study: COIN, CCL, UPST, FCEL, OKTA).

Exit rules. TP 5%, SL 2%, or forced exit near the close. The wide TP/tight SL explains the low win rate — the system wins on asymmetry (2.5:1 RR).

Risk management. Earnings gaps can run 20–30% (UPST) — the stop is survival. Slippage on the open is real; the study itself flags execution speed as make-or-break. Risk 0.5–1%.

Backtest evidence. Jan 2022–Dec 2025: COIN 666 trades, 40.4% WR, PF 1.15, +113%; CCL 476 trades, 47.9% WR, PF 1.24, +100%; OKTA 165 trades, 41.8% WR, PF 1.50, +90.5%, DD only 9.4%; FCEL +79.9%; UPST +98%. All beat buy-and-hold, including on names whose B&H was –60 to –94%.

Works / fails. Works on high-beta, retail-heavy, gap-prone stocks. Fails on low-gap mega caps (no signals) and when the gap is a genuine re-rating (index inclusion, M&A).

Backtesting notes. The published parameters are illustrative, not optimal — re-optimize TP/SL per name. Include delisted names to dodge survivorship bias; model 200–500ms execution delay.

Variations. Fade only gaps $>$ 3% (stronger overreaction); VWAP-confirmation entry; PEAD (post-earnings-announcement drift) is the *opposite* documented anomaly — test both directions per name and keep what's true.

S76 Earnings Gap-and-Go (Momentum Side)

Market: Large/mid-cap stocks • **Timeframe:** M5–M15, open + 2 hours • **Style:** Event momentum • **Evidence:** Grade B (PEAD literature — post-earnings drift is academically robust; intraday version practitioner-documented)

Win rate	Typical RR	Filter	Horizon	Style
45–55%	1.5–2.5:1	Surprise quality + guidance	Intraday → multi-day	Momentum

Concept & edge. Post-earnings-announcement drift (PEAD) is one of academia's most replicated anomalies: stocks with genuine earnings surprises keep drifting in the surprise direction for days. The intraday version trades the first hours of that drift in the highest-quality gap-ups/downs.

Setup. Earnings reaction quality: surprise size, guidance, conference-call tone, RVOL, and — critically — the stock holding its gap through the first 30–60 minutes.

Entry rules. Long: gap up > 4% on beat-and-raise, first hour holds above VWAP, enter on the first M5/M15 consolidation break upward. Short: mirror on miss-and-cut.

Exit rules. Stop below the opening consolidation (or VWAP). Targets: prior all-time/swing levels, or trail M15 swings. The PEAD literature supports holding 1–5 days for the full drift.

Risk management. The gap that can't hold VWAP is not a gap-and-go — it's S75's fade. Let the first hour classify the event. Risk 0.75–1%.

Backtest evidence. PEAD is documented across decades and markets (surprise-sorted drift, significant for weeks post-event). Intraday continuation systems on beat-and-raise large-caps: 45–55% win rates at 1.5–2.5:1 in practitioner tests.

Works / fails. Works in earnings season on liquid names with clean stories. Fails in macro-panic tapes (stock-specific news drowns) and on crowded high-short-interest names (squeeze mechanics distort signals).

Backtesting notes. You need historical surprise/guidance data to filter honestly; price-only backtests of "all big gaps" mix S75 and S76 populations and teach you nothing.

Variations. Multi-day PEAD swing version (hold 3–10 days); sector-sympathy plays (trade the laggard peer); options-defined-risk version.

S77 Weekend Gap Monday Fade (Forex & Crypto)

Market: Forex majors (Sunday open), BTC/ETH • **Timeframe:** Sunday/Monday session • **Style:** Gap mean reversion • **Evidence:** Grade B (weekend-gap statistics; practitioner backtests)

Win rate	Typical RR	Frequency	Best conditions	Style
55–70% (small gaps)	0.8–1.2:1	~weekly	Quiet weekends, small gaps	Gap fill

Concept & edge. Forex closes Friday and reopens Sunday; weekend headlines gap the open. Like equity gaps (S06/S07), *small* forex weekend gaps fill at high rates while large gaps (real news) trend. Crypto never closes, but its Monday-morning "CME gap" (BTC futures close Friday) creates a famous analogous magnet.

Setup. Friday close vs Sunday open; gap-size classification ($< 0.3\%$ small, $> 0.8\%$ large for majors). For CME-gap: Friday CME close vs Sunday futures reopen level.

Entry rules. Small gap + no weekend-breaking news: enter toward the Friday close at the Sunday open (or after the first 15–30 minutes for confirmation). Large gaps: stand down or trade *with* them.

Exit rules. Target: gap fill (Friday close). Stop: $1\times$ gap size beyond the open. Time stop: by Monday NY lunch.

Risk management. Weekend geopolitical shocks (wars, exchange hacks for crypto) gap *and keep going* — the news filter is not optional. Risk 0.5–0.75%.

Backtest evidence. Practitioner gap statistics: small forex weekend gaps fill ~55–70% of the time; CME-gap folklore has partial statistical support (gaps "fill" frequently simply because BTC revisits most levels eventually — be precise about causality in your own test).

Works / fails. Works on quiet weekends with positioning-driven gaps. Fails on genuine weekend news and in crisis regimes (Sunday opens gap repeatedly in one direction).

Backtesting notes. Bucket results by gap size and by weekend-news flag; the size threshold where fill rate collapses is your trading line. For CME gaps, define "fill" rigorously (touch within 48h, not eventually).

Variations. Trade the fill in two stages (half at open, half at midpoint); CME-gap + funding combo; Monday-Asia momentum instead (S52) when the gap is large.

PART VIII

Seasonality & Calendar Strategies

Calendar effects are the oldest documented anomalies — and the most honest ones, because the rules are public and decades of data are free. Several (turn-of-month, Halloween) are among the statistically strongest edges in this book. They trade infrequently; treat them as overlays, not meal tickets.

S78	Turn of the Month Effect
S79	Halloween Indicator (Sell in May)
S80	Day-of-Week Effects (Monday/Friday)
S81	First Trading Day of the Month
S82	Power Hour Momentum (Final Hour)
S83	Overnight vs Intraday Split Trading
S84	Gold Session Rotation (Asia Accumulate / London-NY Distribute)
S85	BTC Worst-Hours Short Window (03:00–04:00 UTC)

S78 Turn of the Month Effect

Market: S&P 500 (global; 19 of 20 markets documented) • **Timeframe:** ~4 days/month • **Style:** Calendar anomaly • **Evidence:** Grade A (multiple academic studies; S&P futures 1980–2024 replication)

Annual capture	Exposure	Max DD	Global robustness	Significance
≈ B&H minus 0.89%	~4 days/month (~16%)	~20%	19/20 countries	"Only calendar effect statistically & economically significant" (study conclusion)

Concept & edge. Own the index only from the last trading day of the month through the first ~3–4 days of the new month. Fund flows (401k contributions, month-end rebalancing, salary cycles) concentrate buying at the month boundary. Historically, this ~16% exposure captured nearly the entire market return.

Setup. Trading calendar. Entry: close of the day before month-end (some variants: last trading day close).

Entry rules. Buy at the close before month-end.

Exit rules. Sell at the close of the 3rd trading day of the new month (variants: 4 days).

Risk management. Exposure is tiny; risk per window is a bad macro week at month-end. Size as an overlay; combine with other systems using the freed capital.

Backtest evidence. 1980–2024 S&P replication: the 4-day/month equity curve nearly matched all-other-days' total return, with max DD ~20%. The cited academic study found turn-of-month the *only* persistent, economically significant calendar effect in S&P futures; a 2014 study confirmed it in 19 of 20 countries.

Works / fails. Works on equity indices globally. Fails (mildly) in regimes where month-end flows turn negative (forced liquidations cluster at month-end too — 2008).

Backtesting notes. One of the easiest tests in this book (daily data, 10 lines). Test window variants (–1 to +3, –2 to +2, last day + first 4) and index variants (SPY, QQQ, IWM, international ETFs).

Variations. "Turn of month + RSI(2) oversold" (quality overlay); QQQ-only version; deploy freed capital into S53 funding arb for a combined return stream.

S79 Halloween Indicator (Sell in May)

Market: S&P 500, Russell 2000 • **Timeframe:** 6-month rotation • **Style:** Seasonal allocation • **Evidence:** Grade A (Dzhaharov & Ziemba, LSE Systemic Risk Centre; Fidelity/CFRA 1945–2026 data)

S&P 500 result	Russell 2000 result	Nov–Apr avg	May–Oct avg	Period
SIM \$9.28 vs B&H \$6.71 (+38%)	SIM \$12.61 vs B&H \$6.67 (+84–89%)	~7%	~2% (negative since 1990 segment)	1993–2019 study; 1945–2026 stats

Concept & edge. November–April historically delivers ~7% average S&P gains vs ~2% for May–October. Going to cash (or defensive sectors) May–October beat buy-and-hold by ~38% (S&P) and ~84–89% (Russell 2000) in the 1993–2019 futures study — with *lower* volatility.

Setup. Calendar: exit on the first trading day of May; re-enter on the 6th trading day before the end of October.

Entry/exit rules. Long Nov 1st-ish → May 1st; flat (or defensive rotation) May → late October.

Risk management. This is allocation, not trading: the risk is opportunity cost in strong summers (2017–2019 showed SIM $\approx$ B&H, statistically indistinguishable in hot streaks). Election-year summers historically perform better.

Backtest evidence. Ziemba study (futures, 1993–2019, one-tail $p \approx 0.05$ S&P, $p = 0.03$ Russell): SIM beat B&H by $0.5\times$ – $2\times$ final wealth with lower risk. Fidelity/CFRA (1945–2026): Nov–Apr ~7% vs May–Oct ~2%; cyclicals outperform in the strong half, defensives in the weak half.

Works / fails. Works as a long-run tilt. Fails in strong-summer years and the effect weakens if over-crowded; best treated as a defensive rotation signal rather than all-cash.

Backtesting notes. Test sector-rotation variant (cyclical ↔ defensive swap) vs full cash. Note the study's election-year interaction.

Variations. Sell-in-May for crypto (untested rigorously — your research project); defensive-half = TLT/gold instead of cash; "Stay in May when above 200-SMA" hybrid.

S80 Day-of-Week Effects (Monday/Friday)

Market: S&P 500, forex, BTC • **Timeframe:** Daily • **Style:** Calendar tilt • **Evidence:** Grade A/B (classical calendar literature; QuantifiedStrategies monthly/day tables)

Pattern	Strength	Best use	Status	Data
Monday weakest / Friday strong (classical); Wednesday–Friday best gap-fade days (documented)	Modest, regime-shifting	Overlay filter	Partially arbed, still useful	1960+ daily data

Concept & edge. Classical day-of-week effects (Monday weakness, Friday strength) have faded as standalone trades but remain useful as *conditional* filters: the documented gap-fade statistics (S06) show Wednesday–Friday systematically outperforming Monday–Tuesday, and the Turnaround Tuesday edge (S08) is the refined descendant of Monday weakness.

Setup. Day-of-week tagging on any daily dataset.

Entry rules. As overlay: bias mean-reversion longs to Monday–Wednesday weakness windows; bias momentum longs to Thursday–Friday; avoid initiating short mean-reversion trades on strong calendar days.

Exit rules. N/A (overlay).

Risk management. Day-of-week edges are second-order — never size up on calendar alone.

Backtest evidence. Documented S&P monthly/day tables (1960+): April/December strongest months (72–73% up), September worst (45% up, PF 0.66); gap-fade day-of-week table: Wednesday +22.7% cumulative vs Monday +12.1% over the study. Friday gap-fades: steadiest equity curve of the week.

Works / fails. Works as a Bayesian prior layered on other strategies. Fails as a standalone system (edges are small and unstable across regimes).

Backtesting notes. Always check day-of-week interaction when building any daily strategy — if your system's P&L concentrates in one weekday, you need to know why.

Variations. Monthly-effect overlay (avoid fresh longs in September; press in November/December); crypto day-of-week (S51's weekend effect is the crypto version).

S81 First Trading Day of the Month

Market: S&P 500 • **Timeframe:** Daily • **Style:** Calendar anomaly • **Evidence:** Grade B (calendar literature; related to turn-of-month flows)

Pattern	Mechanism	Strength	Use	Data
First trading day historically strong	Monthly inflows hit on day 1	Moderate	Overlay/day trade bias	Decades of daily data

Concept & edge. New-month inflows (retirement contributions, fund deployments) hit on the first trading day, giving it a historically positive skew — the flip side of the turn-of-month window (S78). Day traders use it as a directional prior for the first session of the month.

Setup. Calendar; SPY/ES intraday chart on the first trading day.

Entry rules. Intraday: bias longs (ORB longs, VWAP-pullback longs) on the first trading day, especially after a weak month-end. Standalone: buy the open, sell the close (weak but positive historically).

Exit rules. Session close.

Risk management. First-day-of-month also carries ISM/PMI releases at 15:00 UTC — event risk embedded in the calendar edge. Normal sizing.

Backtest evidence. Calendar research documents the first trading day's positive skew (a component of the broader turn-of-month window's strength); day-trader replication shows the effect clearest in the morning session.

Works / fails. Works in normal flow regimes. Fails when month-start macro data shocks (the ISM window) or in distribution regimes.

Backtesting notes. Separate the first trading day's return from the full turn-of-month window to see what it contributes in the current era.

Variations. First-day + month-end-weakness combo (stronger prior); first Friday (NFP) excluded version.

S82 Power Hour Momentum (Final Hour)

Market: SPY, QQQ, ES • **Timeframe:** 15:00–16:00 ET • **Style:** Intraday session momentum • **Evidence:** Grade B/A (intraday momentum literature — S27's academic base covers this window)

Win rate	Typical RR	Best days	Mechanism	Style
55–62% (trend days)	0.8–1.2:1	Strong trend days	MOC flow + short covering	Final-hour continuation

Concept & edge. The last hour concentrates closing flows: MOC orders, ETF rebalancing, day-trader flattening, short covering. On established trend days, the final hour continues the day's direction with documented predictability (the first-half-hour → last-half-hour academic link).

Setup. By 14:30 ET: classify the day (trend vs range — VWAP slope, % of bars above VWAP). M5 chart for execution.

Entry rules. Trend-day long: enter 15:00–15:30 ET on an M5 pullback hold above VWAP, in the day's direction. Target: close. Range days: stand down or play the return-to-VWAP version.

Exit rules. MOC exit (the point is the closing flow). Stop: M5 structure against the trade.

Risk management. The last hour also carries reversal risk on failed trend days (the 15:50 "reversal window") — if the day's leaders break VWAP at 15:00, flip bias fast. Risk 0.5%.

Backtest evidence. Academic intraday-momentum results (first half-hour predicts last, R^2 rising with volatility) cover the final window; practitioner power-hour systems: 55–62% continuation on classified trend days.

Works / fails. Works on high-conviction trend days, expiry Fridays (amplified flows). Fails on choppy days and when late news hits (Fed minutes at 14:00 ET days).

Backtesting notes. Condition everything on your day-type classifier; unconditioned last-hour returns are ~flat historically.

Variations. Closing-imbalance trading (NYSE publishes MOC imbalances ~15:50 — advanced); 15:50 reversal-scalp variant; options 0DTE version (defined risk).

S83 Overnight vs Intraday Split Trading

Market: SPY, QQQ, IWM • **Timeframe:** Daily decomposition • **Style:** Structural anomaly harvesting • **Evidence:** Grade A (academic overnight-return literature; multiple replications)

Finding	Implication	Long-term result	Risk character	Style
~All long-term S&P gains accrue overnight; intraday ≈ flat	Hold overnight, avoid intraday	Overnight-only ≈ B&H with ~1/3 exposure	Gap risk concentrated	Decomposition strategy

Concept & edge. Decompose returns: buy-and-hold = overnight returns + intraday returns. Research across decades shows the overnight component delivers essentially all of the index's long-run gain while the intraday component nets near zero. Owning only the overnight session (S09) is the trading expression; this page is the portfolio-level version.

Setup. Daily open/close data; decompose: overnight = $\text{open}_t / \text{close}_{t-1}$, intraday = $\text{close}_t / \text{open}_t$.

Entry rules. Portfolio version: hold index exposure overnight only (buy close, sell open) for the premium-harvesting variant. Trading version: prefer entering mean-reversion longs at the close (capturing overnight) rather than midday.

Exit rules. Sell at the open (MOO).

Risk management. Concentrated gap risk: single overnight events (−5% opens) land entirely on you. Size accordingly; avoid event nights (CPI/FOMC/earnings for single names).

Backtest evidence. Multiple academic replications (US and international): cumulative overnight returns dominate cumulative intraday returns over long samples, in some periods exceeding total returns (intraday negative). QuantifiedStrategies' published overnight edges (S09 stats: 60–77% WR, PF 1.5–4.0) are the filtered trading versions.

Works / fails. Works on equity indices (US strongest). The effect is weak/reversed in many other asset classes — test before assuming.

Backtesting notes. Produce your own cumulative overnight-vs-intraday chart for your instrument — it's the single most instructive chart in index trading research.

Variations. Weekend-overnight (Friday close → Monday open) as a separate bucket; single-stock overnight premium around earnings (avoid — different beast); combined with S78 turn-of-month windows.

S84 Gold Session Rotation (Asia Accumulate / London-NY Distribute)

Market: XAU/USD • **Timeframe:** Session-based daily • **Style:** Session seasonality • **Evidence:** Grade B (gold market microstructure documentation; practitioner session statistics)

Pattern	Asian session	London/NY session	Use	Style
Session-character split	Quieter, range-bound, accumulation character	Directional, volatile, distribution/breakout character	Session-appropriate tactics	Dual playbook

Concept & edge. Gold's behavior splits by session more cleanly than most instruments: Asian hours (dominated by physical/jewelry demand flows) tend toward quiet accumulation ranges, while London/NY (COMEX dominance) drives the day's directional moves. Two different games — play each with its own rules.

Setup. Session clock (Asia 00:00–07:00 UTC, London 07:00–12:00, NY 13:00–21:00 UTC). Asian range marked for London reference.

Entry rules. Asia: range-fade tactics (S20 logic) at Asian-session extremes with RSI stretch. London/NY: breakout/continuation tactics (S19/S22 logic) using the Asian range and session VWAP as references.

Exit rules. Asia fades: midline, hard flat before London. London/NY breaks: 1.5R or session-close exits.

Risk management. Gold respects levels but wicks violently around US data (13:30 UTC) — flat or bracket-protected through releases. Risk 0.5–0.75% per tactic.

Backtest evidence. Practitioner session statistics consistently show gold's true range concentrating in London/NY hours, with Asian ranges forming meaningful reference levels for the later sessions (the same structural logic as S19's forex version, with COMEX open as the key catalyst window).

Works / fails. Works in normal macro regimes. Fails during Asian-hours shocks (PBoC gold policy, Middle East headlines at Asian open) and rate-decision days (all sessions trend).

Backtesting notes. Compute gold's range and directional bias by session over 2+ years; confirm the split exists in your data before deploying the dual playbook.

Variations. COMEX-open momentum (13:30 UTC) as a standalone; Asian-range breakout for London only when range < 0.5× daily ATR; gold/Silver relative session strength (S100 link).

S85 BTC Worst-Hours Short Window (03:00–04:00 UTC)

Market: BTC • **Timeframe:** Hourly • **Style:** Intraday seasonality (short side) • **Evidence:** Grade A (same Padyšák & Vojtko dataset as S50)

Pattern	Hours	Counterpart	Use	Caution
Worst average hourly returns	03:00–04:00 UTC	Long 21:00–23:00 (S50)	Short/hedge window	Weak in bulls, decay risk

Concept & edge. The mirror of S50: the same peer-reviewed hourly analysis found BTC's *worst* (most negative/insignificant) returns concentrated in the 03:00–04:00 UTC window — late US evening, pre-Asia, the day's liquidity trough. A short-bias window completes the daily map.

Setup. Hourly UTC data; the twin windows: short-bias 03:00–04:00, long-bias 21:00–23:00.

Entry rules. Short BTC at 03:00 UTC, cover at 04:00–05:00 UTC. (Or use it defensively: never initiate spot longs during this window.)

Exit rules. Time exit. Strict.

Risk management. Shorting BTC in a structural bull market is fighting the tape: the edge is smaller than S50's and regime-dependent. Small size; better used as an execution-timing rule (avoid buying at 03:00–04:00) than an aggressive short.

Backtest evidence. Padyšák & Vojtko: 03:00 and 04:00 UTC were the worst-performing hours of the day in the 2015–2022 Gemini sample (no statistically *negative* hour, but these were closest); the paired long-window strategy delivered ~33% annualized — the short side is the weaker twin.

Works / fails. Works as a timing filter and in bear/neutral regimes as a short. Fails in bull runs (every hour is up) and as a primary system.

Backtesting notes. Same code as S50 with sign flipped — verify post-2022 persistence and exchange robustness before allocating real capital.

Variations. Paired daily rotation (long 21:00–23:00, short 03:00–04:00, flat otherwise); execution-timing-only usage (schedule your spot buys/sells around the windows).

PART IX

Market-Neutral & Statistical Arbitrage

The most professional strategies in this book: edges that don't depend on market direction at all. They demand more infrastructure (two legs, hedge ratios, more data), but they're the closest thing to all-weather strategies — and the academic evidence base is strong.

S86	Pairs Trading (Cointegration)
S87	Partial Cointegration Pairs (PCI)
S88	Sector ETF Pairs (Documented Winners & Losers)
S89	Perp–Spot Basis Trade (Term Structure)
S90	Lead-Lag Correlation Trade (Correlated Assets)
S91	Index Futures–ETF Basis Scalp
S92	Triangular Arbitrage (Forex) — Reality Check
S93	Liquidation-Cascade Reversal (Crypto Perps)

S86 Pairs Trading (Cointegration)

Market: Stock pairs, ETF pairs • **Timeframe:** Daily (H1 possible) • **Style:** Statistical arbitrage • **Evidence:** Grade A (Gatev et al. lineage; Quantpedia-curated studies; Springer 2025 ETF study)

Sharpe (documented)	Annual excess return	Beta	Key risk	Win profile
0.8–1.4 (best studies)	Up to 16.4%/yr (Brazil study)	≈ 0 (market neutral)	Pair breakdown	High WR, small gains

Concept & edge. Two cointegrated assets share a long-run equilibrium; when the spread deviates $> 2\sigma$, short the outperformer / buy the underperformer and wait for reversion. Returns are uncorrelated with the market by construction — a true alternative return stream.

Setup. Screen pairs by cointegration (Engle-Granger/Johansen) on a *formation period*; estimate hedge ratio via OLS; compute spread z-score on a rolling 60-day window.

Entry rules. $z > +2$: short y, long hedge-ratio $\times$ x. $z < -2$: opposite. Only trade pairs validated out-of-sample.

Exit rules. Exit at z crossing ± 0.5 . Stop-loss at $|z| > 4$ (relationship breaking). Time stop 20–30 days.

Risk management. Documented killers: in-sample overfitting (formation/trading split is mandatory), transaction costs eating the small per-trade edge, borrow costs on the short leg, and permanent pair breakdown on corporate events (exit, don't average, on fundamental news).

Backtest evidence. Brazil 2005–2012 cointegration portfolio: 16.38%/yr excess, Sharpe 1.34, low market correlation. ETF study (2025): dynamic monthly selection + VIX regime overlay $\rightarrow$ Sharpe 0.81, beta -0.14 , alpha 12.8%, vs static pairs mostly flat-to-negative — *selection and regime handling are the edge*, not the z-score trigger. UK study: returns largely explained after risk/liquidity controls — honest tempering.

Works / fails. Works on sector peers (HYG/JNK, BND/AGG documented stable), same-industry stocks. Fails on lookalike-but-unrelated pairs (QQQ/XLK documented: -10.3% , 21.8-yr DD recovery!) and in structural regime breaks.

Backtesting notes. Formation/trading split, rolling re-estimation, costs, borrow, delisting handling. The 860,000-pair persistence study found cointegration is *not* persistent year-to-year — re-select pairs periodically.

Variations. Partial cointegration (S87 — stronger); distance-method (Gatev original, simpler); intraday H1 pairs on sector ETFs.

S87 Partial Cointegration Pairs (PCI)

Market: Stock pairs (Italian market study; generalizable) • **Timeframe:** Daily • **Style:** Advanced stat-arb • **Evidence:** Grade A (LUISS thesis replicating Clegg-Krauss PCI; out-of-sample tested)

Sharpe	Cumulative return	vs Classical cointegration	Trades	Max DD
1.38 (portfolio OOS)	53.15% (≈3 yrs)	Classical: ~0 Sharpe in same tests	~10/yr	-8.2%

Concept & edge. Classical cointegration assumes the spread is purely mean-reverting — false when a permanent random-walk component hides inside. PCI (Clegg-Krauss) uses a Kalman filter to separate the mean-reverting part from the random-walk part and trades only the former. Result in the published replication: Sharpe 1.38 vs ~0 for the classical approach on identical data.

Setup. Two-stage: estimate PCI parameters on training data; Kalman-filter the residual to isolate the mean-reverting component; z-score it.

Entry rules. Long the spread when the mean-reverting component $< -1\sigma_M$; short when $> +1\sigma_M$. Pair selection filters: AR coefficient 0.5–0.8, mean-reversion variance share $> 50\%$, hedge ratio 0.75–1.25.

Exit rules. Close at $\mp 0.5\sigma_M$ crossing.

Risk management. The thesis itself warns: two-pair concentration was a weakness — run 5+ qualified pairs. Event-risk exit rule identical to S86.

Backtest evidence. Out-of-sample: SRG-TRN +23.6%, HER-TRN +85.7%; portfolio 53.15% cumulative vs benchmark 25.15%; Sharpe 1.38 vs 0.51; max DD -8.2%; Sortino 2.14. Simulated robustness: PCI Sharpe distribution centered ~1, classical ~0.

Works / fails. Works wherever classical pairs *almost* work (spreads with hidden permanent components — most real pairs). Fails with the same event-risk and selection fragility, plus model complexity risk.

Backtesting notes. This is a genuine quant build (Kalman filter, PAR model selection via AIC). The payoff: it's one of the few published upgrades that *dominates* its classical parent out-of-sample.

Variations. Sector-constrained pair universe; ETF-pair PCI; crypto-pair PCI (BTC/ETH-adjacent alts — your research frontier).

S88 Sector ETF Pairs (Documented Winners & Losers)

Market: US sector/bond ETFs • **Timeframe:** Daily • **Style:** ETF stat-arb • **Evidence:** Grade A (Springer 2025 study — unusually honest: publishes losers too)

Best pairs	Worst pairs	Portfolio Sharpe	Overlay	Lesson
HYG/JNK (+1.6%, Sharpe 0.52, DD −0.08%), BND/AGG (Sharpe 0.19)	QQQ/XLK (−10.3%), XME/PICK (−11.4%), SLV/GLD (−0.1%)	0.81 (dynamic + VIX overlay)	Volatility-regime switch	Stable pairs are boring pairs

Concept & edge. The tradable pairs aren't the famous ones. High-yield-bond twins (HYG/JNK) and aggregate-bond twins (BND/AGG) are nearly identical instruments — tiny, reliable divergences. Famous "logical" pairs (gold/silver miners, QQQ/XLK) documented as portfolio poison. Add the study's killer feature: a VIX z-score > 2 overlay that *switches to momentum* during vol spikes, converting crash drawdowns into profit.

Setup. Monthly dynamic pair selection from the ETF universe; z-score 2 entry / 0.75 exit; VIX 5-day SMA z-score as the regime gate.

Entry rules. Equal-share (or beta-adjusted) spread entries at $|z| > 2$ on selected pairs; max 5 concurrent pairs.

Exit rules. $|z| < 0.75$ exit. Regime override: when VIX $z > 2$, flip from fade-the-spread to trade-the-divergence.

Risk management. Published DDs are tiny on the good pairs (−0.08%!) — the risk concentrates in selection turnover and the regime flip. Equal-weight across pairs; monthly re-selection.

Backtest evidence. Dynamic portfolio: Sharpe 0.81, alpha 12.8% (significant), beta −0.14, correlation −0.18 to SPY; the VIX overlay worked in 2015/2018/2020/2022 stress windows. Static famous pairs: mostly negative or flat — read the losers table before choosing yours.

Works / fails. Works on near-duplicate instruments (same index, different issuer) and genuinely-linked sectors. Fails on thematically-related-but-different pairs — the study's central warning.

Backtesting notes. Replicate the loser analysis first: if you understand *why* QQQ/XLK fails (drifting mega-cap weights), you'll never force a bad pair again.

Variations. Leveraged-ETF twins; cross-listing pairs (same company, two exchanges); the VIX-overlay concept ported to S86/S87.

S89 Perp–Spot Basis Trade (Term Structure)

Market: BTC/ETH dated futures vs spot, perps • **Timeframe:** Days–weeks holds • **Style:** Basis arbitrage • **Evidence:** Grade B/A (standard futures basis literature applied to crypto; exchange-published basis data)

Typical annualized	Risk	Capacity	Best regime	Structure
5–20% (calm) to 40%+ (euphoria)	Low (execution/counterparty)	Large (institutional)	Contango expansions	Cash & carry

Concept & edge. Dated futures trade at a premium (contango) in bullish regimes. Buy spot, short the future, lock the basis as yield at convergence. Unlike funding arb (S53), the yield is *fixed at entry* — the most bond-like trade in crypto.

Setup. Spot + futures accounts; basis monitor (annualized premium = $(F-S)/S \times 365/\text{days}$).

Entry rules. When annualized basis exceeds your threshold (e.g., > 10% after costs): buy spot, short equal-notional future. Hold to expiry (or roll).

Exit rules. Convergence at expiry closes naturally. Early exit if basis compresses early (take the yield early and recycle).

Risk management. Margin on the short leg through adverse mark-to-market (the basis can widen before converging — 2021 saw 40%+ premiums); exchange counterparty risk; liquidation discipline identical to S53.

Backtest evidence. Historical annualized bases: 5–15% in calm regimes, 20–40%+ in 2021/2024-style euphoria; basis harvest is a documented institutional strategy (it's why CME open interest explodes in bull markets).

Works / fails. Works in contango regimes (most bull markets). Fails in backwardation (the trade inverts: short spot is hard/expensive) and on counterparty failure.

Backtesting notes. Basis history is publicly available; model margin interest, fees, and the funding differential if substituting perps for dated futures.

Variations. Calendar spread (long near future, short far future); cross-venue basis; tri-trade: basis + funding + staking yield stacking on ETH.

S90 Lead-Lag Correlation Trade (Correlated Assets)

Market: BTC ↔ COIN/MSTR, NQ ↔ mega-caps, gold ↔ miners • **Timeframe:** M1–M15 • **Style:** Cross-asset microstructure • **Evidence:** Grade B (lead-lag microstructure literature; practitioner systems)

Win rate	Horizon	Signal	Requirement	Style
55–65% (documented practitioner)	Seconds–minutes	Leader moves, laggard hasn't	Fast data on both	Latency-sensitive

Concept & edge. Strongly-coupled assets don't move simultaneously: BTC moves first, crypto-equities (COIN, MSTR) follow seconds later; index futures lead their heaviest components; gold leads miners. Trading the laggard after the leader's move is one of the oldest prop-desk edges.

Setup. Two live feeds (leader + laggard); rolling correlation/beta; the leader's short-window return as the signal.

Entry rules. Leader moves $\geq$ threshold (e.g., BTC +0.3% in 1 min) while laggard hasn't moved proportionally: trade the laggard in the leader's direction. Exit when the laggard "catches up" (beta-adjusted).

Exit rules. Target: the beta-implied catch-up move. Stop: leader reverses through its trigger level. Time stop: 5–15 minutes.

Risk management. Decay is brutal: everyone sees the same lag, and the window shrinks yearly (HFT arbs it). Your feed speed relative to the marginal arbiter decides viability. Risk 0.25–0.5%.

Backtest evidence. Lead-lag effects are well documented in market-microstructure literature (futures lead stocks; ADRs lead/lag home markets; ETF leads components at short horizons). Practitioner BTC → COIN systems: 55–65% short-horizon win rates with tight targets.

Works / fails. Works during high-volatility leader moves (news, liquidations). Fails in quiet tape (lag compresses below costs) and for slow feeds.

Backtesting notes. Measure the lag structure first (cross-correlation at multiple horizons) — if the lag is $<$ your round-trip latency, you don't have a trade, you have a donation.

Variations. ETF-vs-components basket version; ADR/home-market overnight version (no speed needed — time zones create the lag); sector-ETF leader vs laggard stock.

S91 Index Futures–ETF Basis Scalp

Market: ES vs SPY, NQ vs QQQ • **Timeframe:** Seconds–minutes • **Style:** Basis microstructure • **Evidence:** Grade C (institutional practice; retail-viable only with premium tools)

Edge size	Hold	Requirement	Retail viability	Style
Basis points per trade	Seconds	Fast dual feed + low fees	Low-marginal	Pure arbitrage

Concept & edge. Futures and their ETF shadow track the same index; transient basis dislocations (beyond fair-value bands) revert within seconds-minutes. Institutions arb this continuously; retail can only nick the wider, slower dislocations (open/close, volatility bursts).

Setup. Compute fair value continuously (futures price adjusted for carry/rates/dividends); plot the basis spread vs its normal band.

Entry rules. When the basis exceeds the normal band by a threshold covering your full round-trip costs: long the cheap leg, short the rich leg, equal beta.

Exit rules. Exit both at band re-entry (basis normalizes). Time stop: minutes. Never hold inventory.

Risk management. Legging risk (one leg fills, other moves) — use spread orders where possible. This strategy's real lesson for retail: understanding the basis tells you *which instrument is expensive* when you trade directionally. Risk small; edge per trade is bp-scale.

Backtest evidence. Institutional literature documents basis mean reversion within fair-value bands; retail-scale published results are essentially absent — treat as Grade C and primarily educational.

Works / fails. Works at opens/closes and vol bursts with fast infrastructure. Fails as a retail day-in-day-out system.

Backtesting notes. Requires synchronized tick data for both legs — most retail data isn't timestamp-aligned. If you can't verify alignment, results are artifacts.

Variations. Triple-witching expiry-basis plays; cash-index arb via options replication; educational use only: basis as an execution-cost signal for S14/S33 systems.

S92 Triangular Arbitrage (Forex) — Reality Check

Market: Forex crosses (EUR/USD × USD/JPY vs EUR/JPY) • **Timeframe:** Milliseconds • **Style:** Pure arbitrage • **Evidence:** Grade C — included as an honest negative

Edge today	Era	Retail viability	Why included	Lesson
~None for retail	Pre-2010s: real	~Zero	Most-requested "risk-free" strategy	Latency IS the strategy

Concept & edge. If $\text{EUR/USD} \times \text{USD/JPY} \neq \text{EUR/JPY}$ (beyond costs), a three-leg loop locks riskless profit. It existed; banks and HFTs killed it with co-location. Documented reality: deviations now last milliseconds and are smaller than retail spreads.

Setup (academic). Compute the synthetic cross continuously; compare to the quoted cross; the loop executes all three legs simultaneously when deviation > costs.

Entry rules. N/A for retail — included so you stop looking for it in the wrong place.

Exit rules. N/A.

Risk management. The risk isn't market risk, it's fantasy risk: weeks building a system that was dead before you started. If you want the *transferable* version, see S90 (lead-lag) and S54 (cross-exchange funding) — same instinct, live edges.

Backtest evidence. Market-efficiency literature: triangular parity holds within transaction costs essentially continuously in modern electronic forex; historical studies show the opportunity window collapsed as electronic trading spread.

Works / fails. Works only with bank-grade co-location and prime-broker spreads. Fails everywhere else.

Backtesting notes. If you must verify: you need synchronized tick quotes for all three legs with your *actual* spreads — you'll find the deviation smaller than your costs, which is the answer.

Variations. Crypto-venue triangular loops (slightly more alive due to fragmented venues — still latency-gated); DEX triangular/cyclic arbitrage (real but MEV-bot-dominated).

S93 Liquidation-Cascade Reversal (Crypto Perps)

Market: BTC/ETH perpetuals • **Timeframe:** M1–M15 • **Style:** Forced-flow fade • **Evidence:** Grade B (liquidation data publicly documented; practitioner systems)

Win rate	Typical RR	Signature	Data	Style
55–65% (documented ranges)	1.5–2:1	Cascade + funding extreme + OI flush	Liquidation feed, OI, funding	Capitulation fade

Concept & edge. Perp liquidations are *forced* market orders — pure price-insensitive flow. Cascades (clusters > \$100M+ in an hour on majors) mark local extremes with documented regularity: when forced sellers are exhausted, price reverts. Unlike pattern-based reversals, this one has a mechanical driver you can watch in real time.

Setup. Liquidation feed (Coinglass-style), open interest, funding rate. Context: price down 3–5%+ fast, long liquidations spiking, OI dropping sharply (leverage flushed).

Entry rules. Long after a long-liquidation cascade when: liquidation rate peaks and declines, price prints a stall/reversal M5 candle, and OI has dropped (leverage gone). Enter on the reversal confirmation. Short: mirror on short-cascades (rarer, faster).

Exit rules. Target: retrace to the cascade's origin (or 50%). Stop below the cascade low. Time stop: a few hours.

Risk management. Cascades chain (one triggers another at the next liquidation shelf) — never catch the *first* flush candle; wait for exhaustion evidence. Funding at extreme positive readings adds confirmation for long-fades. Risk 0.5–1%.

Backtest evidence. Liquidation-cascade dynamics are documented in crypto-microstructure research (forced-flow impact, OI-reset behavior); practitioner fade systems on majors report 55–65% win rates at 1.5–2:1 with the exhaustion filter.

Works / fails. Works on majors with deep liquidation data. Fails in deleveraging *regimes* (multiple cascade waves over days — fade the first wave, respect the second) and on thin alts (data lies, wicks lie more).

Backtesting notes. Historical liquidation data is available (exchange APIs, aggregators); align cascade timestamps with price and measure forward returns — a genuinely novel dataset most retail backtests ignore.

Variations. Cascade + S49 sweep confluence; funding-extreme-only entries; OI-reset momentum continuation (trade *with* the post-flush trend instead).

PART X

Gold, Oil & Commodity Intraday

Commodities trend better than equities and respect technical levels more violently. Gold and crude are the most liquid non-financial intraday vehicles on earth — with their own event calendars (EIA inventories), session personalities, and macro drivers. The published evidence here includes one of the cleanest open-source 5-minute systems available.

S94	Gold Pullback-Window (5-Minute System)
S95	Gold London/NY Breakout
S96	Safe-Haven Spike Fade (Gold/Silver)
S97	Crude Oil EIA Inventory Trade
S98	Crude Opening Range Momentum (CL)
S99	Commodity Trend-Day Continuation (Second-Day Momentum)
S100	Gold–Silver Ratio Mean Reversion

S94 Gold Pullback-Window (5-Minute System)

Market: XAU/USD • **Timeframe:** M5 • **Style:** Volatility-expansion pullback • **Evidence:** Grade A (open-source Backtrader project, 5-year verified backtest 2020–2025)

Total return	Win rate	Profit factor	Max DD	Sharpe
+44.75% (5 yrs)	55.43%	1.64	5.81%	0.89

Concept & edge. A 4-phase state machine: volatility-expansion channel signals the trend, then the system *waits for the pullback window* and enters on the breakout of the pullback — never chasing. ATR-based dynamic risk throughout. One of the rare fully-documented, code-available gold intraday systems with 175 trades over 5 years.

Setup. M5 XAU/USD; volatility-expansion channel; ATR(14) for stops/targets; 4-phase state logic (idle → trend detected → pullback window → entry).

Entry rules. Trend signal fires → wait for pullback into the channel zone → enter on the breakout of the pullback bar in the trend direction. ~3 trades/month — patience is coded in.

Exit rules. ATR-dynamic stop (initial ~1× ATR beyond pullback extreme); ATR-scaled targets; average win \$1,187 vs average loss \$913 (1.3:1 documented).

Risk management. The documented max DD of 5.81% comes from strict ATR sizing and low frequency — ~3 trades/month means losing streaks are survivable by design. Expectancy \$251/trade documented.

Backtest evidence. July 2020–July 2025, \$100k start: +44.75% total, 175 trades (97W/78L), PF 1.64, Sharpe 0.89, max DD 5.81%, ~8.95%/yr — with full code and metrics published for verification.

Works / fails. Works on gold's trending M5 structure in normal-to-high volatility. Fails in dead ranges (few signals — fine) and during headline whipsaws (the pullback window offers some protection).

Backtesting notes. The repo is a template for professional backtesting practice: state machine, dynamic risk, full metrics documentation. Clone it, verify it, then modify one variable at a time.

Variations. Port to silver (higher beta, wider ATR); H15 version for fewer-noise signals; add session filter (London/NY only).

S95 Gold London/NY Breakout

Market: XAU/USD • **Timeframe:** M15–H1 • **Style:** Session breakout (gold-tuned) • **Evidence:** Grade B (session-structure documentation; practitioner backtests)

Win rate	Typical RR	Key windows	Reference levels	Style
48–58%	1.5–2:1	London 08:00, COMEX 13:30 UTC	Asian range, pre-London range	Dual-session breakout

Concept & edge. Gold's two ignition windows: the London open (physical/LBMA flows) and the COMEX open + US data window (13:30 UTC). Range breaks at these windows, referenced against the Asian range (S84), carry gold's most reliable intraday momentum.

Setup. Mark the pre-London range (Asian session) and the pre-COMEX range (London morning). ATR context; US calendar flagged.

Entry rules. London play: break of the Asian range after 07:30 UTC with M15 close confirmation. COMEX play: break of the London-morning range after 13:30 UTC, direction confirmed by DXY/yields if possible.

Exit rules. Stop at the opposite side of the reference range. Targets: 1.5–2× range height; trail H1 swings on strong days. Flat by NY afternoon.

Risk management. Gold wicks: require *close-based* triggers, not wick touches. On US data mornings, either wait 15 min post-release or switch to the S73 CPI playbook. Risk 0.5–0.75%.

Backtest evidence. Gold session-breakout practitioner backtests: 48–58% win rates at 1.5–2:1, with COMEX-window breaks outperforming London-only entries in recent high-rate-volatility years. Structurally identical to the documented London-breakout family (S19) with gold-specific tuning.

Works / fails. Works in rate-uncertainty regimes (gold trends). Fails in dead summer tape and on fake-out-heavy FOMC days (stand down or use S72).

Backtesting notes. Test the two windows separately — they have different drivers and may deserve different parameters. DXY correlation filter: measure its standalone value.

Variations. Asian-range fade in London (when range is wide); COMEX + data-surprise combined entry; weekly-range projection (Monday range → week).

S96 Safe-Haven Spike Fade (Gold/Silver)

Market: XAU/USD, XAG/USD • **Timeframe:** M15–H1 • **Style:** Headline-overreaction fade • **Evidence:** Grade C/B (event-fade logic; geopolitical-premium decay literature)

Win rate	Typical RR	Trigger	Danger	Style
55–65%	1–1.5:1	1%+ headline spike stalling	Escalation (real)	Event fade

Concept & edge. Geopolitical headlines spike gold instantly; when the event doesn't actually escalate (most cases), the fear premium decays within hours. Fading the stalled spike sells insurance back to the panicked.

Setup. News feed; M15 chart; spike size ($> 1\%$ or $> 1.5 \times \text{ATR}(H1)$) and, critically, *no follow-through headlines* for 30–60 minutes.

Entry rules. After the spike stalls (M15 lower high / failure to make new highs for 3+ bars), short with stop above the spike high. Only when the headline is rhetorical/anticipated, not kinetic escalation.

Exit rules. Target: 50–100% of the spike retrace. Time stop: end of session. If escalation headlines resume — out instantly, regardless of price.

Risk management. This is judgment trading: misreading a genuine escalation (invasion, strike) as rhetoric is catastrophic. Hard stop always; half size; skip if you can't watch the news feed live. Risk 0.5% max.

Backtest evidence. Geopolitical-risk-premium research documents rapid decay of event premiums when events don't escalate; practitioner spike-fade systems on gold: 55–65% win rates with strict stall-confirmation.

Works / fails. Works on rhetorical escalations, drills, diplomatic spats. Fails on true kinetic events and structural regime breaks (those spikes trend for days).

Backtesting notes. Nearly impossible to backtest honestly (headline classification is human). Forward-test with a news log: record headline, classification, outcome — build your own dataset.

Variations. Silver version (amplified both ways); fade via DXY instead (cleaner sometimes); pairs version: short gold / long silver on fear-spikes (ratio normalization, see S100).

S97 Crude Oil EIA Inventory Trade

Market: WTI (CL), Brent • **Timeframe:** M1–M15, Wednesdays 15:30 UTC • **Style:** Scheduled-event volatility • **Evidence:** Grade B (EIA-impact documentation; practitioner systems)

Win rate	Typical RR	Frequency	Key detail	Style
50–60% (filtered)	1.5–2:1	Weekly	API vs EIA divergence as pre-signal	Event momentum/fade

Concept & edge. The weekly EIA inventory report is crude's NFP. Playbook: (1) the prior evening's API report previews direction — big API/EIA divergences get faded; (2) genuine surprise → first-15-minute direction persists. Structured, scheduled, weekly repetition makes it ideal for systematic practice.

Setup. API (Tue evening) and EIA (Wed 15:30 UTC) expectations; CL M5/M15; prior day's range.

Entry rules. Momentum case (genuine surprise vs expectations): enter with the first-5-minute direction on the first pullback. Fade case (in-line EIA but big API-driven pre-move): fade back toward pre-API levels.

Exit rules. Targets 1.5–2R; stop at the post-release range midpoint/extreme. Flat within 2 hours — inventory effects are fast.

Risk management. OPEC headlines override inventory data entirely — check the calendar. CL ticks are \$10/contract; slippage hurts; size accordingly. Risk 0.5–0.75%.

Backtest evidence. Practitioner EIA systems: 50–60% win rates on classified setups; inventory-surprise impact on WTI is well documented in energy-market literature. Unclassified "trade every report" versions degrade toward breakeven after costs.

Works / fails. Works in inventory-driven regimes (balanced markets). Fails when macro (dollar, rates, OPEC) dominates oil pricing and on report-day technical glitches/delayed prints.

Backtesting notes. Build the surprise database (API, EIA consensus, actual, 5/15/60-min forward returns) — 52 events/year accumulate fast into a real edge map.

Variations. Refined-products (RBOB/HO) relative play; the "Wednesday range" as reference for the rest of the week; options straddle for defined risk.

S98 Crude Opening Range Momentum (CL)

Market: WTI futures (CL) • **Timeframe:** M5, first 30–60 min of NY • **Style:** Commodity ORB • **Evidence:** Grade B (ORB family; commodity-session documentation)

Win rate	Typical RR	Best days	Reference	Style
45–55%	1.5–2:1	Post-EIA, OPEC weeks, macro trends	First 30-min range	ORB variant

Concept & edge. Crude's NY floor hours concentrate commercial hedging flow; the opening 30 minutes set the day's value area. CL trends harder than equity indices on macro days — its ORB skews more trend-day-friendly than ES's.

Setup. First 30-min range of the NY session (14:00–14:30 UTC summer); daily ATR and macro calendar as context.

Entry rules. Break of the 30-min range on M5 close with volume; trend-aligned preferred (above/below daily pivot or 20-day SMA).

Exit rules. Stop: opposite range edge. Targets: 1.5R and 2.5R with trail; CL trend days travel 2–3× the opening range regularly.

Risk management. CL's tick value makes stops expensive — use the micro (MCL) for accounts under \$25k. Avoid range entries on EIA mornings (merge with S97 instead). Risk 0.5–0.75%.

Backtest evidence. Commodity-session studies document CL's trend-day frequency exceeding equity indices in macro-active years; ORB-family statistics (S14–S18) transfer with wider ranges and faster follow-through.

Works / fails. Works in OPEC/active-macro regimes. Fails in storage-report-driven chop and thin summer rebalancing periods.

Backtesting notes. Compare CL ORB stats by macro-regime buckets (contango/backwardation, OPEC meeting proximity) — regime conditioning matters more in commodities than equities.

Variations. 60-min range for fewer/cleaner signals; Brent/WTI spread as confirmation; overnight (Globex) range as the reference instead.

S99 Commodity Trend-Day Continuation (Second-Day Momentum)

Market: Gold, crude, silver, copper • **Timeframe:** Daily context → intraday entry • **Style:** Multi-day momentum • **Evidence:** Grade B/A (time-series momentum literature is *strongest* in commodities)

Win rate	Typical RR	Evidence base	Hold	Style
50–58%	1.5–2:1	TSMOM literature (Moskowitz et al.)	1–3 days	Continuation

Concept & edge. Time-series momentum — assets continue in their recent direction — is academically documented as *most reliable in commodities* (supply/demand adjustments are slow; inventory cycles persist). The intraday expression: after a big commodity trend day, trade next-day continuation with session-level timing.

Setup. Day 1: commodity closes directionally with range $> 1.5 \times \text{ATR}(14)$. Day 2: use session structure (ORB, VWAP pullback) for entry timing in day-1's direction.

Entry rules. Day-2 entry on the first session pullback/consolidation break in day-1's direction (combine S23/S33 mechanics). No entry if day 2 opens with a full reversal gap.

Exit rules. Stop at day-2 structure. Targets: 1.5–2R or day-1 range projected. Time stop: 2–3 days.

Risk management. Commodity gaps (overnight limit moves in ags/energy) can jump stops — size with gap assumptions, not just stop distance. Risk 0.75–1%.

Backtest evidence. Moskowitz-Ooi-Pedersen time-series momentum: significant continuation across 58 instruments with commodities among the strongest; 1–12-month horizon canonical, with documented short-horizon (daily) persistence in energy and metals.

Works / fails. Works in supply-shock and macro-trend regimes (the 2021–2022 commodity run was a golden era). Fails in mean-reverting inventory-balanced markets.

Backtesting notes. Test on a *basket* of commodities (the literature's edge is cross-sectional + time-series combined). Single-market versions are noisier.

Variations. Weekly continuation (position-trading version); sector-basket confirmation (energy complex agrees); combine with S66's daily coil.

S100 Gold–Silver Ratio Mean Reversion

Market: XAU/USD vs XAG/USD (or GC/SI futures) • **Timeframe:** H4–Daily • **Style:** Cross-commodity stat-arb • **Evidence:** Grade B (ratio's historical behavior documented; practitioner systems)

Win rate	Typical RR	Ratio context	Structure	Style
55–65% (extreme readings)	1.5–2:1	Trade extremes, not mid-range	Long cheap leg / short rich leg	Metals pair

Concept & edge. The gold/silver ratio oscillates in long waves (historically ~50–100+). At extremes, the metals converge: silver's higher beta overcorrects in both metals rallies and panics. Trading the ratio's extremes = pairs trading (S86) applied to the oldest monetary pair on earth.

Setup. Ratio = gold price / silver price; 2–5-year context; z-score or percentile of the ratio. Both legs executed (futures or CFDs).

Entry rules. Ratio at > 90th percentile of its 2-year range: short gold / long silver (betting compression). < 10th percentile: opposite. Enter progressively (⅓ now, ⅓ deeper, ⅓ at historical extreme).

Exit rules. Target: ratio return to its 1-year median. Stop: ratio exceeding the 5-year extreme (regime break). Time horizon: weeks — this is the book's slowest "intraday-adjacent" strategy.

Risk management. Ratio trends can run for months (2020's spike to 120+) — progressive entries and the 5-year-extreme stop are survival gear. Leg sizing by volatility parity (silver moves ~1.5–2× gold daily). Risk 0.75% per tranche.

Backtest evidence. Practitioner ratio systems at documented extremes: 55–65% win rates over multi-week holds; the ratio's long-run range-bound-then-regime character is well documented (dozens of public long-run charts/analyses).

Works / fails. Works at true extremes with patience. Fails mid-range (no edge — most traders' mistake) and during monetary-regime breaks (silver demonetization narratives, CB gold buying waves).

Backtesting notes. Use ratio percentile, not absolute levels (the "normal" ratio has drifted across eras — 15 in 1900, ~60 in 2000s, 80+ in 2020s).

Variations. Miner-based version (GDX/GDXJ or gold-miner/silver-miner pairs); platinum/palladium ratio as a second pair; options-structured version (ratio call/put spreads).

PART XI

Appendices

The four appendices turn the 100 pages into a working system: a code-ready backtesting checklist, the research loop for generating new strategies, the risk-management constants that keep you alive, and an honest account of how strong the evidence behind each page really is.

Appendix A The Backtesting Blueprint (Code-Ready Checklist)

Appendix B Generating New Strategies (The Research Loop)

Appendix C Risk Management Constants

Appendix D Honest Notes on Evidence

Appendix A The Backtesting Blueprint (Code-Ready Checklist)

Use this exact workflow for every strategy page in this book. Skipping steps is how fake edges are born.

- 1. Translate rules to pseudo-code first.** If any rule contains "strong", "clean", "obvious" or "good volume" — stop. Mechanize it (e.g., "strong" → $\text{body} > 0.6 \times \text{ATR}(14)$). Unmechanizable rules can't be validated.
 - 2. Data discipline.** Use the resolution the strategy trades (M1 systems need M1 or tick data). Include delisted instruments where relevant (survivorship bias). Verify timezone handling — session strategies break on DST bugs more than any other single cause.
 - 3. Cost modeling (non-negotiable).** Per-instrument spread (time-varying if possible), commission, slippage (1–3 ticks stress test), borrow/funding for shorts and perps. Published disasters to remember: 3 pts of slippage turned a winning DAX ORB negative; 0.1% fees killed a viral 5-min crypto system.
 - 4. Validation stack.** In-sample optimize → out-of-sample verify → walk-forward re-optimize → paper trade 30 sessions. Demand: parameter *plateaus* (neighbor parameters also profitable), not peaks.
 - 5. Metrics that matter.** Expectancy per trade in R (target $> +0.2R$ after costs); profit factor (> 1.3 viable); max drawdown (size for $1.5\times$ backtested); exposure-adjusted return ($\text{CAGR} \div \text{time-in-market}$); trade count (> 100 minimum, > 300 preferred); losing-streak distribution (can you survive the 95th-percentile streak psychologically and financially?).
 - 6. Kill criteria (decide in advance).** E.g., "retire if live expectancy $< 50\%$ of backtested for 100 trades" or "retire if drawdown exceeds $1.5\times$ backtested max". Written kill criteria prevent zombie strategies.
- Minimal Python skeleton.** pandas for bars; vectorized signal column; event loop for fills (next-bar open, never same-bar close unless you model MOC); costs subtracted per trade; metrics via quantstats. Every strategy in this book fits this skeleton.

Appendix B Generating New Strategies (The Research Loop)

This book's pages are starting points. The loop that created them works for you:

- 1. Start from an anomaly, not an indicator.** Every durable edge here traces to a *behavioral or structural* cause: forced flow (liquidations, MOC), hedging pressure (overnight premium), calendar flows (turn-of-month), incentive structures (VWAP benchmark), or psychology (3-day panic). Ask "who is forced to trade, and why?" — that's where edges live.
- 2. Measure the base rate before building entries.** P(up day | 3 down days, above 200 SMA) vs unconditional P(up day). P(gap fill | gap < 0.2%) vs size distribution. If the conditional base rate doesn't beat the unconditional by a wide margin, no entry trick will save the idea.
- 3. Condition on regime.** Most strategies here are regime-conditioned (ADX gates, day-type classifiers, volatility percentiles, session windows). When your system fails, the first research question is "which regime filter would have prevented those trades?" — this loop is documented throughout the book (S30, S34, S45).
- 4. Combine orthogonal edges.** The documented blends outperform: momentum + mean reversion (Sharpe 1.71 vs 1.0/2.3 alone, crypto study), IBS + RSI (78% vs 68% WR), ORB + daily filter (0.04% → 0.27% per trade). Orthogonality test: do the two signals fire on the same days? If yes, you have one signal, not two.
- 5. Publish to yourself.** Log every test — including failures (like S41's Stochastic autopsy). A documented negative result saves you from re-testing the same dead idea every six months and teaches more than another cherry-picked winner.

Appendix C Risk Management Constants

- **Risk per trade:** 0.25–0.5% (scalping/high frequency), 0.5–1% (intraday), 1% max (low frequency). Never more because a setup "looks certain" — certainty is not a variable in expectancy math.
- **Daily loss limit:** 2–3R. Hit it, stop. Chop days punish persistence more than they reward it.
- **Portfolio heat:** max 4–6% total open risk; correlated positions count double (3 gold-longs = 1 position).
- **Sizing formula:** Position size = (equity × risk%) ÷ stop distance. The stop is set by the chart (structure/ATR), never by the size you want to trade.
- **Drawdown protocol:** at –10% equity, halve size; at –15%, stop and re-validate every live strategy against fresh data. Edges decay; regimes change; the protocol survives both.
- **Leverage ceiling (crypto/perps):** $\leq 3\times$ on directional systems, $\leq 2\text{--}3\times$ with collateral buffer on delta-neutral. Liquidation is not a stop loss — it's a fee plus a stop loss plus a margin call.

Appendix D Honest Notes on Evidence

- **Grade A** pages cite named published backtests, peer-reviewed papers, or books (Connors & Alvarez 2008; Crabel 1990; Faith 2007; Lucca & Moench; Padyšák & Vojtko 2022; Gao-Han-Li-Zhou intraday momentum; ScienceDirect 2025 funding-arb study; PMC 2023 turn-of-candle; Dzhabarov & Ziemba 2019; the 1.17M-trade ORB Setups study; the 14-year DAX study; QuantifiedStrategies' public backtest archive; the 2020–2025 XAU/USD open-source backtest; the 2022–2025 five-stock gap-fill study). Figures are quoted as published, with their sample periods.
- **Grade B** pages report consensus ranges from multiple practitioner/documentated sources; no single authoritative study exists, and your own validation is expected.
- **Grade C** pages are frameworks used professionally with sparse public statistics; the numbers shown are hypotheses to test, not results to trust.
- **Every** number in this book is historical. Markets compete away published edges — assume decay, re-validate annually, and never risk money on a page you haven't personally re-tested. This book is for education and research; it is not financial advice.